

Ophthalmic Agents, Topical

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The following section lists the majority of veterinary-labeled ophthalmic topical products and some of the more commonly used human-labeled products in veterinary medicine; written by Alison Clode, DVM, DACVO, based on previous work by Gigi Davidson, DICVP, and Michael Davidson, DVM, DACVO. Section editor: James A. Budde, PharmD, DICVP. Drugs are listed in alphabetical order.

For additional information, an excellent review on veterinary ophthalmic pharmacology and therapeutics can be found in the following textbooks: *Slatter's Fundamentals of Veterinary Ophthalmology*, 6th Edition. Maggs DJ, Miller PE, and Ofri R, Eds. Saunders, 2017 and *Veterinary Ophthalmology*, 5th Edition; Gelatt KN, Gilger BC, Kern TJ, Eds. Wiley-Blackwell, 2013.

Routes of Administration for Ophthalmic Drugs

The route of administration selected to deliver therapy for an ocular condition is critical to successful therapy. The following table lists advantages and disadvantages of each route of administration for ocular medications. **NOTE:** Unless otherwise indicated, **small animals** (dogs and cats) receive 1 drop of solutions per administration, or $\frac{1}{4}$ – $\frac{1}{2}$ inch of ointments per administration; **horses** receive 1 – 2 drops (administered directly to the eye) or 0.1 mL (administered via subpalpebral lavage (SPL) catheter or squirted directly into the eye) of solutions, or $\frac{1}{4}$ – $\frac{1}{2}$ inch of ointments per administration.

Route	Target tissues	Dosage forms	Advantages	Disadvantages	Comments
Topical	Conjunctiva, cornea, anterior uvea, lids	Solutions, suspensions	- General ease of administration - Minimal interference with vision - Low incidence of contact dermatitis - Minimal toxicity to interior of eye in presence of penetrating wound	- Imprecise dosing - Potential for contamination - Short contact time with eye - Requires more frequent application than ointment - Diluted by tearing - Systemic absorption - Difficult to administer to horses without SPL catheter placement	- Doses greater than 1 drop rarely indicated in small animals (maximum tear capacity is 10 – 20 μLs, volume of a drop is 25 – 50 μLs) - Allow 5 minutes between drops - Instill in order of least viscous to most, solutions prior to suspensions, and aqueous base prior to oil base - Owners should be counseled to avoid contact of applicator tip with eye.
		Ointments	- Longer contact time - Less frequent administration than drops - Protect cornea from drying - Not diluted by tearing	- Contribute to volume of ocular discharge - Temporarily blur vision - More difficult to administer - Contact dermatitis - Should not be applied to penetrating corneal wounds as vehicles cause granulomatous uveitis - Difficult to determine exact dose - Metal tubes may fatigue and split before all medication is used.	- Owners should be counseled to avoid contact of application tube with eye. - Observe patient for brief period after application due to temporarily blurred vision - Instill solutions and suspensions prior to ointments
Subconjunctival injection	Cornea, anterior uvea	Sterile solutions and suspensions	- High local concentrations with smaller total dose minimizes systemic effects - Circumvents poor patient compliance with topical administration	- Uncertain drug disposition following administration - May create scar tissue - Drug cannot be removed once applied. - Drug vehicle residues may persist. - Inadvertent globe perforation	- Indicated for poorly compliant owners and/or uncooperative patients - Indicated for drugs with poor corneal penetration
Intracameral injection	Anterior segment	Sterile, non-preserved solutions and suspensions	Allows very high local drug concentrations	- Risk for infection - Risk for hemorrhage, retinal detachment, cataract formation, and retinal degeneration - Risk for intraocular toxicity	- Indicated to dissolve fibrin and blood clots in the anterior chamber (eg, tPA) - Indicated for intraocular surgery (local anesthetics, viscoelastic substances) - With caution, indicated for severe intraocular infections

Route	Target tissues	Dosage forms	Advantages	Disadvantages	Comments
Intravitreal injection	Posterior segment	Sterile, non-preserved solutions and suspensions	Allows very high local drug concentrations	- Risk for infection - Risk for hemorrhage, retinal detachment, cataract formation, and retinal degeneration - Risk for intraocular toxicity	- Indicated to manage end-stage glaucoma (gentamicin/steroid) - With caution, indicated for severe intraocular infections - With caution, indicated for intractable uveitis
Retrolbulbar and peribulbar injection	Posterior segment, optic nerve	Sterile solutions and suspensions	Allows very high local drug concentrations	- Risk for infection - Risk for globe perforation, optic nerve trauma, retrolbulbar hemorrhage	- Indicated prior to enucleation (local anesthetics) - Indicated for intraocular surgery when neuromuscular blockade is not possible
Systemic drugs (oral, SC, IM, IV)	Lids, posterior segment, optic nerve, anterior uvea (occasionally)	Tablets, capsules, sterile solutions and suspensions	Allows drug penetration to areas topical therapy will not reach	- Systemic effects, including toxicities, occur - Limited penetration into the cornea	See specific monographs for use of systemic agents.

Principles of Compounding Ophthalmic Products

GIGI DAVIDSON, BSPharm, DICVP

Physiochemical Considerations for Compounding Ophthalmic Preparations

The availability of suitable commercially available products for every veterinary ophthalmic indication is highly unlikely. Many agents used in veterinary ophthalmology are no longer or were never commercially available. Examples of agents that are commonly used by veterinarians but are not currently commercially available include acetylcysteine, amphotericin B, chloramphenicol, cidofovir, tacrolimus, and voriconazole ophthalmic solutions and/or ointments. Even if commercially available, products may be of inappropriate concentration to achieve a therapeutic effect in a given patient (eg, cyclosporine) or may have agents and excipients that may have adverse effects in animal patients (eg, polymyxin B in cats). In other cases, no product is commercially available and must be compounded from other nonophthalmic drugs or from bulk chemicals (eg, acetylcysteine ophthalmic solution and disodium edetate ophthalmic solution). For these reasons, pharmacists are frequently called upon to compound products to be used in the animal eye. These products may be administered topically in the form of solutions, suspensions, or ointments or administered by periocular or intraocular injection, by drug-implanted collagen shields, or by drug-impregnated disposable contact lens delivery systems. The quality and sterility of these products are critical, and preparations for injection into the eye must meet endotoxin limits. To ensure adequate stability, uniformity, and sterility, both the American Society of Health-Systems Pharmacists and the United States Pharmacopoeia (USP) Convention have published standards (eg, USP General Chapter <797> *Pharmaceutical Compounding—Sterile Preparations*) for pharmacy-prepared ophthalmic products. These standards address the following areas of concern.

Validation of Formulation

Before compounding any preparation for ophthalmic use, the pharmacist should obtain documentation that substantiates the stability, safety, and benefit of the requested formulation. Pharmacists may call the manufacturer of the drug, refer to primary literature, call regional eye centers, or call professional compounding organizations to obtain such information. If no such documentation is available, the pharmacist must employ professional judgment in determining

a suitable formulation for ophthalmic administration and then apply the most conservative beyond-use dates established in USP <797>. Factors to consider when making this judgment include sterility, tonicity, endotoxin burden, pH and buffering, the toxicity of the drug, need for preservatives, solubility, stability in the chosen vehicle, particle size in suspensions, viscosity, packaging, and any precautions necessary to keep drug residues from occurring in any food-producing animals.

Documentation

A readily retrievable document (electronic or written), known as a Master Formulation Record, must be maintained for each ophthalmic preparation compounded. This Master Formulation Record must indicate the name of the preparation, the dosage form, the specifications and source of each ingredient used, the weights and measures of each ingredient used, the equipment required, a complete description of each step in the compounding process with a special notation of aseptic techniques to be utilized and which method of sterilization is employed, beyond-use dating, storage requirements, specific packaging requirements, sample label and auxiliary labeling, quality control testing that must be performed (eg, pH, sterility, endotoxin testing for injections), and references for the formula. The expected visual appearance of the final preparation must also be described in the Master Formulation Record. Compounding Records for each compound prepared must also be made (electronic or written) and must include the date of compounding, lot or batch number assigned, the quantity of each ingredient used to prepare the compound, the manufacturer, lot number, and expiration date of each ingredient used, results of any quality testing, a sign-off provision for compounder and checker, the amount compounded, and the projected beyond-use date for the compounded preparation.

Sterility

Ophthalmic dosage forms must be compounded in aseptic conditions. Sterile compounding standards must be consulted in the United States Pharmacopoeia General Chapter <797>. Sterility is the most important consideration for ophthalmic products. Contaminated ophthalmic products can result in eye infections leading to blindness or even loss of the eye, especially if pathogens, such as *Pseudomonas*,

are present. Eye infections from contamination can also lead to systemic infections requiring hospitalization and may even result in death. All ophthalmics should be compounded in a primary engineering control (PEC; eg, laminar flow hood or compounding aseptic isolator) that maintains an ISO 5 level of air quality. PECs must be certified at least annually for acceptable performance and maintenance of expected environmental quality. The compounding pharmacist must also be appropriately garbed as described in USP <797> and must use impeccable aseptic technique when handling products intended for use in the eye. All products must be rendered sterile after formulation in the laminar flow hood and maintain sterility throughout their labeled beyond-use date.

Sterilization of the final preparation for ophthalmic solutions is most easily achieved through filtration through vehicle-compatible 0.2 µm sterilizing filters, which also remove particulate matter. Note that this method is not suitable for ophthalmic suspensions or ointments. Ophthalmic suspensions and ointments must be sterilized by other means (such as autoclaving, dry heat, or irradiation) to avoid filtering out active drugs. Preservatives may also be added to prevent bacterial growth, especially if the container is intended for multiple uses. FDA has established maximum potency per unit dose thresholds for preservative agents in FDA-approved ophthalmic products, which can be found at <https://www.accessdata.fda.gov/scripts/cder/iig/index.cfm>, USP <797>, places limits on beyond-use dates for nonpreserved aqueous ophthalmic preparations. The preservative selected must be compatible with the active drug and excipients as well as nontoxic to the eye or to the patient.

Clarity

Drugs prepared as ophthalmic solutions should meet requirements for particle size and quantity as standardized in USP <789> *Particulate Matter in Ophthalmic Solutions*. Drugs prepared as ophthalmic suspensions cannot be filtered but must be of a particle size that does not irritate or scratch the cornea, typically less than 25 µm. A general rule of thumb is that particles that can be felt when rubbing a drop of the preparation between the thumb and forefinger are too large to be used in the eye. The use of an ointment mill is highly recommended to decrease the particle size for ophthalmic ointments.

Tonicity

Ophthalmic products do not need to be isotonic if the contact time with the cornea is only a few minutes. The eye can tolerate a range of 200 to 600 mOsm/L for short periods of time. Some ophthalmic preparations are intentionally hypertonic in order to reduce corneal edema. For ointments, irrigations, and products that will remain in contact with the eye longer than a few minutes, isotonic products should be used. Hypotonic agents may cause corneal edema, and hypertonic agents may dehydrate the cornea and cause pain. Tear fluid and normal saline have identical osmotic pressures, making 0.9% sodium chloride an excellent vehicle for ophthalmic products. For products that are hypotonic, sodium chloride equivalencies can be used to determine how much sodium chloride is necessary to render the product isotonic. An extensive list of sodium chloride equivalents for drugs has been published.¹

Buffering and pH

Ophthalmic preparations are generally buffered in a range from 4.5 to 11.5. Buffering is necessary to provide maximal stability of the drug or for the comfort and safety of the patient. Alkaloids, such as atropine and pilocarpine, are usually buffered. If the activity and stability of the drug are not pH-dependent, and the pH of the product is not irritating, then buffers may be omitted from the formulation. Commonly used buffers for ophthalmic preparations include Pal-

itsch buffer, boric acid buffer, boric acid/sodium borate buffer, sodium acetate/boric acid buffer, Sorensen's modified phosphate buffer, Atkins and Pantin buffer solution, Feldman buffer, and Gilford ophthalmic buffer. Formulations for these solutions and ratios required to achieve the desired pH have been published.²

Viscosity Enhancers

Because tears and blinking reflexes reduce the total amount of drug available for penetration, an increase in residence time in the eye will increase drug absorption. Increasing the viscosity of the drug is the most common way to prolong contact time. Methylcellulose derivatives (eg, carboxymethylcellulose, hydroxyethylcellulose, and hydroxypropylmethylcellulose) are the most commonly used agents. Polyvinyl alcohol and polyvinylpyrrolidone have also been used. The FDA has established maximum potency per unit dose thresholds for carboxymethylcellulose and for viscosity agents in FDA-approved ophthalmic products, which can be found at <https://www.accessdata.fda.gov/scripts/cder/iig/index.cfm>.

Quality Control

Finished preparations should be thoroughly inspected visually for clarity and uniformity of suspension. The pH of the final preparation should always be checked, and the value should be recorded on the master formula record for that batch. Most compounded preparations should have a pH of 5 to 7 unless otherwise indicated for stability or penetration of ocular tissue. Sterility testing and endotoxin testing should be performed where required.

Packaging

Ophthalmic preparations should be packaged in sterile dropper bottles (glass or plastic), or individual doses can be placed in sterile syringes with sterile tip caps. Ointments should be packaged in sterile ointment tubes and heat-sealed. Preparations that are sensitive to light and moisture should be packaged in tight, light-resistant containers.

Beyond-Use Dating

Beyond-use dates must be assigned according to the limits specified in the most current official USP Chapter <797>. Limits are assigned based on the risk for contamination according to the method of preparation, use of nonsterile ingredients, and presence or absence of preservatives.

Considerations for the Use of Ophthalmics in Veterinary Patients

Veterinary patients experience many of the same ophthalmic diseases and conditions as humans, and treatments are often based on human therapy. Animals, however, have a variety of species-related characteristics that might cause human-designed therapies to fail or be toxic. Behavioral characteristics, such as grooming, may significantly reduce the contact time of ophthalmic agents with the eye and increase systemic exposure through ingestion. Anatomical differences, such as size and tear capacity volume, must be considered when formulating ophthalmic therapies. Ease of administration must also be considered. Horses and other large animals may simply elevate their eyes out of a caregiver's reach if ophthalmic treatments are objectionable. Specialized delivery devices have been created to treat these patients. Subcutaneous palpebral lavage systems are tunneled under the skin over the animal's brow into the upper eyelid and allow for the passage of medication through long catheters that are easily reachable by caregivers. Food-producing animals require special consideration, as systemic absorption of ophthalmic agents in

animals could result in violative drug residues in food intended for human consumption. Finally, the use of active pharmaceutical ingredient powders (ie, bulk drug substances) to prepare compounds for animals is allowed only under the regulatory discretion provided by CVM FDA's Guidance for Industry #256.

General Principles of Ocular Penetration

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Corneal penetration

Drugs must generally be administered topically to treat corneal and intraocular conditions. Although the eye would appear to be an easy target for topical administration, the eye has several anatomical barriers to prevent penetration by foreign substances. Instantaneous tear production, strong blinking reflexes, and alternating layers of lipophilic and hydrophilic tissue all work in conjunction to prevent the entry of foreign substances. The clear tissue known as the cornea covers the visible outer surface of the eye between the lids. The cornea must be clear to allow for vision, and nature has accomplished this by omitting blood vessels in the cornea. Because of this lack of vascular tissue, systemically administered drugs rarely penetrate the cornea. Additionally, the cornea is composed of several layers of lipophilic (outer layers) and hydrophilic (inner layer) tissue. For a topically administered drug to fully penetrate the cornea, the drug must be able to exist in ready equilibrium between both ionized and nonionized forms in order to cross these layers (eg, chloramphenicol, atropine, and pilocarpine). Most antibiotics are water-soluble and will not penetrate the lipophilic outer layer of the cornea unless ulcers are present. Small molecular weight (less than 350) and high local concentration of drugs will increase penetration, however, even if the drugs are ionized and hydrophilic. Thus, topical administration is ideal as it allows for very high local concentrations of the drug on the cornea. For a topically administered drug to reach the anterior chamber and bind to intraocular structures (eg, ciliary body, iris, aqueous humor) to exert a clinical effect, the drug must pass through the cornea. Drugs may also reach the anterior chamber to some extent by passive absorption through the conjunctiva.

Key points for corneal penetration of drugs:

- Lipophilic
- Equilibrium between ionized and nonionized forms
- Small molecular weight (less than 350)
- High local concentrations

Intravitreal Penetration

Topically administered drugs reach the vitreous only in very small concentrations. To treat severe conditions of the anterior chamber (uveitis) as well as intravitreal conditions, drugs may be administered by periocular or intraocular injection. The periocular routes include subconjunctival injection and sub-Tenon's membrane injection, while the intraocular routes are intracameral injection (directly into the aqueous humor) or intravitreal injection (directly into the vitreous humor). Periocular injections can be administered under sedation and topical anesthesia. Intraocular injections are usually only performed while the patient is very heavily sedated or under general anesthesia. Both periocular and intraocular routes bypass the outermost chemical and physical ocular defenses and allow for a better concentration of drugs in the vitreous. The volume of administration for these routes is relatively small: periocular injections should not exceed 0.5 to 1 mL in small animals and 2 mL in large animals, and intraocular injections should not exceed 0.1 mL in

References

For the complete list of references, see wiley.com/go/budde/plumb

small animals and 0.25 mL in large animals because of the risk for increased intraocular pressure. Drugs injected into the eye should be free of preservatives and buffers and must be sterile and meet endotoxin limits for injections.

Route of Therapy for Given Ocular Target

Target tissue	Routes of administration
Eyelids	Topical, systemic
Corneal surface	Topical
Anterior segment	Topical if good penetration or mild disease Systemic if poor penetration or severe disease
Posterior segment	Systemic or intraocular injection (rarely)

Questions to Ponder Prior to Compounding Ophthalmic Products

- Where is the target of therapy? (eyelids, corneal surface, cornea, anterior segment, posterior segment)
- What is the character of the drug?
 - Lipophilic? Hydrophilic?
 - What is the molecular weight?
 - What is the inherent toxicity of the drug to the eye (gentamicin)? To the caregiver (chloramphenicol)? To the patient (polymyxin B in cats)?
 - Are there data to support what concentration is necessary for corneal penetration?
 - Is the drug soluble in a vehicle that is not toxic to the eye?
 - If not soluble, will the particle size of the suspension or the ointment scratch the corneal or conjunctiva?
 - What is the pH of the final product? Is this in an acceptable range to avoid irritation (4.5 to 11.5)?
 - What is the tonicity of the final product? Hypertonic? Hypotonic? How long will the product be in contact with the cornea if not isotonic?
 - Will the viscosity need to be enhanced to prolong contact with the eye? Which agent is compatible?
 - What is the duration of therapy? Will the product require preservation if used long-term and/or multiple times? Which preservative is compatible?

Topical Ophthalmic Agents Listed by Class/Indication

NOTE: Some agents are listed in more than one category.

Analgesics (see also NSAIDs and Anesthetics, Ocular)

Morphine
Nalbuphine

Anesthetics, Ocular

Benoxinate
Lidocaine
Proparacaine
Tetracaine

Antibacterials, Single-Agent

Amikacin
Azithromycin
Besifloxacin
Chloramphenicol
Ciprofloxacin
Erythromycin
Gatifloxacin
Gentamicin
Levofloxacin
Moxifloxacin
Ofloxacin
Povidone Iodine
Silver Sulfadiazine
Sulfacetamide
Tobramycin
Vancomycin

Antibiotic Combinations

Bacitracin/Polymyxin B
Neomycin/Polymyxin B/Bacitracin
Neomycin/Polymyxin B/Gramicidin
Oxytetracycline/Polymyxin B

Antibiotic Corticosteroid Combinations

Gentamicin/Prednisolone
Neomycin/Polymyxin B with Dexamethasone
Neomycin/Polymyxin B with Hydrocortisone
Neomycin/Polymyxin B/Bacitracin with Hydrocortisone
Sulfacetamide/Prednisolone
Tobramycin/Dexamethasone
Tobramycin/Loteprednol etabonate

Anticollagenase Agents

Acetylcysteine
Edetate Disodium

Antifungals

Amphotericin B
Itraconazole
Miconazole
Natamycin
Silver Sulfadiazine
Voriconazole

Antihistamines (see also Mast Cell Inhibitors)

Alcaftadine
Azelastine
Bepotastine
Cromolyn
Epinastine
Ketotifen
Lodoxamide
Nedocromil sodium
Olapatadine

Antineoplastics

Fluorouracil
Mitomycin

Antiseptics

Povidone Iodine
Silver Sulfadiazine

Antivirals

Acyclovir
Cidofovir
Ganciclovir
Idoxuridine
Trifluridine

Chelating Agents

Edetate Disodium

Corticosteroids, Single-Agent

Dexamethasone
Difluprednate
Fluorometholone
Hydrocortisone
Loteprednol
Prednisolone

Decongestants

Phenylephrine

Diagnostics

Fluorescein Sodium
Lissamine Green
Phenylephrine
Rose Bengal

Fibrinolytics

Alteplase (Tissue Plasminogen Activator [tPA])

Glaucoma, Ocular Hypertension Agents

Alpha-Adrenergic Agonists

Apraclonidine
Brimonidine

Beta-Blockers

Betaxolol
Carteolol
Levobunolol
Timolol

Carbonic Anhydrase Inhibitors

Brinzolamide
Dorzolamide

Miotics, Cholinesterase Inhibitors

Demecarium

Miotics, Direct-Acting

Pilocarpine

Prostaglandin Analogues

Bimatoprost
Latanaprost
Tafluprost
Travoprost

Fixed-Dose Combinations

Brimonidine/Timolol
Brinzolamide/Brimonidine
Dorzolamide/Timolol

Hyperosmotic Agents

Sodium Chloride, Hypertonic

Immunologics

Cyclosporine
Pimecrolimus
Tacrolimus

Irrigating Solutions

Mast Cell Stabilizers

Cromolyn
Nedocromil

Mydriatics and Cycloplegic

Mydriatics

Atropine
Cyclopentolate
Phenylephrine (mydriatic)
Tropicamide

Mydriatic Combinations

Cyclopentolate/Phenylephrine
Hydroxyamphetamine/
Tropicamide

Nonsteroidal

Anti-Inflammatory Drugs (NSAIDs)

Bromfenac
Diclofenac
Flurbiprofen
Ketorolac
Nepafenac

Ocular Lubricants

Surgical Adjuncts

Fluorouracil (5-FU)
Mitomycin

Acetylcysteine Ophthalmic

(a-*see*-tuhl-*sis*-teen)

Anticollagenase Agent

Prescriber Highlights

- ▶ Mucolytic agent with antioxidant and anti-inflammatory effects; used to stop digestion of the cornea by collagenases and proteases
- ▶ Used to improve tear film stability
- ▶ Foul odor is usual and does not affect drug activity

Uses/Indications

Acetylcysteine is used topically to prevent corneal melting caused by proteinases and collagenases that are present when bacterial and/or fungal infection of ulcers occurs. Acetylcysteine at concentrations greater than 0.5% reduces such enzymatic activity in vitro.^{1,2} Additionally, acetylcysteine is beneficial in stabilizing the tear film in quantitative and qualitative tear film deficiency disorders.³

Pharmacology/Actions

Acetylcysteine inhibits collagenase activity by reducing disulfide bonds and chelating the cofactors zinc and calcium. Additionally, acetylcysteine inhibits matrix metalloproteinase-9 (MMP-9) production by corneal epithelial cells.⁴ Although MMPs play a role in initial corneal wound healing, down-regulation is necessary to prevent corneal digestion and allow corneal wound healing.

Contraindications/Precautions/Warnings

Acetylcysteine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Acetylcysteine exposure to the pulmonary tree may result in severe bronchospasm, so this medicine should be used with caution in asthmatic patients.

Adverse Effects

Adverse effects reported in humans or animals include irritation, redness, and tearing of the eyes. If inhaled, acetylcysteine can rarely cause bronchoconstriction and tracheal irritation. Concentrations of acetylcysteine greater than 10% are irritating to the eye.

Acetylcysteine may decrease corneal epithelial migration.⁴

Dosages

Corneal ulcers (extra-label): 1 drop to the affected eye(s) every 1 to 2 hours for the first 24 hours, followed by 3 to 4 times daily until clinical improvement occurs

Dry eye (extra-label): 1 drop to the affected eye(s) every 12 to 24 hours as long as clinical improvement is present

Monitoring

- Clinical efficacy as demonstrated by control of corneal melting and subsequent corneal wound healing, or signs of improved tear film quantity and quality

Client Information

- Use this medication as directed by your veterinarian.
- Wait 5 to 10 minutes after applying this medication before applying any other medications to the eye(s).
- Foul, rotten-egg odor is normal with this drug and does not affect how well it works.
- Store solutions in the refrigerator. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product in the refrigerator (2°C–8°C [35.6°F–46.4°F]) away from moisture and sunlight; do not freeze. Compounded 5% or 10%

solutions of acetylcysteine are reported as stable for 60 days in the refrigerator, protected from light.⁵

Compatibility/Compounding Considerations

Acetylcysteine ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Acetylcysteine ophthalmic solution (1% to 5%) must be compounded by a qualified compounding pharmacist.

References

For the complete list of references, see wiley.com/go/budde/plumb

Acyclovir Ophthalmic

(a-*sye*-klo-veer)

Antiviral

For systemic use of acyclovir, refer to *Acyclovir*

Prescriber Highlights

- ▶ Used as an ophthalmic ointment in cats to treat feline herpesvirus-1 (FHV-1) with variable reported efficacy
- ▶ Requires frequent administration (ie, 5 to 6 times daily)
- ▶ This medication should NOT be used systemically in cats because of myelotoxicity, poor bioavailability, and poor efficacy against FHV-1.
- ▶ In the United States, acyclovir is not commercially available as an ophthalmic preparation; it must be compounded.

Uses/Indications

Acyclovir has been used topically as an ophthalmic ointment in cats with feline herpesvirus-1 (FHV-1), although inferior in vitro efficacy against herpesvirus-1 (FHV-1) and requirements for frequent administration limit its clinical usefulness.^{1–4}

Pharmacology/Actions

Acyclovir is an acyclic nucleoside analogue that inhibits viral replication through the inhibition of feline herpesvirus-1 (FHV-1) DNA polymerase. In the first of 3 activation steps, acyclovir is selectively phosphorylated by viral thymidine kinase, with the 2 subsequent steps performed by host cell enzymes. Poor efficacy against FHV-1 is due to poor selectivity of viral thymidine kinase for FHV-1.

Contraindications/Precautions/Warnings

Acyclovir is contraindicated in patients that have a history of hypersensitivity to it or drugs like it (valacyclovir) or to any component of the formulation. Systemic use of acyclovir is contraindicated in cats as it could lead to leukopenia, acute anemia, and nephrotoxicity.⁵

Adverse Effects

When acyclovir is used systemically, adverse effects reported in humans or animals include bone marrow suppression, nephrotoxicity, and hepatotoxicity. When acyclovir is applied topically to the eyes, mild stinging and blepharitis may occur after application.

Dosages

Feline herpesvirus-1 (extra-label): ¼ – ½ inch strip to the affected eye(s) 5 times daily⁴; decreased frequency of administration is associated with decreased clinical effect.

Monitoring

- Resolution of clinical signs associated with corneal ulceration, corneal vascularization, conjunctivitis
- Baseline and periodic CBC and serum renal chemistries with prolonged topical therapy

Client Information

- This medicine must be given 5 times daily to be effective.
- Store acyclovir according to package instructions. Discard this medicine if the color changes or if it becomes cloudy.
- Do not allow cats to lick this medicine when grooming.

Storage/Stability

Because this compound must be compounded (in the United States), storage recommendations can vary; follow storage requirements listed on the packaging.

Compatibility/Compounding Considerations

Acyclovir ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Acyclovir is not approved in an ophthalmic dosage form, but sterile ophthalmic ointment may be compounded by qualified compounding pharmacists.

References

For the complete list of references, see wiley.com/go/budde/plumb

Alcaftadine Ophthalmic

(al-*kaf*-ta-deen) Lastacast®

Antihistamine

Prescriber Highlights

- ▶ H1 receptor antagonist and mast cell stabilizer
- ▶ Indicated for allergic conjunctivitis
- ▶ May cause ocular burning, stinging, and irritation

Uses/Indications

Alcaftadine may be used in animals to alleviate the clinical signs of allergic conjunctivitis. No reports of clinical efficacy or tolerability in animals have been published.

Pharmacology/Actions

Alcaftadine is a histamine-1 (H₁) receptor antagonist and mast cell stabilizer. Inhibition of histamine receptor activation decreases the early-phase response in allergic conjunctivitis. Mast cell stabilization controls the late-phase allergic response in the conjunctiva by inhibiting the release of cytokines and lipid mediators. In humans, itching is prevented within 3 minutes after administration and this effect lasts up to 24 hours.¹

Contraindications/Precautions/Warnings

Alcaftadine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans include irritation, burning and stinging on application, redness, and pruritus. Evaluation of adverse effects has not been performed in animals. Discontinue use if pruritus worsens. Alcaftadine has demonstrated less cytotoxicity in corneal epithelial cell cultures compared with other topical antiallergic medications.²

Dosages

Allergic conjunctivitis (extra-label): 1 drop to the affected eye(s) once daily

Monitoring

- Clinical efficacy: resolution of ocular pruritus

Client Information

- Use this medication as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the solution. Keep the dropper bottle tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes, becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product at controlled room temperature (15°C-25°C [59°F-77°F]) away from moisture and sunlight.

Compatibility/Compounding Considerations

Alcaftadine ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Alcaftadine 0.25% Ophthalmic Solution (containing alcaftadine 0.25%; benzalkonium chloride 0.005%; edetate disodium; sodium phosphate, monobasic; purified water; NaCl; NaOH and/or HCl to adjust pH to 7, with osmolality of 290 mOsm/kg); 3 mL in sterile ophthalmic dropper bottle; Lastacast®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

**Alteplase Ophthalmic
Tissue Plasminogen Activator (tPA)**

(ahl-teh-plays) Activase®

Fibrinolytic

Prescriber Highlights

- ▶ Used via intracameral (anterior chamber) injection to lyse intraocular fibrinohemorrhagic clots
- ▶ Does not lead to fibrinolysis when administered topically
- ▶ Toxic to the corneal endothelium in intracameral doses greater than 50 µg
- ▶ Not approved for intraocular injection; must be prepared from the injectable form intended for intracatheter administration

Uses/Indications

Alteplase ophthalmic is most commonly injected intracamerally to lyse intraocular fibrin and blood clots. Intraocular fibrin clots, which form as a result of intraocular inflammation and/or bleeding, can induce synechiae formation and obstruct aqueous humor flow and drainage, thus predisposing affected eyes to the development of glaucoma. Intracameral injection of alteplase in humans following cataract surgery dissolves fibrin clots and reduces posterior synechiae without resulting in re-bleeding.¹

Immediate postoperative intracameral injection of alteplase in dogs undergoing cataract surgery does not significantly impact postoperative development of anterior chamber fibrin or postoperative ocular hypertension.² Topical application of alteplase does not lyse intraocular clots.³ In dogs and cats, intracameral injection of alteplase 25 µg results in fibrinolysis without corneal toxicity, whereas alteplase doses of 50 µg lead to corneal endothelial toxicity.^{3,4}

Pharmacology/Actions

Alteplase (tissue plasminogen activator) is an enzyme (serine protease) that binds to fibrin in clots, which converts the plasminogen component of the clot to plasmin, thus inducing local fibrinolysis.

Contraindications/Precautions/Warnings

Tissue plasminogen activator is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. The most frequent adverse reaction associated with thrombolytics in approved indications is bleeding. Caution should be exercised with patients that have thrombocytopenia, other hemostatic defects (including those secondary to severe hepatic or renal disease), or any condition for which bleeding constitutes a significant hazard.⁵ Intracameral doses of 50 µg have resulted in toxicity to the corneal epithelium in dogs and cats.^{3,4}

Adverse Effects

Adverse effects reported in humans or animals include increased risk for bleeding following administration. Corneal toxicity is likely with intracameral alteplase doses of 50 µg but does not occur with doses of 25 µg.^{3,4}

Dosages

Intraocular fibrinolysis (extra-label): Intracameral injection of 0.1 mL of a 0.25 mg/mL sterile solution (25 µg total dose)

Monitoring

- Clinical efficacy
- Resolution of the intraocular clot, which generally occurs within 24 to 48 hours.⁶ Observe the patient for signs of re-bleeding.

Client Information

- This medication is not administered outside of veterinary clinics.

Storage/Stability

Refer to individual packaging for storage information; generally, this product is kept under refrigeration (2°C–8°C [36°F–46°F]). Protect the lyophilized material during extended storage from excessive exposure to light. After reconstitution, the drug may be stored between 2°C and 30°C (36°F–86°F) and used within 8 hours. Discard any unused solution after administration is complete. Alteplase contains no preservatives. Store aliquots of reconstituted and further diluted solutions in an ultralow (–80°C [–112°F]) freezer for 45 days unless extended stability and sterility testing has been conducted. Do not re-freeze aliquots once thawed. Discard this product if the color changes, if it becomes cloudy, or if there are particles in the solution.

Compatibility/Compounding Considerations

Alteplase ophthalmic preparations should not be mixed directly (in vitro) with any other ophthalmic medications. It is not available as an ophthalmic injection. For intraocular injection, *Cathflo Activase*® is further diluted with normal saline to a final concentration of 0.25 mg/mL.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Alteplase is not approved for ophthalmic use.

Cathflo Activase® is available (containing alteplase 2.2 mg [which includes a 10% overfill], L-arginine 77 mg, polysorbate-80 0.2 mg, and phosphoric acid for pH adjustment. Each reconstituted vial will deliver 1 mg/mL of alteplase at a pH of ≈7.3). For intraocular injection, this solution is further diluted with normal saline to a final concentration of 0.25 mg/mL.

Activase® is available as 50 and 100 mg lyophilized powder vials with diluent (sterile water for injection) included.

References

For the complete list of references, see wiley.com/go/budde/plumb

Amikacin Ophthalmic

(am-uh-**kay**-sin)

Aminoglycoside Antibiotic

For systemic use, see **Amikacin**

Prescriber Highlights

- ▶ Topical ophthalmic aminoglycoside antibiotic used for susceptible ocular bacterial infections that are resistant to other aminoglycoside antibiotics
- ▶ Amikacin is not commercially available as an ophthalmic preparation in the United States; it must be compounded.
- ▶ Cats may be more subject to toxic effects of amikacin than other species.

Uses/Indications

Although an amikacin ophthalmic formulation is not commercially available, it can be compounded for use in ophthalmic infections caused by susceptible strains of *Pseudomonas* spp, *Escherichia coli*, *Proteus* spp, *Klebsiella* spp, and other gram-negative bacteria.¹ Amikacin has activity against penicillinase and nonpenicillinase-producing species of *Staphylococcus* spp, but low activity against other gram-positive organisms. Amikacin is less toxic to the posterior segment compared with other aminoglycoside antibiotics and the injectable dosage form is sometimes used as an intravitreal injection to treat bacterial endophthalmitis.

The World Health Organization (WHO) has designated amikacin as a Critically Important, High Priority antimicrobial for human medicine.² The Office International des Epizooties (OIE) has designated amikacin as a Veterinary Critically Important Antimicrobial (VCIA) Agent in equine species.³

Pharmacology/Actions

Amikacin sulfate, an aminoglycoside, is actively transported into the bacterial cell. It binds to a specific receptor protein on the 30S subunit of bacterial ribosomes. It consequently interferes with an initiation complex between mRNA (messenger RNA) and the 30S subunit, inhibiting protein synthesis by inducing misreading of the mRNA. Aminoglycosides are bactericidal, whereas most other antibiotics that interfere with protein synthesis are bacteriostatic. Although topical administration does not produce substantial aqueous humor concentrations, subconjunctival injections of amikacin solutions may achieve therapeutic concentrations in the cornea and the aqueous humor.⁴

Contraindications/Precautions/Warnings

Amikacin is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

When amikacin is used systemically, it can cause nephrotoxicity and neurotoxicity (ototoxicity and vestibular toxicity). Cats are more susceptible to the toxic effects of amikacin, and it is recommended that all patients receiving topical amikacin be monitored for systemic toxicity. Although uncommon, topical aminoglycoside preparations have been reported to cause local reactions such as lid itching, swelling, and conjunctival erythema in humans.

Dosages

Susceptible ocular infections (extra-label): Apply 1 drop to the affected eye(s) every 1 to 2 hours for the first 24 hours, followed by every 4 to 6 hours, pending clinical improvement

Monitoring

- Clinical efficacy: cytology and culture to evaluate for the presence of bacteria. Clinical resolution of signs of ocular surface infection
- Adverse effects (eg, signs of vestibular or auditory toxicity or nephrotoxicity)
- Urinalysis with sediment to check for casts in the urine, which may be an early indicator for nephrotoxicity

Client Information

- Give this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when this medicine is not in use.
- Wait for 5 minutes after applying this medicine before applying any other medications to the eye.
- Store this medicine according to the directions on the product label, protected from light. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are visible particles in the solution.

Storage/Stability

Store product as directed on the product label, protected from light.

Compatibility/Compounding Considerations

Amikacin sulfate ophthalmic solution should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Amikacin is not approved in an ophthalmic dosage form, but sterile solutions of amikacin 1% to 5% may be compounded by qualified compounding pharmacists.

References

For the complete list of references, see wiley.com/go/budde/plumb

Amphotericin B Ophthalmic

(am-foe-**ter**-i-sin **bee**)

Polyene Antifungal

Prescriber Highlights

- ▶ Antifungal agent used to treat ocular surface fungal infections
- ▶ Not commercially available in an ophthalmic dosage form
- ▶ Solutions made from the formulation for IV injection may be used topically or injected subconjunctivally.

Uses/Indications

Amphotericin B is used primarily in horses to treat fungal keratitis. It is effective against yeast and dimorphic fungal organisms, including *Histoplasma* spp, *Coccidioides* spp, *Candida albicans*, *Blastomyces* spp, *Cryptococcus* spp, *Sporothrix* spp, and *Mucor* spp, with variable activity against filamentous organisms (more efficacy for *Aspergillus* spp than *Fusarium* spp).

Amphotericin B has demonstrated in vitro ability to reduce biofilm formation by *Candida* spp, thus decreasing virulence.¹ Topical administration is minimally effective. There is evidence supporting the beneficial effect of subconjunctival and intrastromal injections in humans.^{2,3}

Pharmacology/Actions

Amphotericin B acts by binding fungal sterols in cell membranes, resulting in leakage of intracellular constituents. Amphotericin B is usually fungistatic but can be fungicidal in higher concentrations.

Contraindications/Precautions/Warnings

Amphotericin B is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans or animals include mild burning, stinging, irritation, or redness. Topical application may be toxic to the corneal epithelium,⁴ whereas subconjunctival injection of nonliposomal forms may induce ocular irritation.⁵

Dosages

Topical administration (extra-label; 0.15% to 0.5% solution): 1 – 2 drops in the affected eye(s), or 0.1 mL in the subpalpebral lavage (SPL) catheter, every 2 to 6 hours, depending upon severity of infection

Subconjunctival injection (extra-label): Inject 0.3 mL of a 0.5 mg/mL solution under the bulbar conjunctiva every 48 to 96 hours.²

Monitoring

- Clinical efficacy and evaluation for clinical signs suggestive of continued infection
- Corneal scraping to determine the resolution of fungal infection

Client Information

- Give this medication to your animal as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication in a refrigerator protected from light.
- Discard this medication if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store the product in the refrigerator (2°C–8°C [35.6°F–46.4°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Amphotericin B should not be mixed with salt-containing solutions and should not be mixed directly (in vitro) with any other drugs. Amphotericin B is not available as an ophthalmic preparation but may be prepared as a topical solution (0.15% to 0.5% in dextrose 5% or sterile water) or as a subconjunctival injection (0.5 mg/mL in sterile water). Historically, only the nonliposomal injection was recommended for compounding topical solutions; however, studies demonstrate that topical liposomal amphotericin B ophthalmic solutions are equally effective and less irritating.⁶ Although stability is concentration-dependent, most solutions are chemically stable for 7 days in the refrigerator protected from light. Additionally, recent evidence suggests that ophthalmic solutions prepared from liposomal amphotericin B retain chemical potency for up to 6 months.⁷

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS:

None are available, but amphotericin B may be compounded by reconstituting the injection as a 0.15% to 0.5% solution and used as an ophthalmic solution. See **Compatibility/Compounding Considerations**.

HUMAN-LABELED PRODUCTS: NONE

References

For the complete list of references, see wiley.com/go/budde/plumb

Apraclonidine Ophthalmic

(a-pra-**cl**oe-ni-deen) *Iopidine*®

Alpha-Adrenergic Anti-Glaucoma Agent

Prescriber Highlights

- ▶ Used to manage glaucoma
- ▶ Uncommonly used in veterinary species
- ▶ Do not use in cats, as bradycardia and vomiting may occur.
- ▶ Results in dry nose and mouth, headaches, cardiac arrhythmias, and fatigue in humans
- ▶ Tachyphylaxis may occur with long-term use.

Uses/Indications

Apraclonidine ophthalmic may be used as a short-term adjunct for patients that are maximally treated with other anti-glaucoma agents but still require additional lowering of intraocular pressure (IOP). Although reports of the use of apraclonidine in veterinary patients are sparse, one report documented control of IOP for ≈8 months in a dog in conjunction with dorzolamide/timolol ophthalmic medications.¹

Pharmacology/Actions

Apraclonidine is a relatively selective alpha-2-adrenergic agonist. Activation of presynaptic alpha-2 receptors leads to inhibition of norepinephrine release, and activation of postsynaptic alpha-2 receptors on the nonpigmented ciliary body epithelial cells leads to decreased intracellular cyclic adenosine monophosphate (cAMP), both of which decrease aqueous humor production and intraocular pressure (IOP). Additionally, aqueous humor outflow may be increased by a direct effect on uveoscleral outflow, as well as structural alterations to conventional outflow tract that decrease resistance to outflow. Tachyphylaxis (loss of effect) may occur over time, usually in less than 1 month in humans.

Contraindications/Precautions/Warnings

Apraclonidine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Apraclonidine should not be used in patients that are receiving clonidine or monoamine oxidase inhibitors (MAOIs). Do not use apraclonidine in cats, as it can cause bradycardia, hypersalivation, diarrhea, and vomiting.² Apraclonidine should be used with caution, if at all, in animals with hepatic or renal insufficiency.

Adverse Effects

Adverse effects reported in humans or animals include conjunctival blanching, eyelid elevation, and mydriasis. Dogs may experience vomiting and diarrhea.³ Rarely, ophthalmic use of apraclonidine may cause dry mouth, dry eye, tachycardia, hypertension, and tachypnea.

Dosages

Glaucoma (extra-label): 1 drop to the affected eye(s) every 8 to 12 hours

Monitoring

- Clinical efficacy
- Intraocular pressure (IOP)

Client Information

- Use this medication as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication as directed on the product label away from moisture and sunlight. Do not freeze.

- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store product in the refrigerator or at controlled room temperature (2°C-25°C [36°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Apraclonidine ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Apraclonidine 0.5% Ophthalmic Solution (containing benzalkonium chloride 0.01%, NaCl, sodium acetate, NaOH and/or HCl [pH 4.4 to 7.8], and purified water) in 5 mL and 10 mL sterile ophthalmic dropper bottles; *Iopidine*®; (Rx)

Apraclonidine 1% Ophthalmic Solution (containing benzalkonium chloride 0.01%, NaCl, sodium acetate, NaOH and/or HCl [pH 4.4 to 7.8], and purified water), 0.1 mL in plastic ophthalmic dispensers, packaged 2 per pouch enclosed in foil overwrap; *Iopidine*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Atropine Ophthalmic

(ah-troe-**peen**) *Isopto Atropine*®

Mydriatic/Cycloplegic

For systemic use, refer to **Atropine**

Prescriber Highlights

- ▶ Used therapeutically to dilate the pupil, paralyze the ciliary body, and stabilize the blood–aqueous barrier in animals with anterior uveitis
- ▶ Used to perform cycloplegic refraction; not intended to induce mydriasis for diagnostic evaluation of the posterior segment as mydriasis will persist for days to weeks
- ▶ Use with caution in affected and unaffected eyes of animals diagnosed with primary glaucoma.

Uses/Indications

Atropine ophthalmic is primarily used to reverse miosis and treat ciliary body spasm associated with anterior uveitis. It stabilizes the blood–aqueous barrier and lessens the occurrence of pupillary adhesions by minimizing intraocular exudation of proteins and cells caused by uveal inflammation.

Pharmacology/Actions

Atropine is a nonselective muscarinic receptor antagonist, leading to parasympatholytic activity. In the eye, anticholinergic activity produces mydriasis and cycloplegia by blocking stimulation of the iris sphincter muscle and the ciliary body accommodative muscle. It induces mydriasis in ≈1 hour, and the effects may persist up to 120 hours in dogs, 144 hours in cats, and days to weeks in horses.¹

Contraindications/Precautions/Warnings

Atropine ophthalmic is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Atropine ophthalmic should also not be used in patients with primary glaucoma or in dogs that are predisposed to developing glaucoma, as elevation in intraocular pressure (IOP) may result.² Atropine ophthalmic may induce colic in horses secondary to systemic absorption, but this is rare.³ Avoid using the ointment formulation on

penetrating corneal wounds because of the potential for intraocular damage associated with exposure of the ointment's inactive ingredients to the intraocular environment.

Adverse Effects

Adverse effects reported in animals include sensitivity to bright light or sunlight and burning and irritation upon application. Cats and dogs may experience transiently decreased tear production; thus, concurrent administration of topical tear replacers is appropriate. Hypersalivation, which may occur because of drainage of the bitter-tasting solution into the nasopharynx, can be avoided by use of ointment formulations. Atropine ophthalmic causes tachycardia in healthy dogs.⁴

Dosages

Mydriasis and cycloplegia (extra-label): Initially, 1 drop or a ¼ – ½ inch strip to the affected eye(s) 2 to 3 times daily and then once daily or every other day afterward to achieve mydriasis

Monitoring

- Clinical efficacy: confirm continued mydriasis and degree of patient comfort

Client Information

- Wash your hands prior to and after applying this medicine to your animal's eye(s) to prevent human exposure and unintended exposure of your animal's eye.
- Excessive drooling may occur (especially in cats).
- Use proper administration techniques to avoid contamination of the medication. Do not touch the dropper (or ointment) tip or allow it to touch your animal's eye or any other surface to prevent contamination.
- Keep the cap tightly closed when this medicine is not in use.
- This medicine may cause your animal's tear production to decrease. Your veterinarian may also recommend the use of an over-the-counter (OTC) tear replacement solution or ointment for your animal's eye(s) to alleviate this problem.
- Wait for 5 to 10 minutes after applying atropine before applying any other medications to the eye.

Storage/Stability

Store at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Atropine sulfate ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Atropine Sulfate 1% Ophthalmic Ointment USP (containing atropine sulfate 10 mg; white petrolatum; mineral oil; lanolin oil; purified water) in 3.5 g ophthalmic tubes; generic; (Rx)

Atropine Sulfate 1% Ophthalmic Solution, USP (containing atropine sulfate 20 mg; benzalkonium chloride 0.01%, dibasic sodium phosphate; edetate disodium; hypromellose (2910); monobasic sodium phosphate; NaOH and/or HCl to adjust pH to 3.5 to 6; water for injection), 2 mL, 5 mL, and 15 mL in sterile ophthalmic dropper bottles; *Ispto Atropine*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Azelastine Ophthalmic

(a-za-**las**-teen)

Antihistamine

Prescriber Highlights

- ▶ Used for the treatment of allergic conjunctivitis
- ▶ Rapid onset of action (3 minutes) and long duration of action (8 hours) in humans

Uses/Indications

Azelastine is used to relieve pruritus caused by allergic conjunctivitis. No studies describing the efficacy and safety of azelastine after topical administration have been performed in veterinary patients.

Pharmacology/Actions

Azelastine is an H₁- and H₂-receptor antagonist, which also inhibits the release of histamine from mast cells and the production of leukotrienes involved in the allergic response.¹ Azelastine has a rapid onset of action (3 minutes) and a long duration of antipruritic action (8 hours) in humans.

Contraindications/Precautions/Warnings

Azelastine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans include transient burning or stinging upon instillation in the eye (greater than experienced with other ophthalmic solutions, possibly related to the moderate pH of 5 to 6.5) and a bitter taste. Asthma, conjunctivitis, dyspnea, eye pain, fatigue, influenza-like symptoms, pharyngitis, pruritus, rhinitis, and temporary blurring are other reported adverse effects (some of these effects were indistinguishable from the disease being treated).²

Dosages

Allergic conjunctivitis (extra-label): 1 drop to the affected eye(s) 2 times daily

Monitoring

- Clinical efficacy. Resolution of pruritus and conjunctivitis

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medicine in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use if the color changes, becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store upright between 2°C and 25°C (36°F–77°F) away from moisture and sunlight.

Compatibility/Compounding Considerations

Azelastine ophthalmic preparations should not be mixed directly (in vitro) with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Azelastine HCl 0.05% Ophthalmic Solution (containing 0.5 mg azelastine hydrochloride; disodium edetate dehydrate; hypromellose; sorbitol solution; NaOH; water for injection; benzalkonium chloride

0.125 mg; pH of ≈ 5 to 6.5; osmolarity of $\approx 271 - 312$ mOsm/kg), 6 mL in a sterile ophthalmic dropper bottle; generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Azithromycin Ophthalmic

(uh-**zith**-roe-**mye**-sin) Azasite®

Macrolide Antibiotic

For systemic use, refer to *Azithromycin*

Prescriber Highlights

- ▶ Used to treat ocular surface bacterial infections
- ▶ May cause reversible phospholipid accumulation in the cornea
- ▶ Prolonged use of antibiotic preparations may result in overgrowth of nonsusceptible organisms, including fungi.

Uses/Indications

Azithromycin ophthalmic solution is used to treat susceptible strains of *Haemophilus* spp, *Streptococcus* spp, and *Staphylococcus* spp. In vitro efficacy against some *Chlamydia* spp and *Mycoplasma* spp has also been reported.¹ Reports of clinical efficacy and/or toxicity/tolerability of azithromycin ophthalmic in animals are not available.

The World Health Organization (WHO) has designated azithromycin as a Critically Important, Highest Priority antimicrobial for human medicine.²

Pharmacology/Actions

Azithromycin is a bacteriostatic macrolide antibiotic that binds to the 50S ribosomal subunit of susceptible microorganisms and interferes with microbial protein synthesis.

Contraindications/Precautions/Warnings

Azithromycin ophthalmic solution is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Azithromycin ophthalmic solution is indicated for topical ophthalmic use only; it should not be administered systemically, injected subconjunctivally, or introduced directly into the anterior chamber of the eye.¹

Adverse Effects

Adverse effects reported in humans include burning, stinging, and irritation on application. Systemic azithromycin has also caused phospholipidosis (intracellular phospholipid accumulation) in dogs, mice, and rats, and cytoplasmic microvacuolation of the cornea was observed in rabbits following ocular administration, which resolved when the medication was discontinued.¹

Dosages

Bacterial conjunctivitis (extra-label): 1 drop to the affected eye(s) 2 times daily the first 2 days, followed by once daily for 5 days

Monitoring

- Clinical efficacy: resolution of bacterial conjunctivitis

Client Information

- Give this medicine exactly as your veterinarian has prescribed.
- Wash your hands before applying this medicine to your animal's eye(s).
- Be sure you understand how to give this medicine to your animal to avoid contamination of the medicine, which could worsen the eye condition.
- Wait for 5 minutes after applying this medicine before applying any other medicines to the eye.
- Keep the cap on the medicine tightly closed when it is not in use.
- Store this medicine in the refrigerator. After opening, the medicine can be stored at room temperature. Discard the medicine 14

days after opening or if the color of the solution changes, if it becomes cloudy, or if there are particles.

Storage/Stability

Store this product in the refrigerator (2°C-8°C [36°F-46°F]) away from moisture and sunlight; do not freeze. Discard this product 14 days after opening.

Compatibility/Compounding Considerations

Azithromycin ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Azithromycin 1% Ophthalmic Solution (containing azithromycin 10 mg [1%]; mannitol; citric acid; sodium chloride; poloxamer 407; polycarbophil; edetate disodium; water for injection; NaOH to adjust pH), 2.5 mL in a sterile ophthalmic dropper bottle; *Azasite*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Benoxinate Ophthalmic Oxybuprocaine Ophthalmic

(bin-**ak**-sin-ate) Altafluor®, Fluress®, Flurox®

Topical Ocular Anesthetic

Prescriber Highlights

- ▶ Rapid-acting topical anesthetic used for ocular diagnostic procedures
- ▶ Should never be prescribed for therapeutic administration
- ▶ Only available in combinations with the vital dye fluorescein sodium
- ▶ Chronic use results in tear film instability and corneal epithelial and stromal damage, which may progress to vision loss.
- ▶ Less burning, stinging, and chemosis than tetracaine

Uses/Indications

Like other topical anesthetic agents, benoxinate is indicated for use in ophthalmic diagnostic procedures, including measurement of intraocular pressure (tonometry), ocular ultrasonography, gonioscopy, and corneal and conjunctival sampling for cytology and culture. Benoxinate is also indicated for therapeutic procedures in the hospital, including removing foreign bodies and sutures from the cornea, eyelid and conjunctival surgery, corneal grafting procedures, and cataract extraction. It is only available in ophthalmic solutions combined with the vital dye fluorescein. In dogs, benoxinate causes fewer adverse reactions (burning, stinging, and chemosis) than tetracaine.^{1,2} Benoxinate has demonstrated comparable corneal anesthesia to that achieved by topical proparacaine in normal horses and donkeys.³

Pharmacology/Actions

Benoxinate induces local anesthesia by inhibiting neuronal depolarization, thus preventing the generation of an action potential. The action occurs due to the binding of the local anesthetic molecule to the intracellular portion of the voltage-gated sodium channel within the neuron, thus inhibiting the intracellular movement of sodium. Variations in the structure of different local anesthetics influence their relative potencies; however, all local anesthetics have decreased efficacy in the acidic environment of inflammation, whereas alkaline environments improve intracellular penetration, thus increasing efficacy. The duration of action is dose- and species-dependent, with a total duration of corneal anesthesia following a single drop of 55

minutes in dogs and 45 minutes in cats.^{1,2}

Contraindications/Precautions/Warnings

Benoxinate is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. It should never be prescribed as an ocular analgesic as prolonged use of a topical ocular anesthetic will produce significant corneal epithelial and stromal disease, as well as tear film instability, and may result in loss of vision.⁴

Adverse Effects

Benoxinate is usually well-tolerated, but local irritation (conjunctival redness, lacrimation, or squinting) may occur several hours after administration. These signs occur less frequently with benoxinate than with tetracaine.³ Although patients that are sensitive to other local anesthetics may be less likely to react to benoxinate, local or systemic sensitivity may still occur and rarely manifest as a pseudoallergic (anaphylactoid) reaction mimicking an IgE-mediated vasodilatory event. Cardiovascular, respiratory, and CNS depressant effects that may result from local anesthetics are extremely rare with appropriate use of topical local anesthetics. However, care should be exercised when administering to small veterinary patients (ie, puppies, kittens, “pocket pets”) or those with pathways that may increase absorption (ie, amphibians). Its use as a therapeutic analgesic leads to superficial punctate keratitis, which may advance to diffuse necrotizing keratitis, severe infiltrative keratitis, persistent epithelial defects, or damage to Descemet’s membrane.⁴

Dosages

Ophthalmic procedures requiring local anesthesia (extra-label): 1 – 2 drops to the target eye(s) prior to examination or performing procedure. May repeat adding 1 drop every 5 to 10 minutes for 2 to 3 additional times. With further applications, the duration of corneal anesthesia can be extended to 55 minutes in dogs. Additional doses should be avoided in small patients (ie, puppies, kittens, “pocket pets”).

Monitoring

- Onset of action as indicated by awake patient, permitting diagnostic procedures to be performed
- Onset of action as indicated by a reduction in anesthetic requirements when administered pre- and intra-operatively
- Damage to corneal epithelium indicated by a positive fluorescein stain test

Client Information

- Animal owners must not administer benoxinate in the home. It is reserved for diagnostic procedures at veterinary clinics.

Storage/Stability

Store the product in the refrigerator (2°C–8°C [36°F–46°F]). Once opened, the medication can be stored at room temperature for up to 1 month.⁵ Keep this product away from moisture and sunlight; do not freeze.

Discard this medication if the color of the solution changes, if it becomes cloudy, or if it develops particles.

Compatibility/Compounding Considerations

Benoxinate ophthalmic preparations should not be mixed directly (in vitro) with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Benoxinate HCl 0.4% and Fluorescein Sodium 0.25% (containing fluorescein sodium 2.5 mg; benoxinate hydrochloride 4 mg; boric acid; povidone; purified water; HCl to adjust pH to 4.3 to 5.3; chlorobutanol 1%); 5 mL in sterile ophthalmic dropper bottles; *Altaflu-*

*or**, *Fluress*®, *Flurox*®; (Rx). No benoxinate ophthalmic products have been approved by the US FDA.

References

For the complete list of references, see wiley.com/go/budde/plumb

Bepotastine Ophthalmic

(be-po-**tas**-teen) *Bepreve*®

Antihistamine

Prescriber Highlights

- ▶ H₁ receptor antagonist used in the management of allergic conjunctivitis

Uses/Indications

Bepotastine is used to relieve pruritus caused by allergic conjunctivitis. The efficacy and safety of topical bepotastine has not been evaluated in veterinary species.

Pharmacology/Actions

Bepotastine is a direct histamine-1 (H₁) antagonist and mast cell stabilizer, with inhibitory effects on other inflammatory mediators, such as leukotrienes and eosinophils. The onset of action (control of itching) in humans is 15 minutes following instillation, with a duration of effect up to 8 hours.¹

Contraindications/Precautions/Warnings

Bepotastine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Use it with caution in pregnant or nursing patients.²

Adverse Effects

Adverse effects reported in humans include transient burning or stinging upon instillation in the eye, nasal irritation, and a bad taste in the mouth.¹

Dosages

Allergic conjunctivitis (extra-label): 1 drop to the affected eye(s) 2 times daily

Monitoring

- Clinical efficacy: resolution of pruritus and conjunctivitis

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store at controlled room temperature (15°C–25°C [59°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Bepotastine ophthalmic preparations should not be mixed directly (in vitro) with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Bepotastine Besilate 1.5% Ophthalmic solution (containing bepotastine besilate 15 mg; benzalkonium chloride 0.005%; monobasic so-

dium phosphate dehydrate; NaCl; NaOH to adjust pH to 6.8; water for injection, USP, with an osmolality of 290 mOsm/kg), 5 mL and 10 mL in sterile ophthalmic dropper bottles; *Bepreve*®; generic (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Besifloxacin Ophthalmic

(be-si-**flocks**-a sin) *Besivance*®

Fourth-Generation Fluoroquinolone Antibiotic

Prescriber Highlights

- ▶ Available in ophthalmic formulation only
- ▶ Used to treat bacterial conjunctivitis or keratitis
- ▶ Physically incompatible with many drugs; therefore, should not be directly mixed with other ophthalmic medications
- ▶ As with other topical antibiotics, prolonged use may result in overgrowth of nonsusceptible organisms.

Uses/Indications

Besifloxacin ophthalmic is used to treat bacterial conjunctivitis and keratitis. It is generally effective against gram-negative, gram-positive, and anaerobic organisms, particularly *Staphylococcus* spp, *Streptococcus* spp, *Haemophilus* spp, and *Pseudomonas aeruginosa*.

Pharmacology/Actions

Besifloxacin is a fourth-generation fluoroquinolone antibiotic that acts, via inhibition of both topoisomerase II (DNA gyrase) and topoisomerase IV, to impair bacterial DNA synthesis. Inhibition of both enzymes decreases the development of resistance to this class of fluoroquinolones in general, and organisms have demonstrated even lower resistance to besifloxacin due to this balanced inhibition.¹ Besifloxacin achieves superior aqueous humor concentrations in normal dogs relative to gatifloxacin and moxifloxacin (other fourth-generation fluoroquinolones)²; however, other studies have demonstrated effective ocular surface concentrations of besifloxacin following topical administration, with lesser aqueous humor concentrations.¹ The currently available formulation of besifloxacin is formulated with a mucoadhesive polymer that increases contact time with the ocular surface, which is favorable for concentration-dependent fluoroquinolone antibiotics.^{1,3}

Contraindications/Precautions/Warnings

Besifloxacin is contraindicated in patients with a history of hypersensitivity to besifloxacin, other fluoroquinolones, or any component of the formulation. For topical ophthalmic use only, not for injection subconjunctivally or into the anterior chamber of the eye.³ Fluoroquinolone-induced retinal toxicity in domestic cats has not been demonstrated with this product.

Adverse Effects

Adverse effects reported in humans or animals include blurred vision, tearing, eye pain, conjunctival redness, itching, and a bad taste in the mouth. Temporary punctate keratitis, which resolves with discontinuation of medication, has also been reported rarely in humans.⁴

Dosages

Conjunctivitis (extra-label): 1 drop to the affected eye(s) 3 times daily

Keratitis (extra-label): 1 drop to the affected eye every 2 to 8 hours

Monitoring

- Clinical efficacy. Corneal cytology with culture and susceptibility testing to evaluate for the persistence of bacteria, and evaluation for clinical signs suggestive of continued infection

Client Information

- Give this medicine to your animal as directed by your veterinarian. Be sure you understand how to safely give to your animal.
- Invert the bottle and shake once before use. Do not give your animal this medicine if the color changes, if it becomes cloudy, or if particles are seen in the solution.
- Crystals may develop on the surface of the eye during treatment but will not cause harm.
- Do not mix this medicine with other eye medications.
- If you are giving your animal other eye medications, wait 5 minutes after applying this medicine before applying any other medications to the eye.
- Do not freeze this medicine. Store this medicine at room temperature, away from light and moisture.

Storage/Stability

Store this product at controlled room temperature 20°C to 25°C (68°F-77°F) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Fluoroquinolone antibiotics interact with many other drugs, including those containing cations (eg, iron, aluminum, magnesium, calcium). Besifloxacin should not be mixed with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Besifloxacin 0.6% (6 mg/mL) Ophthalmic Suspension (containing besifloxacin 0.6%; benzalkonium chloride 0.01%; polycarboxophil; mannitol; poloxamer 407; NaCl; edetate disodium dehydrate; NaOH; water for injection, with an osmolality of ≈290 mOsm/kg), 5 mL in a sterile ophthalmic dropper bottle; *Besivance*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Betaxolol Ophthalmic

(be-**tax**-uh-lol) *Betoptic*®, *Betoptic S*®

Beta-1-Selective Beta-Adrenergic Blocker

Prescriber Highlights

- ▶ Beta-blocker that decreases intraocular pressure (IOP) as a treatment for glaucoma
- ▶ More effective reduction in IOP using 0.5% solution relative to 0.25% solutions in veterinary patients; however, increased risk for adverse effects exists when using the higher concentration
- ▶ Delays onset of glaucoma in the normal eye of dogs with unilateral glaucoma
- ▶ Use with caution in asthmatic patients because of the potential to induce bronchospasm.

Uses/Indications

Betaxolol is a beta-1-selective beta-blocker used to reduce intraocular pressure (IOP) in primary glaucoma and as prophylaxis to prevent glaucoma in the normal eye of animals with unilateral glaucoma.¹ In combination with tafluprost, betaxolol leads to a greater reduction of IOP in normal dogs compared to timolol.²

Pharmacology/Actions

Betaxolol works through the blockade of the beta-receptors in the ciliary body epithelium to reduce intraocular pressure (IOP) by decreasing aqueous humor production. Several mechanisms have been postulated for this effect, including beta-blockade, particularly beta-2 receptors, reducing norepinephrine-induced tonic sympathetic stimulation, leading to decreased activation of cAMP in the

ciliary body; inhibition of Na⁺K⁺ATPase activity; or vasoactive mechanisms on the iris and ciliary body. Betaxolol also inhibits glutamine-induced increases in intracellular calcium and sodium, thereby providing a potential protective effect against ischemic insult to the retina.³ Because of betaxolol's beta-1-selective activity, its hypotensive effect may be less than that of nonselective beta-blockers, as beta-2 blockade is of greater importance to aqueous humor production.

Contraindications/Precautions/Warnings

Betaxolol is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Betaxolol should be used carefully in patients with heart block, bradycardia, heart failure, asthma, chronic bronchitis, or other cardiopulmonary conditions, although systemic cardiopulmonary adverse effects are not likely with ocular use of betaxolol. Beta-blockers should be used with caution in patients with diabetes mellitus, as these drugs may mask signs of acute hypoglycemia. Beta-blockers should be used with caution in patients with hyperthyroidism, as these drugs may mask indications of this condition. Abrupt discontinuation of beta blockers should be avoided.

Adverse Effects

Adverse effects reported in humans or animals include transient burning, stinging, and irritation upon application to the eye, as well as keratitis, anisocoria, and photophobia. Bradycardia, heart block, congestive heart failure, pulmonary distress, insomnia, dizziness, headaches, lethargy, and depression have been reported in humans. Other topically applied beta-blockers produce significant reductions in heart rate in healthy cats.⁴

Dosages

Glaucoma (extra-label): 1 drop to the affected eye(s) twice daily

Monitoring

- Clinical efficacy
- Monitor intraocular pressure (IOP)
- Observe asthmatic and diabetic patients for respiratory difficulty or acute hypoglycemia, respectively.

Client Information

- Use this medication as directed by your veterinarian.
- Suspension formulations must be shaken well before using.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product in the refrigerator (2°C–8°C [36°F–46°F]) or at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Betaxolol ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Betaxolol HCl 0.5% Ophthalmic Solution (containing 5.6 mg betaxolol HCl equivalent to 5 mg betaxolol; benzalkonium chloride 0.01%;

edetate disodium; NaCl, HCl, and/or NaOH to adjust pH; purified water) in 2.5 mL, 5 mL, 10 mL, and 15 mL sterile ophthalmic dropper bottles; *Betoptic*®, generic; (Rx)

Betaxolol HCl 0.25% ophthalmic solution (containing betaxolol HCl 2.8 mg equivalent to 2.5 mg betaxolol; benzalkonium chloride 0.01%; mannitol; poly (styrene-divinyl benzene); edetate disodium; NaOH and/or HCl to adjust pH; purified water) in sterile 10 mL and 15 mL ophthalmic dropper bottles; *Betoptic*®-S; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Bimatoprost Ophthalmic

(bi-mat-oh-prost) Lumigan®, Latisse®, Durysta®

Prostaglandin Analogue Anti-Glaucoma Agent

Prescriber Highlights

- ▶ Used to decrease intraocular pressure (IOP) in patients with primary glaucoma or ocular hypertension
- ▶ Not effective for feline glaucoma, and unlikely to be effective for equine glaucoma
- ▶ Do not use if intraocular inflammation is present, as the drug may worsen the inflammation or lead to pupillary block glaucoma
- ▶ Twice-daily dosing results in less fluctuation of IOP.
- ▶ Lower concentration of benzalkonium chloride than latanoprost or travoprost; therefore, bimatoprost causes less ocular discomfort on administration
- ▶ May increase brown pigment, resulting in discoloration of light-colored irises and increased eyelash growth

Uses/Indications

Bimatoprost is used to lower intraocular pressure (IOP) in the management of glaucoma in dogs. It is indicated as a treatment for primary glaucoma; however, secondary glaucomas (caused by uveitis, lens luxation, or intraocular tumor) are not appropriately treated with bimatoprost. Due to differences in intraocular receptors, bimatoprost is ineffective in lowering the IOP in cats, although it does produce miosis.^{1,2} Although once-daily administration is indicated in humans, twice-daily administration has been demonstrated to more consistently lower IOP in dogs.³ Significant miosis also occurs following administration.³ An intracameral sustained-release bimatoprost implant has demonstrated increased IOP-lowering efficacy in comparison with topical administration of varied concentrations of bimatoprost in normotensive dogs.⁴

Pharmacology/Actions

Bimatoprost, a prostaglandin analogue, is a synthetic structural analogue of prostaglandin derived from a fatty acid precursor, anandamide. It selectively mimics the effects of naturally occurring prostamides. Bimatoprost is believed to lower intraocular pressure (IOP) by increasing the outflow of aqueous humor through both the trabecular meshwork and uveoscleral routes.⁵ In humans, the onset of action is in 4 hours, with a total duration of 8 to 12 hours.

Contraindications/Precautions/Warnings

Bimatoprost is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Bimatoprost should be used with caution in the presence of anterior uveitis and secondary glaucoma, particularly in cases already exhibiting significant miosis, as this treatment may exacerbate miosis, promote synechiae formation, and worsen glaucoma. Bimatoprost should also be used with caution in dogs with primary anterior lens luxation, as the miosis that results may trap the lens and/or vitreous within the

pupillary aperture, thus creating pupillary block glaucoma. Bimatoprost is not contraindicated in dogs with secondary posterior lens luxation, as the miosis may (favorably) trap the lens in the posterior segment.

Adverse Effects

Adverse effects reported in humans or animals include stinging, burning, erythema (conjunctival hyperemia), and irritation on administration. Macular edema has been reported in humans using bimatoprost. Bimatoprost may cause increased eyelash growth, may worsen intraocular inflammation through prostaglandin-mediated mechanism, and/or cause darkening of the iris through direct stimulation of iris melanocytes.

Dosages

Glaucoma (extra-label):

- Solution: Initially, 1 drop to the affected eye(s) once daily in the evening; increase to twice daily as glaucoma progresses. Twice-daily administration of bimatoprost may be more effective in dogs.
- Sustained-release implant: Intracameral injection of the sustained-release implant is performed using the associated ophthalmic drug delivery system. Repeat implantation should not be performed in an eye that has received a prior implant.

Monitoring

- Clinical efficacy measured through sustained reduction of intraocular pressure (IOP)

Client Information

- Use this medication as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store solution in the refrigerator or at controlled room temperature (2°C–25°C [36°F–77°F]) away from moisture and sunlight; do not freeze.

Store implant in the refrigerator (2°C–8°C [36°F–46°F]).

Compatibility/Compounding Considerations

Bimatoprost ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Bimatoprost 0.01% Ophthalmic Solution (containing bimatoprost 0.1 mg/mL; benzalkonium chloride 0.02%; NaCl; sodium phosphate, dibasic; citric acid; purified water; NaOH and/or HCl to adjust pH to 6.8 to 7.8, with an osmolality of 290 mOsm/kg) in sterile 2.5 mL, 5 mL, and 7.5 mL ophthalmic dropper bottles; *Lumigan*®, generic; (Rx)

Bimatoprost 0.03% Ophthalmic Solution (containing bimatoprost 0.3 mg; benzalkonium chloride 0.005%; NaCl; sodium phosphate, dibasic; citric acid; purified water; NaOH and/or HCl to adjust pH to 6.8 to 7.8, with an osmolality of 290 mOsm/kg) in sterile 2.5 mL, 5 mL, and 7.5 mL ophthalmic dropper bottles; *Latisse*®, generic; (Rx). **NOTE:** *Latisse*® is approved for topical use only for inducing eyelash growth and should not be used in the eye.

Bimatoprost 10 µg Ophthalmic Implant (containing 10 µg bimatoprost in a single-use applicator that is packaged in a sealed foil pouch containing desiccant, NDC 0023-9652-01); *Durysta*® (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Brimonidine Ophthalmic

(bri-moe-ni-deen) *Alphagan P*®, *Combigan*®, *Qoliana*®, *Simbrinza*®

Alpha-Adrenergic Agonist; Anti-Glaucoma Agent

Prescriber Highlights

- Used to decrease intraocular pressure (IOP) in patients with glaucoma
- Produces less severe adverse effects than apraclonidine, but it is not a first-line anti-glaucoma agent in veterinary species
- Greenish-yellow color of the solution is a characteristic and not a sign of degradation or contamination

Uses/Indications

Brimonidine may be used to treat elevated intraocular pressure (IOP) in patients with glaucoma or ocular hypertension. Its use in glaucomatous beagles 2 to 3 times daily has demonstrated no consistent clinical effect.¹ Administration of brimonidine to normotensive horses, with or without timolol, has not shown any clinically significant effect.²

Pharmacology/Actions

Brimonidine is a relatively selective alpha-2-adrenergic agonist. Activation of presynaptic alpha-2 receptors inhibits norepinephrine release, and activation of postsynaptic alpha-2 receptors on the non-pigmented ciliary body epithelial cells decreases intracellular cAMP, both of which result in decreased aqueous humor production and decreased intraocular pressure (IOP). Additionally, aqueous humor outflow may be increased by a direct effect on uveoscleral outflow, as well as structural alterations to the conventional outflow tract that decrease resistance to outflow.

Contraindications/Precautions/Warnings

Brimonidine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Brimonidine should be used cautiously, if at all, in patients with vascular insufficiency syndromes (eg, laminitis, cerebral, or coronary vascular insufficiency). Concomitant use of monoamine oxidase inhibitors (MAOIs; eg, amitraz, selegiline) may potentiate the risk for systemic hypotension. Instruct pet owners to keep it out of reach of household animals, as ingestion of brimonidine ophthalmic solution has caused serious cardiovascular toxicities.³ Overdoses of brimonidine can be treated with atipamezole, yohimbine, and other alpha-2 antagonists. Caution should be exercised if brimonidine is used in conjunction with beta-blockers (ophthalmic and systemic), antihypertensives, and/or cardiac glycosides. When brimonidine is used in humans, the clinical effect diminishes over time and requires careful monitoring.

Adverse Effects

Adverse effects reported in humans or animals are numerous and include (but are not limited to) oral dryness, allergic conjunctivitis, conjunctival hyperemia, ocular pruritus, burning, stinging, foreign body sensation, irritation, eyelid and conjunctival edema, dizziness, muscular pain, and hypertension. Use of brimonidine in dogs is associated with miosis and bradycardia.⁴ Administration of brimonidine in cats consistently results in vomiting, which may be ameliorated by premedication with oral maripotent.⁵ Brimonidine in cats also produces dose-dependent sedation, bradycardia, and hypotension.⁶

Dosages

Glaucoma (extra-label): 1 drop to the affected eye(s) 2 to 3 times daily

Monitoring

- Clinical efficacy measured through continued control of intraocular pressure (IOP)
- Baseline and periodic heart rate and blood pressure, particularly in dogs and cats

Client Information

- Use this medicine as directed by your veterinarian.
- Keep this medicine out of reach of household animals and children, as ingestion of brimonidine ophthalmic solution can cause serious heart problems.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Brimonidine ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Brimonidine Tartrate 0.1% Ophthalmic Solution (containing brimonidine tartrate 0.1%; boric acid; calcium chloride; magnesium chloride; potassium chloride; purified water; *PURITE*® 0.005% as preservative; sodium borate; sodium carboxymethylcellulose; NaCl; HCl and/or NaOH to adjust pH to 7.7), 5 mL, 10 mL, or 15 mL in sterile ophthalmic dropper bottles; *Alphagan-P*®; (Rx)

Brimonidine Tartrate 0.15% Ophthalmic Solution (containing brimonidine tartrate 0.15%; boric acid; calcium chloride; magnesium chloride; potassium chloride; purified water; *PURITE*® 0.005% as preservative; sodium borate; sodium carboxymethylcellulose; NaCl; HCl and/or NaOH to adjust pH to 7.7), 5 mL, 10 mL, or 15 mL in sterile ophthalmic dropper bottles; *Alphagan-P*®; (Rx)

Brimonidine Tartrate 0.15% Ophthalmic Solution (containing brimonidine tartrate 1.5 mg/mL; *POLYQUAD*® (polyquaternium-1) 0.01 mg/mL; povidone; boric acid; sodium borate; calcium chloride; magnesium chloride; potassium chloride; mannitol; sodium chloride; purified water; HCl and/or NaOH to adjust pH), 5 mL, 10 mL, or 15 mL in sterile ophthalmic dropper bottles; *Qoliana*®; (Rx)

Brimonidine Tartrate 0.2% Ophthalmic Solution (containing brimonidine tartrate 0.2%; benzalkonium chloride 0.005%; polyvinyl alcohol; NaCl; citric acid; purified water; HCl and/or NaOH to adjust pH to 5.6 to 6.6, with an osmolality of 280 – 330 mOsm/kg) in a 5 mL, 10 mL, and 15 mL dropper bottle; generic; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Brimonidine Tartrate 0.2% and Timolol Maleate 0.5% Ophthalmic Solution (containing brimonidine tartrate 0.2%; timolol 0.5%; benzalkonium chloride 0.005%; purified water; sodium phosphate, dibasic; sodium phosphate, monobasic; HCl and/or NaOH to adjust pH to 6.5 to 7.3, with an osmolality of 280 – 330 mOsm/kg), 5 mL,

10 mL, or 15 mL in sterile ophthalmic dropper bottles; *Combigan*®, generic; (Rx)

Brimonidine Tartrate 0.2% and Brinzolamide 1% Ophthalmic Suspension (containing brinzolamide 10 mg; brimonidine tartrate 2 mg, equivalent to 1.32 mg free base; benzalkonium chloride 0.03 mg; propylene glycol; carbomer 974P; boric acid; mannitol; NaCl; tyloxapol; purified water; HCl and/or NaOH to adjust pH), 8 mL in sterile ophthalmic dropper bottle; *Simbrinza*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Brinzolamide Ophthalmic

(bryn-zoe-la-mide) *Azopt*®, *Simbrinza*®

Carbonic Anhydrase Inhibitor Anti-Glaucoma Agent

Prescriber Highlights

- ▶ Decreases intraocular pressure (IOP) by reducing the production of aqueous humor
- ▶ May be better tolerated than dorzolamide because of less acidic pH
- ▶ Sulfonamide derivative; caregivers with sulfa allergies should avoid direct contact with the medicine.

Uses/Indications

Brinzolamide is used to decrease intraocular pressure (IOP) in glaucoma and ocular hypertension. It may be used in cases of primary glaucoma or secondary glaucoma (caused by uveitis, lens luxation, or intraocular tumors) in dogs, cats, and horses. Brinzolamide effectively lowers the IOP in normal horses but does not decrease the IOP in normal cats.^{1,2} Efficacy of prophylactic brinzolamide administration for preventing manifestation of primary glaucoma in unaffected eyes of dogs with contralateral primary glaucoma has not been demonstrated.³

Pharmacology/Actions

Carbonic anhydrase (CA) is an enzyme found in many body tissues, including the eye. CA inhibitors reduce the formation of hydrogen and bicarbonate ions from carbon dioxide and water by noncompetitive, reversible inhibition of the enzyme CA, thereby reducing the availability of these ions for active transport into secretions. In humans, CA exists as a number of isoenzymes, with CA-II being the most relevant to glaucoma management. Inhibition of CA in the ciliary processes of the eye decreases aqueous humor secretion, presumably by slowing the formation of bicarbonate ions with subsequent reduction in sodium and fluid transport, which results in a reduction in intraocular pressure (IOP).⁴ Brinzolamide is more lipophilic than dorzolamide, resulting in increased intraocular concentrations following topical administration.⁵

Contraindications/Precautions/Warnings

Brinzolamide is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation or drugs like it, such as sulfonamides. Caregivers with sulfonamide allergies should use caution when applying brinzolamide.

Adverse Effects

Adverse effects reported in humans or animals include stinging, burning, and irritation upon application. Corneal edema may occur in patients with low corneal endothelial cell counts. Humans experience headaches and a bitter taste in the mouth after brinzolamide administration. It is possible that animals experience these adverse effects. Carbonic anhydrase (CA) inhibitors, including brinzolamide, have been implicated in the development of significant ulcerative and nonulcerative keratitis following long-term administration.⁶

Dosages

Glaucoma (extra-label): 1 drop to the affected eye(s) 2 to 3 times daily

Monitoring

- Clinical efficacy
- Confirm maintenance of intraocular pressure (IOP) within the normal range for the species being treated
- Monitor for corneal edema

Client Information

- Use this medication as directed by your veterinarian.
- Shake this medication well before using.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes.

Storage/Stability

Store this product in the refrigerator or at controlled room temperature (4°C–30°C [39°F–86°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Brinzolamide ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Brinzolamide 1% Ophthalmic Suspension (containing brinzolamide 10 mg; benzalkonium chloride 0.01%; mannitol; carbomer 974P; tyloxapol; edetate disodium; NaCl; purified water; HCl and/or NaOH to adjust pH to 7.5; osmolality of 300 mOsm/kg), 10 mL or 15 mL in sterile ophthalmic dropper bottles; *Azopt*®; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Brinzolamide 1% and Brimonidine Tartrate 0.2% Ophthalmic Suspension (containing brinzolamide 10 mg; brimonidine tartrate 2 mg, equivalent to 1.32 mg as brimonidine free base; benzalkonium chloride 0.03 mg; propylene glycol; carbomer 974P; boric acid; mannitol; sodium chloride; tyloxapol; purified water; HCl and/or NaOH to adjust pH), 8 mL in sterile 10 mL ophthalmic dropper bottle; *Simbrinza*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

analgesia following surgical removal of cataracts. In a prospective study evaluating inflammation, postoperative ocular hypertension, and posterior capsular opacification (PCO) in dogs following cataract surgery, the effectiveness of prednisolone acetate was compared with artificial lens implants that were impregnated with celecoxib (NSAID) or topically applied bromfenac. Over the long-term follow-up period of 56 weeks, bromfenac was comparable to prednisolone acetate in controlling inflammation and did not lead to an increased incidence of ocular hypertension, and it effectively managed PCO.¹ In contrast, bromfenac was associated with an increased incidence of elevated intraocular pressure (IOP) relative to flurbiprofen in dogs 6 weeks following surgical cataract removal.²

Pharmacology/Actions

Bromfenac is an NSAID that blocks prostaglandin synthesis by inhibiting cyclooxygenase 1 and 2. Prostaglandins mediate many adverse effects that occur because of intraocular inflammation, including disruption of the blood-aqueous humor barrier, vasodilation, increased vascular permeability, leukocytosis, and increased intraocular pressure (IOP).³ Blocking prostaglandin production, therefore, minimizes the adverse short- and long-term effects of intraocular inflammation.

Contraindications/Precautions/Warnings

Bromfenac is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. The potential for cross-sensitivity to acetylsalicylic acid, phenylacetic acid derivatives, and other NSAIDs exists. Topical NSAIDs may result in corneal epithelial breakdown and corneal thinning, erosion, or ulceration.³ Topical NSAIDs may slow or delay corneal wound healing. As topical corticosteroids are also known to slow or delay healing, concomitant use of topical NSAIDs and topical steroids may increase the potential for healing problems. Because of interference with thrombocyte aggregation, ophthalmic NSAIDs may cause increased bleeding of ocular tissues (including hyphema) in conjunction with ocular surgery. It is recommended that bromfenac ophthalmic solution be used with caution in patients with known bleeding tendencies or those receiving other medications that may prolong bleeding times.³

Adverse Effects

Adverse effects associated with bromfenac that have been reported in humans or animals include burning, stinging, and irritation upon application; conjunctival hyperemia; risk for ocular bleeding; and delayed corneal wound healing.

Dosages

Anterior segment inflammation (extra-label): 1 drop to the affected eye(s) every 8 hours initially, then reduced to lowest effective dose

Postoperative ocular analgesia (extra-label): 1 drop to the affected eye(s) every 6 to 8 hours as needed to control pain

Monitoring

- Clinical efficacy through evidence of efficacious analgesia and control of intraocular inflammation

Client Information

- Use this medication as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication at room temperature away from moisture and sunlight; do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solution.

Bromfenac Ophthalmic

(*brome*-fen-ak) *Bromsite*®, *Prolensa*®

Nonsteroidal Anti-Inflammatory Drug (NSAID)

Prescriber Highlights

- ▶ Used to control anterior segment inflammation
- ▶ Longer duration of action than diclofenac
- ▶ May delay ocular wound healing and increase risk for intraocular bleeding

Uses/Indications

Bromfenac is indicated for controlling anterior segment inflammation (eg, conjunctivitis, keratitis, anterior uveitis) and for providing

- Discard compounded gels by the discard date indicated by the compounding pharmacist.

Storage/Stability

Store this product at controlled room temperature (15°C-25°C [59°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Bromfenac ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Bromfenac 0.09% Ophthalmic Solution (containing bromfenac sodium sesquihydrate 0.1035% equivalent to 0.09% bromfenac-free acid; benzalkonium chloride 0.05 mg/mL; boric acid; polysorbate 80; povidone; sodium borate; sodium sulfite anhydrous; NaOH to adjust pH; water for injection), in sterile 2.5 mL and 5 mL ophthalmic dropper bottle; generic; (Rx)

Bromfenac 0.07% Ophthalmic Solution (containing bromfenac sodium hydrate 0.805 mg; benzalkonium chloride 0.005%; boric acid; edetate disodium; povidone; sodium borate; sodium sulfite; tyloxapol; NaOH to adjust pH; water for injection), in sterile 1.6 mL or 3 mL ophthalmic dropper bottle; *Prolensa*®; (Rx)

Bromfenac 0.075% Ophthalmic Solution (containing bromfenac sodium sesquihydrate 0.81 mg equivalent to bromfenac-free acid 0.76 mg; boric acid; sodium borate; citric acid; sodium citrate; poloxamer 407; polycarbophil; NaCl; edetate disodium; water for injection; NaOH to adjust pH; benzalkonium chloride 0.05 mg/0.005%), in sterile 5 mL ophthalmic dropper bottle; *Bromsite*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Carteolol Ophthalmic

(kar-tay-uh-lol)

Nonselective Beta-Blocker

Prescriber Highlights

- ▶ Nonselective (beta-1 and beta-2) beta-blocker that decreases aqueous humor production to decrease intraocular pressure (IOP) as a treatment for glaucoma
- ▶ May delay onset of glaucoma in the normal eye of dogs with unilateral glaucoma
- ▶ Use with caution in asthmatic patients due to the potential to induce bronchospasm

Uses/Indications

Carteolol may be used to reduce intraocular pressure (IOP) in primary glaucoma and as prophylaxis to prevent glaucoma in the normal eye of animals with unilateral glaucoma. The evaluation of its efficacy in animals has not been performed.

Pharmacology/Actions

Carteolol is a nonselective beta-adrenergic blocking agent with associated intrinsic sympathomimetic activity and without significant membrane-stabilizing activity.¹ Blockade of ciliary body beta-receptors (particularly beta-2-receptors) reduces intraocular pressure (IOP) by decreasing aqueous humor production. Several mechanisms of action have been postulated for this effect, including beta-blockade (particularly beta-2-receptors) reducing norepinephrine-induced tonic sympathetic stimulation, which then leads to decreased activation of cAMP in the ciliary body; inhibition of

Na+K+ATPase activity; or vasoactive mechanisms on the iris and ciliary body. Topically administered beta-blockers also inhibit glutamine-induced increases in intracellular calcium and sodium, thereby providing a potential protective effect against ischemic insult to the retina.²

Contraindications/Precautions/Warnings

Carteolol is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Carteolol should be used carefully in patients with heart block, bradycardia, heart failure, asthma, chronic bronchitis, or other cardiopulmonary conditions, although systemic cardiopulmonary adverse effects are not likely with ocular use of carteolol.

Beta-blockers should be used with caution in patients with diabetes mellitus, as these drugs may mask signs of acute hypoglycemia. Beta-blockers should also be used with caution in patients with hyperthyroidism, as these drugs may mask indications of this condition. Abrupt discontinuation of beta-blockers should be avoided.

Adverse Effects

Adverse effects reported in humans or animals include transient burning, stinging, and irritation upon application to the eye. Although not specifically evaluated, it is possible that carteolol could lead to reduction in heart rate following topical administration in veterinary species.³

Dosages

Glaucoma (extra-label): 1 drop to the affected eye(s) 2 times daily

Monitoring

- Clinical efficacy
- Monitor intraocular pressure (IOP)
- Observe asthmatic and diabetic patients for respiratory difficulty or acute hypoglycemia, respectively

Client Information

- Use this medication as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product at controlled room temperature (15°C-25°C [59°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Carteolol ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Carteolol HCl 1% Ophthalmic Solution (containing carteolol HCl 10 mg; benzalkonium chloride 0.05 mg; NaCl; monobasic and dibasic sodium phosphate; NaOH and/or HCl to adjust pH to 6 to 8; purified water), in a 5 mL, 10 mL, and 15 mL dropper bottle; generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Chloramphenicol Ophthalmic

(klor-am-**fen**-uh-kol)

Broad-Spectrum Antibiotic

For systemic use, refer to *Chloramphenicol*

Prescriber Highlights

- ▶ Used to provide antibacterial coverage in the presence of corneal ulcerations or therapeutic antibacterial coverage for susceptible ocular surface infections, particularly feline *Mycoplasma* spp or *Chlamydia* spp infections
- ▶ Chloramphenicol has caused acute toxicity and death in humans; instruct caregiver to wear gloves and avoid contact with this drug.
- ▶ Chloramphenicol ophthalmic preparations are no longer commercially available but may be compounded by a qualified compounding pharmacist.
- ▶ The National Institute for Occupational Safety and Health (NIOSH) classifies chloramphenicol as a hazardous drug; use appropriate precautions when handling.

Uses/Indications

Chloramphenicol is used for the treatment of surface ocular infections involving the conjunctiva and/or cornea caused by chloramphenicol-susceptible organisms.¹

The World Health Organization (WHO) has designated chloramphenicol as a Highly Important antimicrobial for human medicine.² The Office International des Epizooties (OIE) has designated amphenicol as a Veterinary Critically Important Antimicrobial (VCIA) Agent in veterinary species.³

Pharmacology/Actions

Chloramphenicol is a broad-spectrum, bacteriostatic antibiotic that inhibits protein synthesis by interfering with the transfer of activated amino acids from soluble RNA to ribosomes, thus preventing protein elongation. It generally has good activity against gram-positive, gram-negative, and intracellular organisms, common to the ocular surface; the activity against *Pseudomonas* spp is consistently poor. Development of resistance to chloramphenicol is uncommon for *Staphylococcus* spp and *Streptococcus* spp.⁴⁻⁷

Chloramphenicol is found in measurable amounts in the aqueous humor following topical application to the eye.

Contraindications/Precautions/Warnings

Chloramphenicol is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Chloramphenicol may induce dose-dependent bone marrow suppression or idiosyncratic aplastic anemia, both of which have a low risk of occurring following topical ophthalmic administration.^{8,9} Caregivers should be instructed to wear gloves and wash hands after handling this drug. Chloramphenicol is banned for use in food-producing animals.

The National Institute for Occupational Safety and Health (NIOSH) classifies chloramphenicol as a hazardous drug; personal protective equipment (PPE) should be used accordingly to minimize the risk for exposure.

Adverse Effects

Adverse effects reported in animals include occasional burning or stinging on application. Bone marrow suppression and blood dyscrasias have been reported following the use of systemic chloramphenicol.

Dosages

Bacterial conjunctivitis (extra-label): ¼ – ½ inch strip or 1 drop in the affected eye(s) every 6 to 8 hours

Monitoring

- Cytology and culture to evaluate for the presence of bacteria
- Clinical resolution of signs related to ocular surface infection
- Baseline and periodic CBC to monitor for bone marrow suppression

Client Information

- Humans should avoid direct contact with this medicine. Wear gloves during administration and wash hands after handling this medicine.
- Use proper administration techniques to avoid contamination of the medication. Do not touch the dropper (or ointment) tip or allow it to touch your animal's eye or any other surface to prevent contamination.
- Keep cap tightly closed when not in use.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Do not use if the color changes, becomes cloudy, or if there are particles in the solution.
- This medication is considered to be a hazardous drug as defined by the National Institute for Occupational Safety and Health (NIOSH). Talk with your veterinarian or pharmacist about the use of personal protective equipment when handling this medicine.

Storage/Stability

Store at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Chloramphenicol should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Chloramphenicol was formerly available as a 0.5% ophthalmic solution or a 1% ointment but is no longer commercially available. A qualified compounding pharmacist may prepare it if needed.

References

For the complete list of references, see wiley.com/go/budde/plumb

Cidofovir Ophthalmic

(sye-**doe**-fo-veer)

Antiviral

Prescriber Highlights

- ▶ Used as an ophthalmic antiviral agent in cats with feline herpesvirus-1 (FHV-1)
- ▶ Less frequent administration and shorter treatment duration when compared to other ophthalmic antiviral agents
- ▶ Cidofovir has caused hypospermia and birth defects in humans and should not be used in animals intended for breeding.
- ▶ Not commercially available in an ophthalmic dosage form but can be compounded as a 0.5% ophthalmic solution
- ▶ The National Institute for Occupational Safety and Health (NIOSH) classifies cidofovir as a hazardous drug; use appropriate precautions when handling.

Uses/Indications

Cidofovir is compounded into a 0.5% ophthalmic solution for the topical treatment of feline herpesvirus-1 (FHV-1) keratitis in cats. In vitro efficacy of cidofovir against FHV-1 is consistently high.^{1,2}

Because cidofovir is effective when administered twice daily for 10 days,³ it has a significant therapeutic advantage over other antiviral ophthalmic agents (eg, trifluridine, vidarabine, and idoxuridine) which must be administered 4 to 6 times daily for 2 to 3 weeks.

Cidofovir decreases viral shedding in dogs experimentally infected with ocular canine herpesvirus-1 infections; however, it causes local irritation and toxicity.³ Although the efficacy of cidofovir has not been evaluated in horses for treatment of keratitis associated with equine herpesvirus-2, it has demonstrated mild in vitro efficacy versus other equine alpha-herpesviruses.

Pharmacology/Actions

Cidofovir suppresses viral replication by inhibiting the synthesis of viral DNA. Cidofovir diphosphate, the active intracellular metabolite, demonstrates much greater selectivity for viral, rather than human, DNA. Because cidofovir is incorporated into the growing DNA chain, it reduces the rate of viral DNA synthesis, thereby having a virustatic effect.

Contraindications/Precautions/Warnings

Cidofovir is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Do not use cidofovir in animals intended for breeding, as cidofovir has been shown to cause hypospermia and is mutagenic and teratogenic in humans. Following IV administration, cidofovir has caused death by irreversible nephrotoxicity⁴; do not use in animals with pre-existing renal impairment or concurrently with nephrotoxic drugs. Direct intraocular injection is contraindicated.⁴

The National Institute for Occupational Safety and Health (NIOSH) classifies cidofovir as a hazardous drug; personal protective equipment (PPE) should be used accordingly to minimize the risk for exposure.

Adverse Effects

Adverse effects reported in humans or animals following topical administration include burning or stinging on application and nasolacrimal occlusion in rabbits and humans following prolonged (weeks) treatment. Severe adverse effects reported in humans following IV administration include nephrotoxicity (after even a single IV dose), neutropenia, metabolic acidosis, and uveitis/ocular hypotony. Although these reactions reported with IV administration have not been reported following ocular administration, nephrotoxicity is theoretically possible. Topical ocular administration of cidofovir to dogs has led to signs of irritation (eg, conjunctival pigmentation, blepharitis).³

Dosages

Feline herpesvirus-1 keratitis (extra-label): 1 drop to the affected eye(s) 2 times daily for 10 days⁵

Canine herpesvirus-1 ocular infections (extra-label): 1 drop to the affected eye(s) twice daily for 14 days³

Monitoring

- Clinical resolution of signs related to herpesvirus ocular infections (eg, corneal ulceration, corneal vascularization, conjunctivitis)
- Consider baseline and periodic renal chemistry panels

Client Information

- Humans should avoid direct contact with this drug. Wear gloves during administration and wash hands after each treatment is done.
- Use proper administration techniques to avoid contamination of the medication. Do not touch the dropper (or ointment) tip or allow it to touch your animal's eye or any other surface to prevent contamination.

- Keep cap tightly closed when this medicine is not in use.
- Wait 5 minutes after applying this medicine before applying any other medications to the eye.
- Do not use if the color changes or becomes cloudy or if there are particles in the solution.
- This medication is considered to be a hazardous drug as defined by the National Institute for Occupational Safety and Health (NIOSH). Talk with your veterinarian or pharmacist about the use of personal protective equipment when handling this medicine.

Storage/Stability

Store in the refrigerator (2°C–8°C [35.6°F–46.4°F]) away from moisture and sunlight; do not freeze.

The reported chemical stability of this solution is 180 days when stored in glass at 4°C (39.2°F), -20°C (-4°F), and -80°C (-112°F).⁶ Storage times should be determined by the results of individual extended stability and sterility testing or compendial standards.

Compatibility/Compounding Considerations

Cidofovir should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Cidofovir is not approved for use as an ophthalmic dosage form, but a cidofovir 0.5% ophthalmic solution can be prepared by a qualified compounding pharmacist.

References

For the complete list of references, see wiley.com/go/budde/plumb

Ciprofloxacin Ophthalmic

(sip-roe-**flox**-a sin) *Ciloxan*®

Fluoroquinolone Antibiotic

For systemic use, refer to **Ciprofloxacin**

Prescriber Highlights

- ▶ Used to treat bacterial conjunctivitis or keratitis
- ▶ Physically incompatible with many drugs; therefore, should not be directly mixed with other ophthalmic medications
- ▶ Crystalline drug precipitates may be noted in the superficial portion of corneal defects during use.

Uses/Indications

In veterinary patients, ciprofloxacin ophthalmic may be used for the treatment of bacterial conjunctivitis or keratitis caused by susceptible strains of *Staphylococcus* spp, *Streptococcus* spp, *Pseudomonas aeruginosa*, *Chlamydia* spp, and *Haemophilus* spp.¹

The World Health Organization (WHO) has designated ciprofloxacin as a Critically Important, Highest Priority antimicrobial in human medicine.²

Pharmacology/Actions

Ciprofloxacin is a bactericidal and concentration-dependent antibiotic, with susceptible bacteria cell death occurring within 20 to 30 minutes of exposure. Ciprofloxacin has demonstrated a significant postantibiotic effect in both gram-negative and gram-positive bacteria and is active in both the stationary and growth phases of bacterial replication. Its mechanism of action is not thoroughly understood, but it is believed to act by inhibiting bacterial DNA-gyrase (a type-II topoisomerase), thereby preventing DNA supercoiling and DNA synthesis.

Although ciprofloxacin is considered to have greater efficacy against gram-negative organisms as compared with gram-positive, studies involving ocular isolates obtained from dogs and horses with

bacterial keratitis indicate good efficacy with minimal resistance against *Streptococcus* spp, *Staphylococcus* spp, and *Pseudomonas aeruginosa*.³⁻⁶ However, more recent reports involving isolates from humans with ocular infections indicate lower susceptibility of *Staphylococcus* spp to ciprofloxacin.

Contraindications/Precautions/Warnings

Ciprofloxacin is contraindicated in patients that have a history of hypersensitivity to it, other fluoroquinolones, or any component of the formulation. Ophthalmic preparations are for topical ophthalmic use only. They are not for injection subconjunctivally or into the anterior chamber of the eye. Fluoroquinolone-induced retinal toxicity in domestic cats has not been demonstrated with a topical formulation of ciprofloxacin.

Adverse Effects

Adverse effects reported in humans or animals include blurred vision, tearing, eye pain, redness, itching, and a bad taste in the mouth.⁷ Crystalline drug precipitates may be observed in the superficial portion of corneal defects during use; however, this effect did not adversely affect clinical and visual outcomes in human clinical studies.⁷ The arthropathy that occurs in immature animals associated with oral administration of quinolones has not been noted following topical ophthalmic administration.

Dosages

Conjunctivitis (extra-label): 1 drop or ¼ – ½ inch strip to the affected eye(s) every 2 to 8 hours

Keratitis (extra-label): 1 drop or ¼ – ½ inch strip to the affected eye every 2 to 8 hours

Monitoring

- Corneal cytology and culture to evaluate for the persistence of bacteria
- Evaluation for clinical signs suggestive of persistent infection

Client Information

- Use this medication as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Do not touch the dropper tip or allow it to touch your animal's eye or any other surface to prevent contamination.
- Wait 5 minutes after applying this medicine before applying any other medicine to your animal's eye(s).
- Crystals may appear on the surface of your animal's eye during treatment. This change is not harmful and will go away once treatment is complete.
- Keep cap tightly closed when not in use. Store this medication in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes, becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product in the refrigerator (2°C–8°C [35.6°F–46.4°F]) or at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Fluoroquinolone antibiotics interact with many other drugs, including those containing cations (eg, iron, aluminum, magnesium, or calcium). Ciprofloxacin should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Ciprofloxacin 0.3% Ophthalmic Solution (containing ciprofloxacin HCl 3.5 mg equivalent to 3 mg base; benzalkonium chloride 0.006%; sodium acetate; acetic acid; mannitol; edetate disodium; HCl and/or NaOH may be used to adjust pH to 4.5; purified water; osmolality is 300 mOsm/kg), 2.5 mL, 5 mL, or 10 mL in sterile ophthalmic drop-per bottles; *Ciloxan*®, generic; (Rx)

Ciprofloxacin 0.3% Ophthalmic Ointment (containing ciprofloxacin HCl 3.33 mg equivalent to 3 mg base; mineral oil; white petrolatum), in 3.5 g sterile ophthalmic ointment tubes; *Ciloxan*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Cromolyn Ophthalmic

(*kroh*-mah-lin)

Mast Cell Stabilizer

Prescriber Highlights

- ▶ Used for seasonal allergic conjunctivitis or atopic conjunctivitis
- ▶ Administer 2 to 6 times daily
- ▶ Mild transient stinging and burning may occur upon application

Uses/Indications

Mast cell-stabilizing ophthalmic agents are used to control the clinical signs of allergic and atopic conjunctivitis. Response to therapy (eg, decreased itching, tearing, redness, and discharge) is usually evident within a few days, but longer treatment for up to 6 weeks may be required. Once improvement has been established, therapy should be continued for as long as needed to sustain improvement.¹ If required, corticosteroids may be used concomitantly with cromolyn ophthalmic solution. Efficacy of cromolyn as treatment for conjunctivitis in veterinary species has not been established.

Pharmacology/Actions

Cromolyn inhibits the degranulation of sensitized mast cells and blocks the release of histamine and slow-releasing substance of anaphylaxis from mast cells following antigen recognition. Cromolyn does not have intrinsic vasoconstrictor, antihistaminic, cyclooxygenase inhibition, or other anti-inflammatory properties. Cromolyn is poorly absorbed when applied topically.

Contraindications/Precautions/Warnings

Cromolyn is contraindicated in patients that have a history of hypersensitivity to cromolyn or any component of the formulation.

Adverse Effects

Adverse effects reported in humans or animals include a transient mild stinging and burning upon application.

Dosages

Allergic conjunctivitis (extra-label): 1 drop to each eye 2 to 6 times daily

Monitoring

- Clinical efficacy: resolution of clinical signs associated with allergic conjunctivitis

Client Information

- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.

- Wait for 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine at room temperature away from moisture and sunlight. Do not freeze.
- Do not use the medicine if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product at room temperature (15°C-30°C [59°F-86°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Cromolyn ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Cromolyn Sodium 4% Ophthalmic Solution (containing cromolyn sodium 40 mg; benzalkonium chloride 0.01%; edetate disodium 0.1%; purified water; HCl and/or NaOH to adjust pH to 4 to 7), 10 mL in sterile ophthalmic dropper bottle; generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Cyclopentolate Ophthalmic

(sy-e-kloe-**pen**-toe-late) Cyclogyl® Cyclomydril®, Akpentolate®, Pentolair®

Anticholinergic Cycloplegic/Mydriatic Agent

Prescriber Highlights

- ▶ Primarily used as a diagnostic agent to facilitate mydriasis and cycloplegia
- ▶ Rapid and intense onset of mydriasis and cycloplegia in 15 to 60 minutes; heavily pigmented irises may require an increased frequency of administration or stronger concentrations. May increase intraocular pressure (IOP); therefore, use cautiously in patients with glaucoma
- ▶ Higher concentrations (*at least 1%*) cause stinging on application.
- ▶ Available as a combination product with phenylephrine (Cyclomydril®) to facilitate mydriasis with less cycloplegia

Uses/Indications

Cyclopentolate is primarily used to induce rapid mydriasis and cycloplegia to enable funduscopic examination and cycloplegic refraction to be performed. It is also available as a combination with phenylephrine to accomplish greater mydriasis with lesser cycloplegia.

Pharmacology/Actions

Cyclopentolate blocks the responses of the sphincter muscle of the iris and the accommodative muscle of the ciliary body to cholinergic stimulation, resulting in pupillary dilation (mydriasis) and paralysis of accommodation (cycloplegia).¹ Cyclopentolate acts rapidly, producing maximal dilation in less than 1 hour in dogs² and cats,³ and within 12 hours in horses,^{4,5} with a total duration of mydriasis from 2 to 4 days. Complete recovery of accommodation usually takes 6 to 24 hours. Heavily pigmented irises may require an increased frequency of administration or stronger concentrations than lightly pigmented irises.¹ When cyclopentolate is combined with phenylephrine (an adrenergic drug), synergistic mydriasis occurs, with less accompanying cycloplegia due to the lower concentration of cyclopentolate in the combination product.

Contraindications/Precautions/Warnings

Cyclopentolate is contraindicated in patients that have a history of

hypersensitivity to it or any component of the formulation. Although a single drop of cyclopentolate did not elevate intraocular pressure (IOP) in normal dogs,² cyclopentolate may cause an increase in IOP and therefore should be used cautiously in patients with glaucoma. Administration to normal cats³ and horses⁴ has been associated with increased IOP in studies; however, another study documented no effect on IOP in normal horses.⁵ Reduction in STT has been reported in cats.³

Adverse Effects

Adverse effects reported in humans or animals include burning and stinging on application.¹ Increased intraocular pressure (IOP), photophobia, blurred vision, irritation, hyperemia, conjunctivitis, blepharoconjunctivitis, punctate keratitis, and synechiae have also been reported.¹ Burning and stinging are usually associated with higher concentrations (*at least 1%*). CNS and cardiovascular disturbances are often noted in human pediatric patients and may occur when used in smaller animals.

Dosages

Mydriasis and cycloplegia (extra-label): 1 drop to the desired eye(s) followed by a second drop 5 minutes later if necessary. **NOTE:** Use of a single drop may avoid adverse systemic effects in smaller animals. Administer this medicine 40 to 50 minutes prior to a diagnostic procedure to ensure full effects.

Monitoring

- Clinical efficacy: desired mydriasis and cycloplegia achieved; full recovery in 24 hours

Client Information

- This medicine is rarely used at home.
- This medicine may make your animal sensitive to light for ≈24 hours after administration. Keep your animal out of bright sunlight until pupils return to normal size.
- Wash your hands after use to prevent accidental contact with eyes (causes pupil dilation).
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Cyclopentolate ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Cyclopentolate HCl Ophthalmic Solution 0.5% (containing cyclopentolate HCl 0.5%; benzalkonium chloride 0.01%; boric acid; edetate disodium; sodium carbonate and/or HCl to adjust pH to between 3 and 5.5; purified water), 15 mL in sterile ophthalmic dropper bottles; Cyclogyl®, generic; (Rx)

Cyclopentolate HCl Ophthalmic Solution 1% (containing cyclopentolate HCl 1%; benzalkonium chloride 0.01%; boric acid; edetate disodium; sodium carbonate and/or HCl to adjust pH to between 3

and 5.5; purified water), 2 mL, 5 mL, or 15 mL in sterile ophthalmic dropper bottles; *Cyclogyl*®, *Pentolair*®, *Akpetolate*®; (Rx)

Cyclopentolate HCl Ophthalmic Solution 2% (containing cyclopentolate HCl 2%; benzalkonium chloride 0.01%; boric acid; edetate disodium; sodium carbonate and/or HCl to adjust pH to between 3 and 5.5; purified water), 2 mL, 5 mL, or 15 mL in sterile ophthalmic dropper bottles; *Cyclogyl*®; (Rx)

Human-Labeled Combination Products:

Cyclopentolate HCl 0.2% and Phenylephrine HCl 1% Ophthalmic Solution (containing cyclopentolate HCl 0.2%; phenylephrine HCl 1%; benzalkonium chloride 0.01%; edetate disodium; boric acid; HCl and/or sodium carbonate to adjust pH; purified water), 2 mL or 5 mL in sterile ophthalmic dropper bottles; *Cyclomydril*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Cyclosporine Ophthalmic

(*sy*-klo-*spore*-in) *Cequa*®, *Ciclosporin*, *Cyclosporin A*, *Optimmune*®, *Restasis*®

Immunosuppressive Agent

See also *Cyclosporine (Systemic)*

Prescriber Highlights

- ▶ Approved as an ophthalmic ointment for the treatment of keratoconjunctivitis sicca (KCS) in dogs and for other ocular surface inflammatory diseases in other species
- ▶ Poor intraocular penetration requires alternate routes of administration to manage intraocular inflammatory diseases
- ▶ Animals that do not respond to the approved (0.2%) ophthalmic ointment may benefit from higher concentrations (1% to 2%) of cyclosporine ophthalmic solution, which must be compounded.
- ▶ Weeks of therapy may be required to cause an increase of tear production; however, discontinuation of therapy can quickly (within days) result in loss of normal tear production.
- ▶ The National Institute for Occupational Safety and Health (NIOSH) classifies cyclosporine as a hazardous drug; use appropriate precautions when handling.

Uses/Indications

Topical ophthalmic cyclosporine is approved for the treatment of keratoconjunctivitis (KCS) in dogs; it may also be used to treat other ocular surface inflammatory conditions, such as chronic superficial keratitis (CSK; pannus), immune-mediated keratitis (IMMK), eosinophilic keratitis, pigmentary keratitis, and postoperative keratitis in dogs and other species.^{1,2} Topical cyclosporine 1.5% has demonstrated efficacy for the management of eosinophilic keratitis in cats.³ Sustained-released cyclosporine implants have been developed for suprachoroidal and episcleral implantation; the former is for the treatment of equine recurrent uveitis (ERU), and the latter for IMMK in horses and KCS in dogs.⁴⁻⁹

Pharmacology/Actions

Cyclosporine, a calcineurin inhibitor, is an immunosuppressive agent that focuses on cell-mediated immune responses, with some humoral immunosuppressive action. Cyclosporine binds to T-cell cyclophilin and blocks calcineurin-mediated T-cell activation. T-helper lymphocytes are the primary target, but T-suppressor cells are also affected. Cyclosporine can also inhibit cytokine production and release (including IL-2 and interferon-gamma in dogs¹⁰⁻¹² and IL-3, IL-4, and tissue necrosis factor-alpha in humans), thereby affecting the

function of eosinophils, mast cells, granulocytes, and macrophages.

The immunosuppressive effects make cyclosporine useful when it is administered locally in animals with ocular diseases that are suspected to be immune-mediated, such as keratoconjunctivitis (KCS) and chronic superficial keratitis (CSK). Cyclosporine increases lymphocyte apoptosis and decreases corneal epithelial apoptosis in dogs with KCS,¹³ thus protecting the ocular surface. However, other mechanisms likely improve tear production, as tear production is increased by topical administration of cyclosporin in dogs without KCS. For this reason, cyclosporine is thought to have direct stimulatory effects on the lacrimal gland, possibly by altering neurotransmitter release.¹⁴ After initiation of cyclosporine therapy, effective tear production takes several weeks; however, cessation of therapy results in the return of clinical signs in a matter of days.¹ Clinical improvement in cases of KCS is not necessarily dependent on an increase in aqueous tear production as measured by the Schirmer tear test (STT) because other components of the tear film that are not assessed by the STT may also improve with topical cyclosporine administration.¹⁵

Because of the frequent need for the chronic daily administration of topical cyclosporine, the sustained-release cyclosporine devices are surgically placed in the episcleral space in horses with immune-mediated keratitis (IMMK). The duration of drug release is ≈18 months.⁴ Due to limited penetration to the intraocular environment, the sustained-release cyclosporine disc devices are surgically implanted suprachoroidally in horses with equine recurrent uveitis (ERU). The duration of drug release is ≈2 to 3 years.⁷

Contraindications/Precautions/Warnings

Cyclosporine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Use in the presence of ocular viral or fungal infections should be undertaken with caution. Cyclosporine is available as an ophthalmic solution for humans (0.05% to 0.09%); however, these formulations are not effective in controlling keratoconjunctivitis (KCS) in dogs. More concentrated solutions of cyclosporine ophthalmic (1% to 2%) may be prepared (compounded) by qualified compounding pharmacists. It is not recommended to administer cyclosporine ophthalmic ointment or solutions compounded in oil via subpalpebral lavage catheter because of the likely irreversible obstruction of the catheter, necessitating replacement.

The National Institute for Occupational Safety and Health (NIOSH) classifies cyclosporine as a hazardous drug; personal protective equipment (PPE) should be used accordingly to minimize the risk for exposure.

Adverse Effects

Adverse effects reported in humans or animals include local irritation, periocular erythema, lid spasm, rubbing of the eye, alopecia, and epiphora. Many adverse effects are believed to be induced by the vehicle preparation used in compounded formulations rather than by cyclosporine itself. In dogs with herpesvirus keratitis, the administration of topical ophthalmic cyclosporine has not been shown to induce recrudescence of clinical signs.

Dosages

Keratoconjunctivitis sicca and chronic superficial keratitis in dogs (label dosage; FDA-approved): ¼ inch strip to the affected eye(s) every 12 hours¹⁷

Inflammatory ocular surface diseases (extra-label):

- a. Ointment: ¼ – ½ inch strip to the affected eye(s) every 12 hours
- b. Cyclosporine solutions (1% to 2%): 1 drop to the affected eye(s) every 12 hours

Immune-mediated keratitis (extra-label): Surgical implantation of a sustained-release device created for episcleral implantation should be

performed by a board-certified veterinary ophthalmologist. The implant contains cyclosporine for continued release over $\approx$ 12 to 18 months.

Equine recurrent uveitis (ERU) (extra-label): Surgical implantation of a sustained-release device created for suprachoroidal implantation should be performed by a board-certified veterinary ophthalmologist. The implant contains cyclosporine for continued release over $\approx$ 2 to 3 years.

Monitoring

- Clinical effect for keratoconjunctivitis (KCS). Evaluate Schirmer tear test, as well as a reduction in mucoid ocular discharge and conjunctival hyperemia.
- Clinical effect for ocular surface inflammatory conditions (topical administration; episcleral device). Evaluate for lack of progression (and possibly regression) of chronic inflammatory changes, such as corneal pigmentation, vascularization, and fibrosis.
- Clinical effect for equine recurrent uveitis (ERU; suprachoroidal device): Evaluate markers of intraocular inflammation (eg, flare, hypopyon, ocular hypotony, miosis, vitreal debris) and frequency of recurrence.

Client Information

- Make sure you understand how to give this medication to your animal. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 to 10 minutes after applying this medicine before applying any other medicines to the eye.
- Store cyclosporine ointment in the refrigerator or at room temperature away from moisture and sunlight. Handle and squeeze the tube gently so the metal will not crack and leak.
- Store cyclosporine solutions at room temperature (do not refrigerate or freeze).
- Do not use this medicine if the color of the solution changes, if it becomes cloudy, or if there are particles in the solution.
- This medication is considered to be a hazardous drug as defined by the National Institute for Occupational Safety and Health (NIOSH). Talk with your veterinarian or pharmacist about the use of personal protective equipment when handling this medicine.

Storage/Stability

Store the commercially available ophthalmic ointment in the refrigerator (2°C – 8°C [35.6°F – 46.4°F]) or at controlled room temperature (20°C – 25°C [68°F – 77°F]) away from moisture and sunlight. Do not refrigerate or freeze compounded cyclosporine solutions.

The long-term stability of a 1% solution in corn oil determined by USP is 180 days. Storage times should be determined by the results of individual extended stability and sterility testing or compendial standards.

Compatibility/Compounding Considerations

Cyclosporine should not be mixed directly (in vitro) with any other drugs.

Do not administer cyclosporine ophthalmic ointment or solutions compounded in oil in a subpalpebral lavage catheter, because of the likely irreversible obstruction of the catheter.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS:

Cyclosporine 0.2% Ophthalmic Ointment (containing cyclosporine 2 mg/g, USP; petrolatum, USP; corn oil, NF; petrolatum and lanolin alcohol), in a 3.5 g ophthalmic ointment tube; *Optimmune*[®]; (Rx). Approved for use in dogs.

Cyclosporine 1% to 2% Ophthalmic Solutions in fixed oils may be compounded by qualified compounding pharmacists.

HUMAN-LABELED PRODUCTS:

Cyclosporine 0.05% Ophthalmic Emulsion (containing cyclosporine 0.05%; glycerin; castor oil; polysorbate 80; carbomer copolymer type A; purified water; NaOH to adjust pH), 30 or 60 sterile, single-use LDPE vials (0.4 mL each), or 5.5 mL sterile multi-dose ophthalmic dropper; *Restasis*[®]; (Rx). **NOTE:** This solution is not effective in controlling keratoconjunctivitis (KCS) in dogs.

Cyclosporine 0.09% Ophthalmic Solution (containing cyclosporine 0.09%; polyoxyl 40 hydrogenated castor oil; octoxynol-40; sodium phosphate, monobasic, dihydrate; sodium phosphate, dibasic, anhydrous; sodium chloride; povidone K90; hydrochloric acid; sodium hydroxide; water), in 6 pouches with 10 single-use vials (0.25 mL each) per pouch; *Cequa*[®]; (Rx). **NOTE:** This solution is not effective in controlling KCS in dogs.

References

For the complete list of references, see wiley.com/go/budde/plumb

Demecarium Ophthalmic

(deh-meh-**kar**-ee-um)

Anticholinesterase Ocular Antihypertensive

Prescriber Highlights

- Used primarily for preventive management of glaucoma in the unaffected eye of canine patients diagnosed with primary glaucoma in the fellow eye
- Product must be compounded as it is no longer commercially available.
- Used 1 to 2 times daily

Uses/Indications

Demecarium ophthalmic is used for preventive management of glaucoma in the contralateral eye of canine patients diagnosed with primary glaucoma in the fellow eye.^{1,2} Demecarium's therapeutic efficacy for managing active glaucoma is minimal relative to other commercially available medications.

Pharmacology/Actions

Demecarium is an anticholinesterase that prolongs the effect of acetylcholine by inactivating the cholinesterases that break down acetylcholine, which has been released at the neuroeffector junction of parasympathetic postganglionic nerves. Demecarium inactivates both pseudocholinesterase and acetylcholinesterase. It causes constriction of the iris sphincter muscle (miosis) and the ciliary body muscle. These actions facilitate the outflow of aqueous humor, leading to a reduction in intraocular pressure (IOP). Demecarium is an organophosphate and may cause severe and even fatal dose-related toxicity associated with a decrease in systemic acetylcholinesterase levels, especially in cats.³ See **Adverse Effects**.

Contraindications/Precautions/Warnings

Demecarium is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Demecarium is contraindicated in pregnant animals; caregivers that are pregnant or intend to become pregnant should be warned to avoid contact with this medication.

Adverse Effects

Adverse effects reported in humans or animals include stinging, burning, and ocular irritation. Demecarium can cause severe and even fatal dose-related systemic toxicity, especially in cats.³ Signs of systemic toxicity include vomiting, anorexia, diarrhea, lethargy, and weakness. Short-term (8 days) induction of aqueous humor flare occurs following the administration of demecarium in normal dogs.⁴

Dosages

Prophylaxis of glaucoma in the normal eye (extra-label): 1 drop to the nonglaucomatous eye 1 to 2 times daily

Monitoring

- Clinical efficacy; maintenance of normal intraocular pressure (IOP) in the contralateral eye of dogs with unilateral glaucoma
- Monitor for signs of systemic toxicity (vomiting, anorexia, diarrhea, lethargy, and weakness)

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this product as directed on the product labeling, away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product as directed on the product labeling away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Demecarium ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Demecarium bromide is no longer commercially available but may be prepared (compounded) as a 0.125% or 0.25% ophthalmic solution by qualified compounding pharmacists.

References

For the complete list of references, see wiley.com/go/budde/plumb

Dexamethasone Ophthalmic

(decks-uh-**meth**-uh-sown) *Maxidex[®], Dexasporin[®], Maxitrol[®], Tobradex[®]*

Corticosteroid

For systemic use, refer to Dexamethasone

Prescriber Highlights

- ▶ Topical corticosteroid used to treat ocular surface and intraocular inflammatory conditions
- ▶ May be combined with antibiotics (eg, neomycin, polymyxin B)
- ▶ Contraindications include a hypersensitivity to dexamethasone; an existing or suspected bacterial, viral, or fungal ocular infection; and in the presence of a corneal abrasion or ulceration.

Uses/Indications

Dexamethasone ophthalmic can be used to treat inflammation following ocular injury or cataract surgery or to treat generalized inflammatory conditions of the anterior segment (eg, conjunctivitis, keratitis, anterior uveitis). Evaluation of relative efficacy of dexamethasone ophthalmic in veterinary patients has not been performed.

Dexamethasone is available in many combinations with antimicrobial agents for topical ophthalmic use.

Pharmacology/Actions

Corticosteroids have broad immunomodulatory and anti-inflammatory actions, which are mediated via direct effects on gene expression, as well as by nongenomic effects on cell membranes and receptors. Ultimately, corticosteroids decrease pro-inflammatory gene transcription, production of pro-inflammatory prostaglandins and leukotrienes, B- and T-cell number and function, cell-mediated immunity, and humoral immunity.¹ Specific mechanisms involved in controlling ocular inflammatory conditions are not definitively known. Dexamethasone is a potent base steroid (relative glucocorticoid potency of 30 versus 4 for prednisolone and 1 for hydrocortisone). Greater intraocular penetration, and therefore potential clinical efficacy in cases of anterior uveitis, is achieved with dexamethasone alcohol formulations relative to sodium phosphate formulations; however, intraocular penetration of dexamethasone alcohol and sodium phosphate preparations is less than that of prednisolone acetate suspensions.

Following the topical administration of dexamethasone sodium phosphate, measurable concentrations of dexamethasone are present in the anterior segment of normal cat and dog eyes.^{2,3}

Contraindications/Precautions/Warnings

Dexamethasone is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Due to inhibition of epithelial wound healing and potentiation of infection, dexamethasone ophthalmic is also contraindicated when corneal ulceration is suspected or present or in the presence of corneal infection (bacterial, fungal, or viral).⁴ Topical administration of glucocorticoids may produce systemic effects. For more, refer to **Glucocorticoid Agents, General Information**. Use dexamethasone ophthalmic with caution in patients with endocrine diseases (eg, diabetes mellitus, acromegaly), infectious diseases, chronic kidney disease, congestive heart failure, systemic hypertension, or gastric ulceration. When use of dexamethasone ophthalmic is necessary to treat a patient with these conditions, the minimal effective dose should be administered for the shortest possible time.

Adverse Effects

Adverse effects reported in humans include development of cataracts (via altered glucose or lenticular metabolic effects), increased intraocular pressure (IOP; via activation of glucocorticoid receptors in the trabecular meshwork), infection (via immunosuppression), decreased wound healing (via increased collagenolytic activity), mydriasis, and calcific keratopathy. In some animals, IOP may increase with chronic use; however, most reports suggest the change is reversible when the treatment is discontinued.^{5,6} Cats with glaucoma have demonstrated a potentially clinically significant increase in IOP with administration of topical dexamethasone.⁷ Systemic effects with topical ophthalmic administration are rare but include hepatopathy, suppression of endogenous glucocorticoid production, and focal alopecia. Concurrent use of systemic steroids increases the risk for these adverse effects.

Dosages

Conjunctivitis, keratitis, anterior uveitis (extra-label): 1 drop or ¼ inch strip to the affected eye(s) as frequently as every 2 hours until clinical improvement occurs, then gradually (over days to weeks) decrease the frequency of administration, provided clinical improvement is maintained.

Post-cataract removal (extra-label): 1 drop to the affected eye(s) every 8 to 12 hours

Monitoring

- Clinical efficacy
- Resolution or development of signs of inflammation
- Intraocular pressure (IOP; especially in cats) and ocular surface

integrity (eg, fluorescein stain) as needed based on clinical presentation

- Blood or urine glucose changes based on development of clinical signs (eg, polyuria, polydipsia) in diabetic animals

Client Information

- This medicine can be given for various lengths of time. Be sure you understand how often and for how long your animal should be given this medicine.
- Shake suspensions well before using.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Contact your veterinarian as soon as possible if your animal's eye condition worsens or does not improve while using this medicine.
- Your veterinarian will need to monitor your animal's eye condition while taking this medicine. Do not miss these important follow-up visits.
- Store this medicine as directed on the product label. Do not freeze.
- Do not use this medicine if the color changes, becomes cloudy, or if there are visible particles in the solution.

Storage/Stability

Storage recommendations can vary; follow storage requirements listed on packaging. Most commonly, this product is recommended to be stored at controlled room temperature 68°F to 77°F [20°C-25°C]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Dexamethasone ophthalmic solutions should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Dexamethasone Sodium Phosphate 0.1% Ophthalmic Suspension (containing dexamethasone 0.1%; benzalkonium chloride 0.01%; hypromellose 0.5%; NaCl; dibasic sodium phosphate; polysorbate 80; edetate disodium; citric acid and/or NaOH to adjust pH; purified water), 5 mL or 15 mL in sterile ophthalmic dropper bottle; *Maxidex*®; generic; (Rx)

Dexamethasone Sodium Phosphate 0.1% Ophthalmic Solution (containing dexamethasone 0.1%; benzalkonium chloride 0.01%; hypromellose 0.5%; NaCl; dibasic sodium phosphate; polysorbate 80; edetate disodium; citric acid and/or NaOH to adjust pH; purified water), 5 mL or 15 mL sterile ophthalmic dropper bottle; generic; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Neomycin and Polymyxin B Sulfates and Dexamethasone Ophthalmic Suspension (containing neomycin sulfate equivalent to neomycin 3.5 mg, polymyxin B sulfate 10,000 units, dexamethasone 0.1%, benzalkonium chloride 0.004%, hypromellose 2910 0.5%, sodium chloride, polysorbate 20, HCl and/or NaOH to adjust pH, purified water), in a 5 mL sterile ophthalmic dropper bottle; *Maxitrol*®, *Dexasporin*®; (Rx)

Neomycin and Polymyxin B Sulfates and Dexamethasone Ophthalmic ointment (containing neomycin sulfate equivalent to neomycin 3.5 mg, polymyxin B sulfate 10,000 units, dexamethasone 0.1%, methylparaben 0.05%, propylparaben 0.01%, white petrolatum, anhydrous liquid lanolin), in a sterile 3.5 g tube; *Maxitrol*®, generic; (Rx)

Tobramycin and Dexamethasone Ophthalmic Ointment (containing

tobramycin 0.3%, dexamethasone 0.1%, chlorobutanol 0.5%, mineral oil, white petrolatum), in a sterile 3.5 g tube; *Tobradex*®; (Rx)

Tobramycin and Dexamethasone Ophthalmic Suspension (containing tobramycin 0.3%, dexamethasone 0.1%, benzalkonium chloride 0.01%, tyloxapol, edetate disodium, NaCl, hydroxyethyl cellulose, sodium sulfate, sulfuric acid and/or NaOH to adjust pH, purified water), in a sterile 2.5 mL ophthalmic dropper; *Tobradex*®, generic; (Rx)

Tobramycin and Dexamethasone Ophthalmic Suspension (containing tobramycin 0.3%, dexamethasone 0.05%, xanthan gum, tyloxapol, disodium edetate, NaCl, propylene glycol, sodium sulfate, benzalkonium chloride 0.01%, NaOH and/or HCl to adjust pH, purified water), in a sterile 5 mL ophthalmic dropper; *Tobradex*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Diclofenac Ophthalmic

(dye-**kloe**-fen-ak) Voltaren®

Nonsteroidal Anti-inflammatory Drug (NSAID)

For topical cream, refer to *Diclofenac, Topical*

Prescriber Highlights

- ▶ Used to control anterior segment inflammation
- ▶ May delay ocular wound healing and increase risk for intraocular bleeding
- ▶ Sulfite preservatives in diclofenac may trigger asthma attacks in susceptible patients.

Uses/Indications

Diclofenac ophthalmic is indicated for controlling anterior segment inflammation (eg, conjunctivitis, keratitis, anterior uveitis), inducing mydriasis during cataract-removal surgery, and for providing analgesia following surgical removal of cataracts. In dogs and cats, diclofenac significantly reduces clinical signs associated with experimentally induced intraocular inflammation^{1,2}; however, it has not demonstrated reliable reduction in corneal sensitivity in normal dogs.^{3,4}

Pharmacology/Actions

Diclofenac is an NSAID that blocks prostaglandin synthesis by inhibiting cyclooxygenase 1 and 2. Prostaglandins mediate many of the adverse effects of intraocular inflammation, including disruption of the blood-aqueous barrier, vasodilation, increased vascular permeability, leukocytosis, and increased intraocular pressure (IOP). Blocking prostaglandin production therefore minimizes the adverse short- and long-term effects of intraocular inflammation. Prostaglandins also appear to play a role in the miotic response produced during ocular surgery by constricting the iris sphincter independently of cholinergic mechanisms. Thus, diclofenac inhibits miosis induced during cataract surgery.⁴

Contraindications/Precautions/Warnings

Diclofenac is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. The potential for cross-sensitivity to acetylsalicylic acid, phenylacetic acid derivatives, and other NSAIDs exists.⁵ Topical NSAIDs may result in corneal epithelial breakdown and corneal thinning, erosion, or ulceration, which could cause delayed corneal wound healing; concurrent use of topical NSAIDs and topical corticosteroids may further delay healing and increase the potential for development of adverse effects.⁵ Due to interference with thrombocyte aggregation, ophthalmic NSAIDs may cause increased bleeding of ocular tissues (eg, hyphema) in conjunction with ocular surgery. It is recommended

that diclofenac ophthalmic solution be used with caution in patients with known bleeding tendencies (eg, severe liver disease) or in those receiving anticoagulant medications (eg, heparins, rivaroxaban), which may prolong bleeding time.⁵

Adverse Effects

Adverse effects reported in humans or animals include transient ocular burning or stinging on administration.⁵ Superficial punctate keratitis has been reported with long-term use. NSAIDs may also cause direct cellular damage, causing delayed healing and/or deterioration of existing corneal wounds. Although unlikely, systemic adverse effects may occur with the use of topical NSAIDs, including increased risk for gastric ulceration, decreased platelet aggregation, worsening of asthma and cardiac disease, and renal toxicity. Concurrent use of systemic NSAIDs and corticosteroids increases the risk for systemic adverse effects. The increased bleeding tendency of ocular tissues in conjunction with ocular surgery has also been reported. A comparison of the anti-inflammatory effects of topically administered diclofenac and prednisolone acetate in dogs with cataracts and well-regulated diabetes did not detect differences in measured parameters indicative of diabetes control (eg, fructosamine, body weight), although the authors concluded more research is warranted.⁶ Although diclofenac administration was not associated with an elevation in intraocular pressure (IOP) in dogs with normal eyes,⁷ it did result in increased IOP in dogs with experimentally induced uveitis.⁸

Dosages

Postoperative cataract surgery (extra-label): 1 drop to the affected eye(s) every 6 hours throughout the first 2 weeks of the postoperative period

Anterior segment inflammation (extra-label): 1 drop to the affected eye(s) every 6 to 12 hours

Monitoring

- Clinical efficacy (eg, decreased or absent anterior segment inflammation)
- Adverse effects (eg, delayed wound healing)

Client Information

- Use this medicine as directed by your veterinarian.
- Do not use this medicine if the color changes, if the solution becomes cloudy, or if there are particles in the solution.
- Diclofenac ophthalmic may cause burning or stinging on administration. It may also increase the risk for bleeding and may slow down healing in the eye.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed on the medicine when it is not in use.
- Wait for 5 minutes after applying this medicine before applying any other medications to the eye.
- Store this medicine at room temperature away from moisture and sunlight. Do not freeze.
- Your veterinarian will need to monitor your animal while it is taking this medicine. Do not miss these important follow-up visits.

Storage/Stability

Store this product at controlled room temperature (68°F to 77°F [20°C–25°C]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Diclofenac ophthalmic solutions should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Diclofenac Sodium 0.1% Ophthalmic Solution (containing 1 mg diclofenac sodium; polyoxyl 35 castor oil; boric acid; tromethamine; sorbic acid (2 mg/mL); edetate disodium (1 mg/mL); purified water), in a 5 mL ophthalmic dropper bottle; *Voltaren*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Difluprednate Ophthalmic

(dye-floo-*pred*-nate) *Durezol*®

Corticosteroid

Prescriber Highlights

- ▶ Used for treatment of anterior segment inflammatory conditions
- ▶ Exerts clinical effect equal to that of prednisolone acetate
- ▶ Do not use in patients with an existing or suspected bacterial, viral, or fungal ocular infection or in the presence of corneal ulceration

Uses/Indications

Difluprednate ophthalmic can be used to treat inflammation following ocular injury or cataract surgery or to treat generalized inflammatory conditions of the anterior segment (eg, conjunctivitis, keratitis, anterior uveitis). It has been shown to exert comparable efficacy to prednisolone acetate in humans¹ and in dogs with experimentally induced anterior chamber inflammation.²

Pharmacology/Actions

Corticosteroids have broad immunomodulatory and anti-inflammatory actions, which are mediated via direct effects on gene expression, as well as by nongenomic effects on cell membranes and receptors. Ultimately, corticosteroids decrease proinflammatory gene transcription, production of proinflammatory prostaglandins and leukotrienes, B- and T-cell number and function, cell-mediated immunity, and humoral immunity.³ Specific mechanisms involved in regulating ocular inflammation are not known.

Contraindications/Precautions/Warnings

Difluprednate is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Due to inhibition of epithelial wound healing and potentiation of infection, difluprednate ophthalmic is contraindicated when corneal ulceration is suspected or present or in the presence of corneal infection (bacterial, fungal, or viral),⁴ infectious diseases, chronic renal failure, congestive heart failure, systemic hypertension, or gastric ulceration. When use of difluprednate is necessary in these conditions, the minimal effective dose should be administered for the shortest possible time.

Adverse Effects

Adverse effects reported in humans include development of cataracts (via altered glucose or lenticular metabolic effects), increased intraocular pressure (IOP; via activation of glucocorticoid receptors in the trabecular meshwork), infection (via immunosuppression), decreased wound healing (via increased collagenolytic activity), mydriasis, and calcific keratopathy. Increased IOP may occur in some animals with long-term use of difluprednate; however, most reports suggest the change is reversible.^{5,6} Systemic effects are rare but include hepatopathy, suppression of endogenous glucocorticoid production, and focal alopecia. Concurrent use of systemic steroids increases the risk for these adverse effects.

Dosages

Conjunctivitis (extra-label): 1 drop in affected eye(s) every 4 to 8 hours, then gradually decrease the dose over days to weeks, provided

clinical improvement is maintained

Keratitis (extra-label): 1 drop in affected eye(s) every 4 to 8 hours, then gradually decrease the dose over days to weeks, provided clinical improvement is maintained

Anterior uveitis (extra-label): 1 drop in affected eye(s) 4 times daily for 14 days, then gradually decrease the dose over days to weeks, provided clinical improvement is maintained

Post-cataract removal (extra-label): 1 drop in the surgical eye 4 times daily, beginning 24 hours after surgery and continuing 2 weeks postoperative, then decrease to 2 times daily for 1 week, then continue to taper dosage based on clinical response

Monitoring

- Clinical efficacy
- Resolution or development of signs of inflammation
- Intraocular pressure (IOP; especially in cats) and ocular surface integrity (eg, fluorescein stain) as needed based on clinical presentation
- Blood or urine glucose changes based on development of clinical signs (eg, polyuria, polydipsia) in diabetic animals

Client Information

- This medicine can be given for various lengths of time. Be sure you understand how often and for how long your animal should be given this medicine.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Contact your veterinarian as soon as possible if your animal's eye condition worsens or does not improve while using this medicine.
- Your veterinarian will need to monitor your animal's eye condition while taking this medicine. Do not miss these important follow-up visits.
- Store this medicine as directed on the product label. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are visible particles in the solution.

Storage/Stability

Store this product at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Difluprednate ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Difluprednate 0.05% Ophthalmic Emulsion (containing difluprednate 0.05%; sorbic acid 0.1%; boric acid; castor oil; glycerin; polysorbate 80; water for injection; sodium acetate; sodium EDTA; NaOH to adjust the pH to 5.2 to 5.8, with tonicity of 304 – 411 mOsm/kg), in 5 mL ophthalmic dropper bottle; *Durezol*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Dorzolamide Ophthalmic

(dor-zoe-la-mide) *Trusopt*®, *Cosopt*®, *Cosopt*® PF
Carbonic Anhydrase Inhibitor Anti-Glaucoma Agent

Prescriber Highlights

- ▶ Decreases intraocular pressure (IOP) by reducing the production of aqueous humor
- ▶ Three-times-daily dosing recommended over twice-daily dosing
- ▶ More stinging on application compared with brinzolamide, due to more acidic pH of dorzolamide
- ▶ Sulfonamide derivative; caregivers with sulfa allergies should avoid direct contact with the medicine.
- ▶ Available as a single agent or in combination with timolol maleate

Uses/Indications

Dorzolamide ophthalmic (alone or in combination with timolol) is used to decrease intraocular pressure (IOP) in patients with glaucoma and ocular hypertension. In dogs with and without glaucoma, dorzolamide leads to a significant IOP reduction, which is more consistent when it is dosed 3 times daily versus 2 times daily.^{1,2} When dorzolamide ophthalmic is administered in combination with a beta-blocker, the IOP-lowering effects are additive.^{3,4} Reduced IOP also occurs in normal cats and horses,⁵ and similar to dogs, improved efficacy may result when dorzolamide is administered in combination with timolol in horses.^{6,7} Efficacy of prophylactic dorzolamide administration for preventing manifestation of primary glaucoma in unaffected eyes of dogs with contralateral primary glaucoma has not been demonstrated.⁸

Pharmacology/Actions

Carbonic anhydrase (CA) is an enzyme found in many body tissues, including the eye. CA inhibitors reduce the formation of hydrogen and bicarbonate ions from carbon dioxide and water by noncompetitive, reversible inhibition of the enzyme CA, thereby reducing the availability of these ions for active transport into secretions.⁹ In humans, CA exists as multiple isoenzymes, with CA-II being the most relevant to glaucoma management. Inhibition of CA in the ciliary processes of the eye decreases aqueous humor secretion, presumably by slowing the formation of bicarbonate ions with subsequent reduction in sodium and fluid transport, which results in a decrease in intraocular pressure (IOP).⁹ Dorzolamide equilibrates between acidic and basic forms, allowing for both lipid and water solubility and greater corneal and scleral penetration.

Contraindications/Precautions/Warnings

Dorzolamide is contraindicated in patients that have a history of hypersensitivity to it or sulfonamides or any component of the formulation.

Adverse Effects

Adverse effects reported in humans or animals include burning and stinging upon application (due to low pH), superficial punctate keratitis, local hypersensitivity reactions, and corneal edema. Systemic adverse effects are not usually observed with the topical application; however, renal tubular acidosis and hypokalemia have been reported in a cat due to topical carbonic anhydrase inhibitor (CAI) administration.¹⁰ CAIs, including dorzolamide, have been implicated in the development of significant ulcerative and nonulcerative keratitis following long-term administration.¹¹

Dosages

Glaucoma (extra-label): 1 drop in the affected eye(s) or 0.1 mL in the subpalpebral lavage (SPL) catheter every 8 to 12 hours

Monitoring

- Clinical efficacy
- Confirm maintenance of intraocular pressure (IOP) within the normal range for the species being treated
- Monitor for corneal edema

Client Information

- Use this medication as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes.

Storage/Stability

Store this product at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight.

Compatibility/Compounding Considerations

Dorzolamide ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Dorzolamide HCl 2% Ophthalmic Solution (containing dorzolamide HCl 22.3 mg equivalent to 20 mg dorzolamide; benzalkonium chloride 0.0075%; hydroxyethyl cellulose; mannitol; sodium citrate dehydrate; NaOH to adjust pH; water for injection), in 5 mL and 10 mL ophthalmic dropper bottles; *Trusopt*®, generic; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Dorzolamide HCl 2% and Timolol maleate 0.5% Ophthalmic Solution (containing dorzolamide HCl equivalent to 20 mg dorzolamide; timolol maleate 6.8 mg equivalent to 5 mg timolol; benzalkonium chloride 0.0075%; sodium citrate; hydroxyethyl cellulose; NaOH; mannitol; water for injection, with a pH of 5.65 and an osmolality of 242 – 323 mOsm/kg), in sterile 10 mL ophthalmic dispenser bottles. *Cosopt*®, generic; (Rx)

Dorzolamide HCl 2% and Timolol maleate 0.5% Ophthalmic Solution, preservative-free (containing 22.3 mg dorzolamide HCl equivalent to 20 mg dorzolamide; 6.8 mg timolol maleate equivalent to 5 mg timolol; hydroxyethyl cellulose; mannitol; sodium citrate; NaOH to adjust pH; water for injection), in 15 sterile single-use containers, 0.2 mL per container; *Cosopt*® PF; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Edetate Disodium Ophthalmic Sodium EDTA

(ed-uh-tayt dye-so-dee-um)

Chelating Agent

Prescriber Highlights

- ▶ Used for dissolution of corneal calcium deposits
- ▶ Also used to decrease corneal digestion due to collagenase activity in infected ulcers
- ▶ Do not mix with calcium-containing drugs
- ▶ No longer commercially available; must be compounded

Uses/Indications

Edetate disodium is used topically to remove calcific corneal deposits that occur in the superficial cornea, which may cause pain or impair vision. Corneal calcium deposits may form in association with chronic keratitis, chronic uveitis, or hypercalcemia. These deposits may be dissolved from the conjunctiva, corneal epithelium, and anterior stroma following EDTA administration. Calcium deposits in the deep stroma are not affected, as edetate disodium is lipid-insoluble and does not penetrate the corneal epithelium. Topical administration of EDTA also decreases collagenase activity in infected corneal ulcers, thus helping to control corneal melting.^{1,2} Edetate disodium has been used as an emergency treatment to decontaminate the eye after injury by zinc chloride, and for emergency management and follow-up treatment of calcium hydroxide burns in the eye.

Pharmacology/Actions

Edetate disodium forms a soluble complex with calcium ions and other divalent and trivalent metals, which leads to the dissolution of calcium deposits from the ocular surface. As calcium (and other) ions serve as cofactors for enzymes responsible for collagenolysis, binding the ions makes them unavailable to the enzymes, thus decreasing enzymatic activity.¹

Contraindications/Precautions/Warnings

Edetate disodium ophthalmic should not be used in patients that are hypersensitive to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans or animals include swelling of the stroma (usually following application of concentrations greater than 1%), stinging on application, and swelling of the eyelids. Toxic effects have been demonstrated in vitro to corneal epithelial cells.³

Dosages

Calcium chelation for corneal deposits and/or collagenolysis (extra-label): 1 drop of the solution (0.35% to 1.85%, but most commonly 1%) to the affected eye(s) initially 4 to 6 times daily and then 3 to 4 times daily as indicated

Zinc chloride injury (extra-label): Topical, to the cornea, as a 1.7% (0.046 M) solution as irrigation for 15 minutes; more likely to be effective if started within 2 minutes of exposure

Monitoring

- Clinical efficacy
- Monitor mineral deposits in the cornea
- Monitor corneal digestion (melting)

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medicine as directed on the product label. Do not freeze.
- Do not use this product if the color changes, if it becomes cloudy, or if there are particles in the solution.
- Discard this product by the date indicated on the label.

Storage/Stability

Store this product as directed on the product labeling; do not freeze.

Compatibility/Compounding Considerations

Edetate disodium ophthalmic preparations should not be mixed directly with any other ophthalmic medications, especially those containing calcium.

Dosage Forms/Regulatory Status**VETERINARY-LABELED PRODUCTS:** NONE**HUMAN-LABELED PRODUCTS:** NONE

Edetate disodium ophthalmic solution must be prepared (compounded) by a qualified compounding pharmacist. It is usually prepared as a 1% solution in an artificial tear solution not containing calcium.

References

For the complete list of references, see wiley.com/go/budde/plumb

Epinastine Ophthalmic(ep-i-*nas*-teen) *Elestat*®

Antihistamine

Prescriber Highlights

- ▶ H₁-selective antihistamine with H₂-blocking and mast cell stabilizing activity
- ▶ Used to relieve signs of allergic conjunctivitis (particularly pruritus)
- ▶ Duration of action allows for twice-daily dosing

Uses/Indications

Epinastine is used to treat and prevent pruritus associated with allergic conjunctivitis.

Pharmacology/Actions

Epinastine is a topically active, direct histamine-1 (H₁) receptor antagonist and an inhibitor of histamine release from the mast cell. Epinastine is selective for the H₁ receptor and has an affinity for the histamine-2 (H₂) receptor. Epinastine also possesses an affinity for the alpha-1, alpha-2, and 5-HT₂ receptors.¹ No evaluation of its use as a topical ophthalmic agent has been performed in veterinary species.

Contraindications/Precautions/Warnings

Epinastine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans include burning sensation in the eye, folliculitis, hyperemia, pruritus, and epiphora.¹ The most frequently reported nonocular adverse reactions include cold symptoms and upper respiratory infections, headache, rhinitis, sinusitis, increased cough, and pharyngitis. Some of these adverse effects may be related to the underlying disease being treated.

Dosages

Allergic conjunctivitis (extra-label): 1 drop in the affected eye(s) every 12 hours

Monitoring

- Clinical efficacy: resolution of signs related to allergic conjunctivitis (eg, pruritus, tearing, squinting)

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in solution.

Storage/Stability

Store at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Epinastine ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status**VETERINARY-LABELED PRODUCTS:** NONE**HUMAN-LABELED PRODUCTS:**

Epinastine HCl 0.05% Ophthalmic Solution (containing epinastine 0.05%; benzalkonium chloride 0.01%; edetate disodium; purified water; NaCl; sodium phosphate, monobasic; NaOH and/or HCl to adjust pH to 7; osmolality range of 250 – 310 mOsm/kg), in a 5 mL sterile ophthalmic dropper bottle; *Elestat*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Erythromycin Ophthalmic(eh-*rith*-rowe-*mye*-sin)

Macrolide Antibiotic

For systemic use, refer to *Erythromycin***Prescriber Highlights**

- ▶ Used to treat susceptible ocular surface infections (eg, *Mycoplasma* spp and *Chlamydia* spp in cats)
- ▶ Only available as an ophthalmic ointment
- ▶ Used up to 6 times daily
- ▶ Use with caution in pocket pets that may ingest the drug via grooming.

Uses/Indications

Erythromycin ophthalmic ointment is used for the treatment of superficial ocular infections involving the conjunctiva and/or cornea caused by organisms that are susceptible to erythromycin. See **Pharmacology/Actions**. Administration of topical erythromycin to normal cat eyes did not induce significant changes in normal ocular surface microflora.¹

The World Health Organization (WHO) has designated erythromycin as a Critically Important, Highest Priority antimicrobial for human medicine.² The Office International des Epizooties (OIE) has designated macrolides as Veterinary Critically Important Antimicrobial (VCIA) Agents in avian, bovine, caprine, equine, lagomorphs, ovine, and swine species.³

Pharmacology/Actions

Erythromycin inhibits protein synthesis by binding the 50S subunit of the bacterial ribosome, thus inhibiting the formation of peptide bonds between amino acids by a variety of specific mechanisms. Erythromycin is usually active against *Streptococcus* spp, *Staphylococcus* spp, *Mycoplasma* spp, and *Chlamydia* spp.⁴

Erythromycin is considered ineffective against *Pseudomonas aeruginosa*.⁵

Contraindications/Precautions/Warnings

Erythromycin is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Topical macrolides should be used carefully, if at all, in pocket pets (eg, hamsters, chinchillas) and rabbits, as these animals may ingest macrolides while grooming, which increases the potential for GI adverse effects.

Adverse Effects

Adverse effects reported in humans or animals include mild ocular stinging, burning, irritation, and redness. Pocket pets ingesting macrolides may suffer from acute and potentially fatal diarrhea.

Dosages

Susceptible ocular infections (extra-label): ¼ – ½ inch strip in the affected eye(s) up to 6 times daily depending on the severity of infection

Monitoring

- Corneal cytology and culture to evaluate for the presence of bacteria
- Clinical efficacy: improvement and resolution of clinical signs associated with bacterial infection
- GI clinical signs; discontinue administration if diarrhea occurs following use on pocket pets.

Client Information

- Use this medicine as directed by your veterinarian
- Do not give this medicine to pocket pets (eg, hamsters, chinchillas) or rabbits without frequent examinations by your veterinarian.
- To prevent contamination, do not touch the dropper tip or allow it to touch your animal's eye or any other surface. Keep the cap tightly closed on the medication when it is not in use.
- If you are administering more than 1 eye medication to your animal, wait 5 minutes between each medication before giving the next. Use eye drops before eye ointments to allow the drops to absorb into the eye.
- Your veterinarian will need to monitor your animal while it is receiving this medicine. Do not miss these important follow-up visits.
- Store this medicine at room temperature; do not freeze. Do not use the medicine if the color or consistency changes.

Storage/Stability

Store erythromycin ophthalmic formulations at controlled room temperature (20°C–25°C [68°F–77°F]); do not freeze.

Compatibility/Compounding Considerations

Erythromycin should not be mixed directly (in vitro) with any other drugs. Erythromycin is not available as an ophthalmic solution.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Erythromycin 0.5% Ophthalmic Ointment (containing erythromycin 5 mg/g in a sterile ophthalmic base of mineral oil and white petrolatum), in a 3.5 g ointment tube; generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Uses/Indications

Topical application of fluorescein solutions to the ocular surface enables identification of corneal ulcers, evaluation of tear film breakup time (an indicator of tear film quality), and determination of patency of the nasolacrimal system (Jones test). Manufactured solutions of fluorescein, with or without topical anesthetics, are available for use, whereas dilution of individually wrapped fluorescein strips with eyewash is an alternative method that also decreases the risk for contamination between patients. Visualization of fluorescein retention by the cornea or conjunctiva, with either white or cobalt blue light, indicates epithelial loss (ulceration). The rapid appearance of breaks in fluorescein within the tear film suggests tear film quality disturbances. The appearance of fluorescein at the nostril following application to the ocular surface indicates functional patency of the nasolacrimal duct system (positive Jones test). IV injection of fluorescein solutions is used to evaluate blood flow patterns within the eye in the diagnostic procedure fluorescein angiography.

Pharmacology/Actions

Fluorescein is a water-soluble dye that is not retained by normal cornea because it does not pass through the hydrophobic epithelium. If the epithelium is absent, fluorescein penetrates the hydrophilic corneal stroma and stains it bright yellow-green. The green color is best visualized utilizing blue light from a cobalt filter attached to a penlight transilluminator, or a Wood's lamp. It may also be visible with ambient white light. Defects in the corneal stroma that extend to the deepest layer (Descemet membrane) will not retain fluorescein in the deepest aspect, as Descemet membrane is hydrophobic. In fluorescein angiography, vessels in the active areas of inflammation and neovascularization demonstrate leakage of fluorescein associated with increased vascular permeability. Fluorescein fluoresces in light with a wavelength between 485 and 500 nm, with maximal emission between 520 and 530 nm.

Contraindications/Precautions/Warnings

Fluorescein is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Fluorescein should not be used during intraocular surgery.

Adverse Effects

Adverse effects reported in humans or animals following the application of the strips include local hypersensitivity reactions and transient discomfort. Temporary staining of the skin and fur may result after use. Adverse effects following IV administration can be severe and include nausea, vomiting, GI distress, headache, syncope, hypotension, and signs of hypersensitivity. In humans, cardiac arrest, basilar artery ischemia, severe shock, convulsions, thrombophlebitis at the injection site, and, in rare cases, death have been reported. Extravasation of the solution at the injection site causes intense pain at the site and dull aching pain in the injected limb.¹

Dosages

Determination of corneal injury: Wet fluorescein strip with 1 to 2 drops of normal saline solution. Apply moistened tip to conjunctiva or fornix. Allow patient to blink several times after application. Wait 60 seconds, rinse with sterile irrigating solution, and examine with the use of a light source; a cobalt filter may be attached to the light surface for better visualization. Defects in the corneal epithelium illuminate the stroma bright green. In deep corneal lesions, the lesion's center may fail to take up the stain and appears black, indicating that a descemetocoele is present.

Tear film breakup time: Wet fluorescein strip with 1 to 2 drops of normal saline solution. Apply moistened tip to conjunctiva or fornix. Allow patient to blink once, and then hold lids open and monitor tear film for breaks, which appear as dark lines within the green.

Fluorescein Stain Ophthalmic

(*flur*-e-seen)

Ocular Diagnostic Agent

Prescriber Highlights

- ▶ Used clinically to delineate full-thickness loss of corneal epithelium (corneal ulcer), to evaluate tear film breakup time, and to determine patency of nasolacrimal outflow system (Jones test)
- ▶ Yellow-green dye binds to corneal stroma but not to the corneal epithelium or Descemet membrane
- ▶ Available in combination with topical anesthetic (oxybuprocaine) to minimize discomfort on application and to facilitate additional diagnostic tests
- ▶ IV injection of its approved formulations is used diagnostically during fluorescein angiography.
- ▶ Fluorescein should **not** be used during intraocular surgery.

Normal values are greater than 20 seconds in dogs, with shorter times indicating qualitative tear film deficiency.

Nasolacrimal outflow patency (Jones test): Wet fluorescein strip with 1 to 2 drops of normal saline solution. Apply moistened tip to conjunctiva or fornix. It is recommended that the patient's nose be tipped downward to facilitate drainage of fluorescein-laden tears through the nasolacrimal system, at which point fluorescein will be visible at the nares; a cobalt filter may be attached to a light source for better visualization. Normal times are 2 to 5 minutes in dogs and up to 10 minutes in cats.

Fluorescein angiography: Rapid IV injection (≈ 1 mL/second) of 2 to 5 mL of fluorescein 100 mg/mL in the antecubital vein is followed by ocular vasculature visualization, using a cobalt filter light source and appropriate magnification. Time to filling and pattern of filling of vasculature within the anterior and posterior segments are noted and correlated with normal versus disease states. (This procedure is performed by board-certified veterinary ophthalmologists.)

Monitoring

- Diagnostic product; no monitoring is required

Client Information

- Do not use fluorescein strips at home.

Storage/Stability

Store this product at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Fluorescein ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Although IV fluorescein solutions have been approved by the US FDA, no topical ophthalmic fluorescein products have been approved.

Fluorescein Sodium 10% Injection (containing fluorescein 100 mg/mL and sterile water for injection; NaOH and/or HCl to adjust pH), in 5 mL single-use glass vials; *Fluorescite*®; (Rx)

Fluorescein Sodium 10% Injection (containing fluorescein 100 mg/mL and sterile water for injection; NaOH and/or HCl to adjust pH), in 5 mL single-dose glass vials; *AK-FLUOR 10%*®; (Rx)

Fluorescein Sodium 25% Injection (containing fluorescein 250 mg/mL and sterile water for injection; NaOH and/or HCl to adjust pH), in 2 mL single-dose glass vials; *AK-FLUOR 25%*®; (Rx)

Fluorescein Sodium Ophthalmic Strips USP 1 mg per strip; individually wrapped strips in cartons of 50, 100, or 300; *Dry Eye Test*®, *Ful-Glo*®, *Bioglo*®, *Glostrips*®; (Rx)

COMBINATION PRODUCTS APPROVED FOR USE IN HUMANS:

Fluorescein Sodium and Benoxinate Hydrochloride Ophthalmic Solution 0.25%/0.4% (containing fluorescein sodium 2.5 mg; benoxinate hydrochloride 4 mg; povidone; hydrochloric acid; boric acid; sodium hydroxide; water for injection; hydrochloric acid and/or sodium hydroxide to adjust pH to 4.3 to 5.3), in sterile 5 mL ophthalmic dropper bottle; *Altafluor Benox*®, generic; (Rx)

Fluorescein Sodium and Benoxinate Hydrochloride Ophthalmic Solution 0.25%/0.4% (containing fluorescein sodium 2.5 mg; benoxinate hydrochloride 4 mg; chlorobutanol 10 mg; boric acid; povidone; sodium hydroxide and/or hydrochloric acid may be added to adjust pH to 4.3 to 5.3; purified water), in sterile 5 mL ophthalmic dropper bottle; *Fluress*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Fluorometholone Ophthalmic

(flew-roe-**meth**-oh-lone) Flarex®, FML®, FML Forte®

Corticosteroid

Prescriber Highlights

- Used to treat anterior segment inflammatory conditions
- Contraindications include the presence of corneal ulceration and existing or suspected bacterial, viral, or fungal ocular infection.
- Use with caution in avian species, as systemic adrenal suppression may result.
- Shake suspension well before using.

Uses/Indications

Fluorometholone ophthalmic can be used to treat inflammation following ocular injury or cataract surgery or to treat generalized inflammatory conditions of the anterior segment (eg, conjunctivitis, keratitis, anterior uveitis). Evaluation of relative efficacy of fluorometholone ophthalmic in veterinary patients has not been performed.

Pharmacology/Actions

Corticosteroids have broad immunomodulatory and anti-inflammatory actions, mediated via direct effects on gene expression, as well as by nongenomic effects on cell membranes and receptors. Ultimately, corticosteroids decrease pro-inflammatory gene transcription, production of pro-inflammatory prostaglandins and leukotrienes, B- and T-cell number and function, cell-mediated immunity, and humoral immunity.¹ The specific mechanisms involved in regulating ocular inflammation are not known. Fluorometholone alcohol preparations are most effective for ocular surface inflammation, rather than intraocular inflammation, due to comparatively poor transcorneal penetration relative to the acetate preparations.

Contraindications/Precautions/Warnings

Fluorometholone is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Ocular corticosteroids should not be used in animals with bacterial, fungal, or viral ocular infections, or if corneal ulceration is present, as they may potentiate infection and decrease epithelial wound healing.

Adverse Effects

Adverse effects reported in humans include development of cataracts (via altered glucose or lenticular metabolic effects), increased intraocular pressure (IOP; via activation of glucocorticoid receptors in the trabecular meshwork), infection (via immunosuppression), decreased wound healing (via increased collagenolytic activity), mydriasis, and calcific keratopathy. Some animals may develop increased IOP with chronic use; however, most reports suggest the change is reversible.^{2,3} Additionally, fluorometholone is associated with less risk of IOP elevation compared with other topical corticosteroids. Systemic effects are rare but include hepatopathy, suppression of endogenous glucocorticoid production, and focal alopecia. Concurrent use of systemic steroids increases the risk for these adverse effects.

Dosages

Conjunctivitis (extra-label): 1 drop or ¼ inch strip in the affected eye(s) as frequently as every 4 hours until clinical improvement occurs, then gradually (over days to weeks) decrease the dosage, provided clinical improvement is maintained

Keratitis (extra-label): 1 drop or ¼ inch strip in the affected eye(s) as frequently as every 4 hours until clinical improvement occurs, then

gradually (over days to weeks) decrease the dosage, provided clinical improvement is maintained

Anterior uveitis (extra-label): 1 drop or ¼ inch strip in the affected eye(s) as frequently as every 4 hours until clinical improvement occurs, then gradually (over days to weeks) decrease the dosage, provided clinical improvement is maintained

Monitoring

- Clinical efficacy through resolution of signs of inflammation
- Monitor intraocular pressure (IOP) and ocular surface integrity
- Blood or urine glucose changes based on development of clinical signs (eg, polyuria, polydipsia) in diabetic animals

Client Information

- Use this medicine as directed by your veterinarian.
- It is important to shake the suspension well before giving this medicine to your animal.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 to 10 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes or if there are particles in the suspension.

Storage/Stability

Store this product in the refrigerator or at controlled room temperature (2°C–25°C [36°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Fluorometholone ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Fluorometholone 0.1% Ophthalmic Suspension (containing fluorometholone 1 mg/mL; benzalkonium chloride 0.01%; sodium chloride; monobasic sodium phosphate; edetate disodium; hydroxyethyl cellulose; tyloxapol; HCl and/or NaOH to adjust pH; purified water), 5 mL or 10 mL in sterile ophthalmic dropper bottles; *FML*®, *Flarex*®; (Rx)

Fluorometholone 0.25% Ophthalmic Suspension (containing fluorometholone 2.5 mg/mL; benzalkonium chloride 0.005%; edetate disodium; polysorbate 80; polyvinyl alcohol 1.4%; purified water; NaCl; sodium phosphate, dibasic; sodium phosphate, monobasic; NaOH to adjust the pH from 6.2 to 7.5), 5 mL or 10 mL in sterile ophthalmic dropper bottles; *FML Forte*®; (Rx)

Fluorometholone 0.1% Ophthalmic Ointment (containing fluorometholone 1 mg/mL; phenylmercuric acetate (0.0008%); mineral oil; petrolatum (and) lanolin alcohol; white petrolatum), in 3.5 g ophthalmic ointment tubes; *FML*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Fluorouracil Ophthalmic

(flew-oh-**your**-uh-sil)

Antineoplastic Agent

For systemic use, see *Fluorouracil*

Prescriber Highlights

- ▶ Antifibrotic agent used adjunctively in periocular surgeries (most commonly in glaucoma implant procedures)
- ▶ Antineoplastic used for susceptible cutaneous tumors
- ▶ **Do NOT use in cats** or in patients with myelosuppression or serious infections.
- ▶ Adverse effects include dose-dependent myelosuppression, GI toxicity, and neurotoxicity.
- ▶ Known teratogen
- ▶ The National Institute for Occupational Safety and Health (NIOSH) classifies fluorouracil as a hazardous drug; use appropriate precautions when handling.

Uses/Indications

Fluorouracil ophthalmic is used as an adjunct in ocular surgeries, particularly in those that treat glaucoma through placement of drainage shunts or alteration of the natural aqueous humor outflow tracts, because of fluorouracil's ability to reduce fibroblastic proliferation and prevent subsequent scarring.¹ Fluorouracil ophthalmic is also used to treat ocular surface neoplastic conditions, such as squamous cell carcinoma and sarcoids.

Pharmacology/Actions

Fluorouracil is a pyrimidine analogue that is converted through intracellular mechanisms into multiple active metabolites which interfere with DNA and RNA synthesis, inhibit cell growth, and promote cell death.²

Contraindications/Precautions/Warnings

Fluorouracil is contraindicated in patients that are hypersensitive to it or any component of the formulation. Fluorouracil is absolutely contraindicated in cats by any route of administration, including ophthalmic administration, as cats develop a severe, potentially fatal neurotoxicity when they are exposed to fluorouracil by any route. Do not use fluorouracil in patients that are pregnant or in those that are myelosuppressed or have concurrent serious infections. Although systemic adverse effects are less likely after topical application of fluorouracil in other species, they can occur.

The National Institute for Occupational Safety and Health (NIOSH) classifies fluorouracil as a hazardous drug; personal protective equipment (PPE) should be used accordingly to minimize the risk for exposure.

Adverse Effects

Although less likely after topical administration, adverse effects reported in humans or animals include myelosuppression, GI toxicity (eg, diarrhea, GI ulceration, sloughing, stomatitis), and neurotoxicity (seizures). The most frequent adverse reactions with ophthalmic fluorouracil administration occur locally and are often related to an extension of the pharmacological activity of the drug.³ These effects include burning, crusting, allergic contact dermatitis, erosions, erythema, hyperpigmentation, irritation, pain, periorbital alopecia, photosensitivity, pruritus, scarring, rash, soreness, and ulceration.

Dosages

NOTE: Multiple protocols exist for use during surgical procedures for correction of glaucoma; these protocols vary with the type of procedure and other patient parameters. As these procedures should be performed by board-certified veterinary ophthalmologists, the protocols are not detailed here.

Ocular surface neoplasia (extra-label): Fluorouracil 1% ophthalmic solution (compounded), 1 drop in affected eye(s) 4 times daily for 1 week; may be repeated in regular cycles.^{4,5} Use of a *topical* 1% fluorouracil ointment applied 4 times daily also has been described.^{6,7}

Monitoring

- Clinical efficacy
- Maintenance of patency of the glaucoma implant, which leads to control of intraocular pressure
- Cell or tissue samples for evaluation to confirm remission of neoplasia

Client Information

- This is a chemotherapy (anticancer) drug. The drug and its by-products can be hazardous to other animals and humans. Pregnant women and sick or immunocompromised individuals should avoid contact.
- Wear protective gloves and a mask when administering this medicine. After administration, seal these gloves, mask, and other waste products in a plastic bag, and then place in the regular trash. Wash your hands immediately after each application.
- Loss of hair around the eyes may occur after use.
- Wait for 5 to 10 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine according to instructions on the bottle or prescription label.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solutions.

Storage/Stability

Store this product as directed on the product label.

Compatibility/Compounding Considerations

Fluorouracil ophthalmic preparations should not be mixed directly with any other ophthalmic medications. Fluorouracil ophthalmic preparations are not commercially available but may be prepared by qualified compounding pharmacists by utilizing the 50 mg/mL injectable solution. Fluorouracil 1% ophthalmic solution prepared this way is reported to be stable for 3 weeks.⁸

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Fluorouracil ophthalmic formulations must be prepared (compounded) by a qualified compounding pharmacist. See *Compounding: How to Find a Quality Assured Pharmacy*.

References

For the complete list of references, see wiley.com/go/budde/plumb

Flurbiprofen Ophthalmic

(flur-bi-*proe*-fin) Ocufer®

Nonsteroidal Anti-inflammatory Drug (NSAID)

Prescriber Highlights

- ▶ Used to control anterior segment inflammation
- ▶ May cause transient increases in intraocular pressure (IOP)
- ▶ May delay ocular wound healing and increase risk for intraocular bleeding

Uses/Indications

Flurbiprofen ophthalmic is indicated for controlling anterior segment inflammation (eg, conjunctivitis, keratitis, anterior uveitis), inducing mydriasis during cataract-removal surgery, and for providing analgesia following surgical removal of cataracts. In dogs, flurbiprofen signifi-

cantly reduced clinical signs associated with experimentally induced intraocular inflammation,¹ whereas it demonstrated poor efficacy in cats.² Flurbiprofen did not decrease corneal sensitivity in normal dogs, suggesting its use as an ophthalmic analgesic may be limited³; however, analgesic effects may improve in the presence of ocular pain.

Pharmacology/Actions

Flurbiprofen is an NSAID that blocks prostaglandin synthesis by inhibiting cyclooxygenase 1 and 2. Prostaglandins mediate many of the adverse effects of intraocular inflammation, including disruption of the blood–aqueous barrier, vasodilation, increased vascular permeability, leukocytosis, and increased intraocular pressure (IOP). Blocking prostaglandin production therefore minimizes the adverse short- and long-term effects of intraocular inflammation. Prostaglandins also appear to play a role in the miotic response produced during ocular surgery by constricting the iris sphincter independently of cholinergic mechanisms. Thus, flurbiprofen inhibits miosis induced during cataract surgery.⁴

Contraindications/Precautions/Warnings

Flurbiprofen is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. The potential for cross-sensitivity to acetylsalicylic acid, phenylacetic acid derivatives, and other NSAIDs exists.⁴ Topical NSAIDs may result in corneal epithelial breakdown and corneal thinning, erosion, or ulceration, which could cause delayed corneal wound healing; concurrent use of topical NSAIDs and topical corticosteroids may further delay healing and increase the potential for development of adverse effects.⁴ Due to interference with thrombocyte aggregation, ophthalmic NSAIDs may cause increased bleeding of ocular tissues (eg, hyphema) in conjunction with ocular surgery. It is recommended that flurbiprofen ophthalmic solution be used with caution in patients with known bleeding tendencies (eg, severe liver disease) or in those receiving anticoagulant medications (eg, heparins, rivaroxaban), which may prolong bleeding time.⁴

Adverse Effects

Adverse effects reported in humans or animals include transient ocular burning or stinging on administration.⁴ Additional adverse reactions reported with the use of flurbiprofen ophthalmic include fibrosis, miosis, and mydriasis. NSAIDs may also cause direct cellular damage, causing delayed healing and/or deterioration of existing corneal wounds. Although unlikely, systemic adverse effects may occur with the use of topical NSAIDs, including increased risk for gastric ulceration, decreased platelet aggregation, worsening of asthma and cardiac disease, and renal toxicity. Concurrent use of systemic NSAIDs and corticosteroids increases the risk for systemic adverse effects. The increased bleeding tendency of ocular tissues in conjunction with ocular surgery has also been reported. Flurbiprofen has been documented to elevate intraocular pressure (IOP) in dogs,^{5,6} and decrease the IOP-lowering effect of latanoprost in normal canine eyes⁶ and canine eyes following surgical cataract removal.⁷ Flurbiprofen increased corneal sensitivity when it was administered to normal canine eyes; therefore, it is suspected to be associated with an ocular irritant effect.⁸

Dosages

Preoperative for cataract surgery (extra-label): 1 drop every 30 minutes for 4 applications beginning 2 hours prior to surgery

Anterior segment inflammation (extra-label): 1 drop in affected eye(s) every 8 hours; frequency may be reduced to once daily after resolution of initial inflammation

Monitoring

- Clinical efficacy (eg, decreased or absent anterior segment inflammation)
- Adverse effects (eg, delayed wound healing)

Client Information

- Use this medicine as directed by your veterinarian.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solution.
- Flurbiprofen ophthalmic may cause burning or stinging on administration. It may also increase the risk for bleeding and may slow down healing in the eye.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed on the medicine when it is not in use.
- Wait for 5 minutes after applying this medicine is before applying any other medications to the eye.
- Store this medicine at room temperature away from moisture and sunlight. Do not freeze.
- Your veterinarian will need to monitor your animal while it is taking this medicine. Do not miss these important follow-up visits.

Storage/Stability

Store as directed on package labeling (15°C-25°C [59°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Flurbiprofen ophthalmic solutions should not be mixed directly with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Flurbiprofen sodium 0.03% Ophthalmic Solution (containing flurbiprofen 0.3 mg/mL; thimerosal 0.005%; citric acid; edetate disodium; polyvinyl alcohol 1.4%; potassium chloride; purified water; NaCl; sodium citrate; HCl and/or NaOH to adjust pH to 6 to 7; osmolality of 260 – 330 mOsm/kg), in 2.5 mL ophthalmic dropper bottles; *Ocufen*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Ganciclovir Ophthalmic

(gan-sye-kloe-veer) *Zirgan*®

Antiviral Agent

Prescriber Highlights

- ▶ Used as an ophthalmic antiviral in cats with feline herpesvirus-1 (FHV-1) and dogs with canine herpesvirus-1 (CHV-1)
- ▶ May cause myelosuppression and nephrotoxicity in cats if ingested
- ▶ In vitro efficacy versus FHV-1 is 2 to 10 times greater than similar antivirals.

Uses/Indications

In veterinary patients, ganciclovir may be used to treat feline herpesvirus-1 (FHV-1)-induced ocular surface disease in cats. In vitro efficacy of ganciclovir versus FHV-1 has been demonstrated at various concentrations¹ and is 2 times greater than cidofovir and penciclovir and 10 times greater than acyclovir.^{2,3} When ganciclovir was administered to cats with healthy eyes 3 times daily, no signs of ocular irritation developed.¹

Evaluation of ganciclovir in experimentally induced canine herpesvirus-1 infections (CHV-1) indicated good efficacy without signs of ocular toxicity.⁴ In vitro evaluation of ganciclovir suggests efficacy against equine alpha-herpesviruses^{5,6}; however, evaluation of ganciclovir's use in vivo has not been performed.

Pharmacology/Actions

Ganciclovir is a guanosine derivative that, upon phosphorylation,

inhibits DNA replication of herpes simplex viruses (HSV). Ganciclovir is transformed by viral and cellular thymidine kinases (TK) to ganciclovir triphosphate, which works as an antiviral agent by competitive inhibition of viral DNA-polymerase and prevents DNA elongation.

Contraindications/Precautions/Warnings

Ganciclovir is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Myelosuppression and nephrotoxicity may occur in cats after systemic treatment. Ganciclovir has been shown to be embryotoxic in rabbits and mice and teratogenic in rabbits following IV administration.⁷ Because safety has not been established in animals, this drug should only be used when the maternal benefits outweigh the potential risks to offspring. Do not allow cats to ingest the topical formulation; an e-collar may be required to avoid unintentional ingestion during grooming.

Adverse Effects

Adverse effects reported in humans or animals include blurred vision, eye irritation, punctate keratitis, chemosis, and conjunctival hyperemia and discharge.^{1,7} Myelosuppression and nephrotoxicity are possible in cats after ingestion by grooming.

Dosages

Herpes keratitis (extra-label): 1 drop in affected eye(s) up to 5 times daily until the corneal ulcer has healed, then 3 times daily for 7 days

Monitoring

- Clinical efficacy: resolution of clinical signs associated with herpes keratitis (eg, corneal ulceration, corneal vascularization, conjunctivitis)
- Baseline and periodic CBC and chemistry panel for cats to detect myelosuppression and nephrotoxicity

Client Information

- Wipe the cat's face after applying. Do not let cats lick this medication, as it can be toxic when ingested.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this product at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the gel.

Storage/Stability

Store this product at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight.

Compatibility/Compounding Considerations

Ganciclovir should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Ganciclovir 0.15% Ophthalmic Gel (containing ganciclovir 1.5 mg/mL; carbopol; water for injection; NaOH (to adjust the pH to 7.4); mannitol; benzalkonium chloride 0.075 mg), in a 5 g ophthalmic gel tube; *Zirgan*®, (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Gatifloxacin Ophthalmic

(gat-i-**flocks**-a sin) Zymaxid®, Zymar®
Fourth-Generation Fluoroquinolone Antibiotic

Prescriber Highlights

- ▶ Fluoroquinolone antimicrobial available in ophthalmic formulation only; used to treat bacterial conjunctivitis or keratitis
- ▶ Appropriately used as a therapeutic antibiotic in cases with known ocular surface infection; not appropriate for prophylactic use
- ▶ Physically incompatible with many drugs; therefore, it should not be directly mixed with other ophthalmic medications
- ▶ Crystalline drug precipitates may be noted in the superficial portion of corneal defects during use.

Uses/Indications

Gatifloxacin ophthalmic may be useful for treating bacterial conjunctivitis or keratitis caused by susceptible strains of *Staphylococcus* spp, *Streptococcus* spp, *Pseudomonas aeruginosa*, *Chlamydomphila* spp, and *Haemophilus* spp. Gatifloxacin has demonstrated good in vitro activity versus *Pseudomonas* spp organisms cultured from dogs with ulcerative keratitis,¹ particularly in relation to the second-generation fluoroquinolone ciprofloxacin.² In comparison to moxifloxacin and besifloxacin, gatifloxacin demonstrated lower corneal and aqueous humor concentrations when administered to dogs with normal eyes.³ Gatifloxacin should be used to treat, rather than prevent, bacterial ocular infections.

Pharmacology/Actions

Gatifloxacin is a fluoroquinolone antibiotic that acts via inhibition of both topoisomerase II (DNA gyrase) and topoisomerase IV to impair bacterial DNA synthesis. Inhibition of these 2 enzymes decreases the development of resistance to this class of fluoroquinolones. In comparison with besifloxacin, gatifloxacin is reported to have slightly greater in vitro efficacy versus gram-negative organisms, with slightly lesser in vitro efficacy versus gram-positive organisms.⁴

Contraindications/Precautions/Warnings

Gatifloxacin is contraindicated in patients that have a history of hypersensitivity to it, other fluoroquinolones, or any component of the formulation. This drug should only be used as a topical ophthalmic. Do not inject gatifloxacin subconjunctivally or into the anterior chamber of the eye. Fluoroquinolone-induced retinal toxicity in domestic cats has not been demonstrated with this product.

Adverse Effects

Adverse effects reported in humans or animals include blurred vision, tearing, eye pain, redness, itching, and a bad taste in the mouth. Crystalline drug precipitates may be noted in the superficial portion of corneal defects during use.

Dosages

Conjunctivitis (extra-label): 1 drop in affected eye(s) every 2 to 8 hours

Keratitis (extra-label): 1 drop in the affected eye every 2 to 8 hours

Monitoring

- Clinical efficacy. Corneal cytology and culture to evaluate for persistence of bacteria, and evaluation for clinical signs suggestive of continued infection

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.
- Crystals may appear on the ocular surface after initiation of treatment.
- This medicine may precipitate when mixed with other ophthalmic

medicines. Therefore, when giving this medicine to your horse through a subpalpebral lavage catheter, it is important to flush the catheter well with air between medications.

- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine at room temperature away from moisture and sunlight. Do not freeze.

Storage/Stability

Store this product at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Gatifloxacin ophthalmic preparations should not be mixed directly with any other ophthalmic medications and is considered physically incompatible with many drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Gatifloxacin 0.3% Ophthalmic Solution (containing gatifloxacin 3 mg/mL; benzalkonium chloride 0.005%; edetate disodium; purified water and NaCl; HCl and/or NaOH to adjust pH to ≈6; osmolality of 260 to 330 mOsm/kg) in 5 mL ophthalmic dropper bottles; Zymar®, generic; (Rx)

Gatifloxacin 0.5% Ophthalmic Solution (containing gatifloxacin 5 mg/mL; benzalkonium chloride 0.005%; edetate disodium; purified water and NaCl; HCl and/or NaOH to adjust pH to ≈6; osmolality of 260 to 330 mOsm/kg), 2.5 mL in a sterile ophthalmic dropper bottle; Zymaxid®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Gentamicin Ophthalmic

(jen-ta-mye-sin sul-fate) Genoptic®, Gentocin®, Gentak®
Aminoglycoside Antibiotic

For systemic use, see **Gentamicin**

Prescriber Highlights

- ▶ Topical ophthalmic aminoglycoside antibiotic used as a treatment for susceptible ocular surface bacterial infections
- ▶ Do not use if a full-thickness corneal wound is present, as gentamicin is toxic to the eye's interior.
- ▶ Fortified solutions of gentamicin sulfate up to 13.6 mg/mL may be compounded for use in susceptible infections of equine bacterial keratitis.

Uses/Indications

Gentamicin ophthalmic is primarily used for patients with ocular infections caused by gram-negative aerobes, such as *Pseudomonas aeruginosa*.^{1,2} Gentamicin does not demonstrate consistent efficacy against versus *Staphylococcus* spp or *Streptococcus* spp isolates.^{1,3} As the spectrum of gentamicin is weighted toward gram-negative organisms, and as it is more toxic to the corneal epithelium, gentamicin should not be considered a first-line prophylactic antibiotic in veterinary ophthalmology.^{4,5}

In horses with end-stage glaucoma, intravitreal injection of gentamicin at higher doses is used to chemically ablate the ciliary body, thus reducing the production of aqueous humor and lowering intraocular pressure (IOP)⁶; lower doses may be used to manage equine recurrent uveitis (ERU).^{7,8} See **Gentamicin**.

The World Health Organization (WHO) has designated gentamicin as a Critically Important, High Priority antimicrobial for human medicine.⁹ The Office International des Epizooties (OIE) has desig-

nated gentamicin as a Veterinary Critically Important Antimicrobial (VCIA) Agent in equine species.¹⁰

Pharmacology/Actions

Gentamicin sulfate, an aminoglycoside, is actively transported into the bacterial cell. It binds to a specific receptor protein on the 30S subunit of bacterial ribosomes. It consequently interferes with an initiation complex between mRNA (messenger RNA) and the 30S subunit, inhibiting protein synthesis by inducing misreading of the mRNA. Aminoglycosides are bactericidal, whereas most other antibiotics that interfere with protein synthesis are bacteriostatic.

Contraindications/Precautions/Warnings

Gentamicin is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Gentamicin should not be used if a full-thickness corneal wound is present, as gentamicin is toxic to the eye's interior.

Adverse Effects

Adverse effects reported in humans or animals include ocular burning and irritation upon drug instillation, nonspecific conjunctivitis, conjunctival epithelial defects, and conjunctival hyperemia. Other adverse reactions that have rarely been reported include allergic reactions, thrombocytopenic purpura, and hallucinations (humans).¹¹ In dogs and cats, although causation has not been proven, intravitreal gentamicin injections for ciliary body ablation may be linked to the development of intraocular tumors.^{12,13} In horses, cataract formation and diffuse retinal degeneration have been reported following the intravitreal injection of low-dosage gentamicin for equine recurrent uveitis (ERU) management.⁸

When gentamicin is used systemically, it can cause nephrotoxicity and neurotoxicity (ototoxicity and vestibular toxicity). Cats are more susceptible to the toxic effects of gentamicin, and it is recommended that all patients receiving topical gentamicin be monitored for systemic toxicity.

Dosages

Susceptible ocular infections (extra-label): 1 drop of the solution or ¼ – ½ inch strip in the affected eye(s) 4 to 6 times daily. The frequency may be increased to every 30 to 60 minutes for a few hours if needed.

Monitoring

- Clinical efficacy: cytology and culture to evaluate for presence of bacteria; clinical resolution of signs of ocular surface infection and or recurrent inflammation
- Intraocular pressure (IOP) when used as adjunctive treatment for equine glaucoma or recurrent uveitis
- Urinalysis with sediment to check for casts in the urine, which may be an early indicator for renal toxicity

Client Information

- Give this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep cap tightly closed when not in use.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medicine according to the directions on the product label, protected from light. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product as directed on the product label, protected from light; do not freeze.

Compatibility/Compounding Considerations

Gentamicin sulfate should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS:

Gentamicin Sulfate 0.3% Ophthalmic solution (containing gentamicin sulfate equivalent to 3 mg gentamicin base; disodium phosphate; monosodium phosphate; NaCl; benzalkonium chloride; aqueous solution with pH 6.8 to 7.3), in sterile 5 mL dropper bottle; generic; (Rx)

VETERINARY-LABELED COMBINATION PRODUCTS:

Gentamicin Sulfate 0.3% and Betamethasone Ophthalmic solution (containing gentamicin sulfate equivalent to 3 mg gentamicin base; 1 mg betamethasone acetate equivalent to 0.89 mg betamethasone alcohol; polyoxyl 40 stearate; polyethoxy 35 castor oil; edetate disodium; 0.1 mg benzalkonium chloride as a preservative; water for injection), supplied in a sterile 5 mL dropper bottle; *Gentocin® Durafilm®*; (Rx)

HUMAN-LABELED PRODUCTS:

Gentamicin 0.3% Ophthalmic Solution (containing gentamicin sulfate equivalent to 3 mg/mL gentamicin base; benzalkonium chloride; edetate disodium; polyvinyl alcohol 1.4%; purified water; NaCl; sodium phosphate, dibasic; HCl and/or NaOH to adjust the pH in an aqueous, buffered solution with a pH of 6.5 to 7.5), in 5 mL ophthalmic dropper bottles; *Genoptic®*, *Gentak®*; (Rx)

Gentamicin 0.3% Ophthalmic Ointment (containing gentamicin sulfate equivalent to 3 mg/mL of gentamicin in a base of white petrolatum and mineral oil, with methylparaben and propylparaben as preservatives), in a 3.5 g ophthalmic ointment tube; generic; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Gentamicin and Prednisolone Acetate Ophthalmic Ointment 0.3%/0.6% (containing gentamicin sulfate equivalent to 0.3% gentamicin base; prednisolone acetate 0.6%; chlorobutanol (chloral derivative) 0.5%; mineral oil; petrolatum (and) lanolin alcohol; purified water; white petrolatum), in a sterile 3.5 g ophthalmic tube; *Pred-G®*; (Rx)

Gentamicin and Prednisolone Acetate Ophthalmic Suspension 0.3%/1% (containing gentamicin sulfate equivalent to 0.3% gentamicin base; prednisolone acetate (microfine suspension) 1%; benzalkonium chloride 0.005%; edetate disodium; hypromellose; polyvinyl alcohol 1.4%; polysorbate 80; purified water; sodium chloride; sodium citrate dehydrate; NaOH and/or HCl to adjust pH (5.4 to 6.6), in a sterile 5 mL ophthalmic dropper bottle; *Pred-G®*; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Hydrocortisone Ophthalmic

(hye-droe-**kor**-ti-zone)

Corticosteroid Anti-inflammatory Agent

Prescriber Highlights

- ▶ Ophthalmic corticosteroid used to treat ocular surface inflammation
- ▶ Only commercially available formulation is in combination with neomycin/polymyxin B +/- bacitracin
- ▶ NOT effective for deep corneal or intraocular inflammation (uveitis)
- ▶ Contraindications include a hypersensitivity to dexamethasone; an existing or suspected bacterial, viral, or fungal ocular infection; and in the presence of a corneal abrasion or ulceration.
- ▶ Use with caution in avian species due to risk for systemic corticosteroid effects.

Uses/Indications

Hydrocortisone is used in combination with topical antibiotics (eg, neomycin, polymyxin, $\pm$ bacitracin) for inflammatory conjunctivitis.

Pharmacology/Actions

Hydrocortisone is a naturally occurring corticosteroid that is produced by the adrenal glands. Corticosteroids have broad immunomodulatory and anti-inflammatory actions, which are mediated via direct effects on gene expression, as well as by non-genomic effects on cell membranes and receptors. Ultimately, corticosteroids decrease pro-inflammatory gene transcription, production of pro-inflammatory prostaglandins and leukotrienes, B- and T-cell number and function, cell-mediated immunity, and humoral immunity.¹ Specific mechanisms involved in controlling ocular inflammatory conditions are not definitively known. Hydrocortisone has a relative glucocorticoid potency of 1 (versus 4 for prednisolone and 30 for dexamethasone and betamethasone), which correlates with weak clinical efficacy.

Hydrocortisone is a weak base steroid with poor intraocular penetration; therefore, it is ineffective for treating intraocular inflammation (uveitis).

Contraindications/Precautions/Warnings

Hydrocortisone is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Due to inhibition of epithelial wound healing and potentiation of infection, hydrocortisone ophthalmic is also contraindicated when corneal ulceration is suspected or present, or in the presence of corneal infection (bacterial, fungal, or viral).^{2,3} Topical administration of glucocorticoids may produce systemic effects. For more information, refer to **Glucocorticoid Agents, General Information**. Use hydrocortisone ophthalmic with caution in patients with endocrine diseases (eg, diabetes mellitus, acromegaly), infectious diseases, chronic kidney disease, congestive heart failure, systemic hypertension, or gastric ulceration. When use of hydrocortisone ophthalmic is necessary to treat a patient with these conditions, the minimal effective dose should be administered for the shortest possible time. Ophthalmic steroids should be used with extreme caution in avian species due to increased risk for systemic adverse effects.

Adverse Effects

Adverse effects reported in humans or animals include burning, stinging, and irritation upon application to the eye.³ Systemic absorption may result in suppression of the hypothalamus-pituitary-adrenal axis. The use of corticosteroids may result in posterior subcapsular cataract formation, glaucoma, and secondary ocular infections due to bacteria, fungi, or viruses.⁴ With the chronic use of hydrocortisone, intraocular pressure (IOP) may increase in some animals; however, most reports suggest the change is reversible.⁵⁻⁷

Dosages

Inflammatory conjunctivitis (extra-label): $\frac{1}{4}$ – $\frac{1}{2}$ inch strip in the affected eye(s) up to 4 times daily

Monitoring

- Clinical efficacy
- Resolution of signs of inflammation
- Intraocular pressure (IOP; especially in cats) and ocular surface integrity (eg, fluorescein stain)
- Blood or urine glucose changes based on clinical signs (eg, polyuria, polydipsia) in diabetic animals

Client Information

- Available only as an ointment form, in combination with antibiotics.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.

- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medicine as directed on the product label. Do not freeze.
- Discard this medicine if the color change occurs.

Storage/Stability

Store at controlled room temperature (68°F–77°F [20°C–25°C]), away from moisture and sunlight.

Compatibility/Compounding Considerations

Hydrocortisone ophthalmic should not be mixed directly (in vitro) with any other drugs. Instruct caregivers to wait for 5 minutes after applying this medication before applying any other medications to the eye.

Dosage Forms/Regulatory Status**VETERINARY-LABELED PRODUCTS:**

No ophthalmic products containing only hydrocortisone are available; available commercially in combination with antimicrobial agents.

Neomycin/Bacitracin/Polymyxin/Hydrocortisone Ophthalmic Ointment (containing neomycin sulfate equivalent to 3.5 mg neomycin base, polymyxin B sulfate equivalent to 10,000 polymyxin B units, bacitracin zinc equivalent to 400 bacitracin units, hydrocortisone 10 mg/g, and white petrolatum), in a 3.5 g ophthalmic ointment tube; *Vetropolycin HC*®, (Rx)

HUMAN-LABELED PRODUCTS:

Neomycin/Bacitracin/Polymyxin/Hydrocortisone Ophthalmic Ointment (containing neomycin sulfate equivalent to 3.5 mg neomycin base, polymyxin B sulfate equivalent to 10,000 polymyxin B units, bacitracin zinc equivalent to 400 bacitracin units, hydrocortisone 10 mg/g, and white petrolatum), in a 3.5 g ophthalmic ointment tube; generic, (Rx)

Neomycin/Polymyxin/Hydrocortisone Ophthalmic Ointment (containing neomycin sulfate equivalent to 3.5 mg neomycin base, polymyxin B sulfate equivalent to 10,000 polymyxin B units, hydrocortisone 10 mg/g, and white petrolatum), in a 3.5 g ophthalmic ointment tube; generic, (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Idoxuridine Ophthalmic

(eye-docks-**yoor**-uh-deen)

Antiviral (Nucleoside Analogue)

Prescriber Highlights

- ▶ Topical antiviral agent used in cats with feline herpesvirus-1 (FHV-1)
- ▶ Must be used 4 to 5 times daily
- ▶ In vitro efficacy is less than trifluridine and comparable to or greater than other antivirals.
- ▶ No longer commercially available; must be compounded as an ointment, solution, or suspension

Uses/Indications

Idoxuridine is used topically in cats to treat conjunctivitis and keratitis associated with feline herpesvirus-1 (FHV-1) infections. Idoxuridine has also demonstrated efficacy in the treatment of canine herpesvirus-1 (CHV-1) keratitis.¹ Although relative efficacy versus other antiviral agents is not known and direct causation of clinical effect is difficult to determine, idoxuridine was associated with improvement in keratoconjunctivitis caused by equine herpesvirus-2 (EHV-2) in foals.²

Pharmacology/Actions

Idoxuridine is chemically similar to thymidine, and its substitution into viral DNA causes misreading of the viral genetic code, thereby inhibiting viral replication. Because idoxuridine is a nonspecific inhibitor of DNA synthesis, it affects any cellular function requiring thymidine and is not suitable for systemic use. Even with topical therapy, corneal toxicity can occur. Idoxuridine is virustatic. Idoxuridine's in vitro efficacy is less than trifluridine, comparable to ganciclovir, greater than cidofovir, and much greater than acyclovir.^{3,4}

Contraindications/Precautions/Warnings

Idoxuridine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans or animals include acute ocular irritation, including burning, corneal stippling, vascularization, and clouding. Following prolonged use of idoxuridine, ocular irritation characterized by follicular conjunctivitis, blepharitis with punctal swelling, bulbar conjunctival hyperemia, and corneal epithelial staining has also been reported. Treatment with idoxuridine should be discontinued if adverse effects occur.

Dosages

Keratitis caused by feline herpesvirus-1 (FHV-1), canine herpesvirus-1 (CHV-1), or equine herpesvirus-2 (EHV-2) (extra-label): 1 drop in affected eye(s) every 2 to 3 hours for 48 hours, then 4 to 5 times daily for a week beyond the resolution of clinical signs

Monitoring

- Resolution of clinical signs of herpetic keratitis (eg, corneal ulceration, corneal vascularization, conjunctivitis)

Client Information

- Give this medicine to your animal as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication as directed on the product label. Do not freeze and protect from light.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in solutions.

Storage/Stability

Because this compound must be compounded (in the United States), storage recommendations can vary; follow storage requirements listed on the packaging. Do not freeze.

Compatibility/Compounding Considerations

Idoxuridine should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

No ophthalmic dosage form of idoxuridine is approved for use; however, Idoxuridine may be compounded as a 0.1% to 0.5% solution, suspension, and/or ointment by a qualified compounding pharmacist.

References

For the complete list of references, see wiley.com/go/budde/plumb

Irrigating Solutions, Ophthalmic

BSS®, Collyrium®

Ocular Irrigant

Prescriber Highlights

- ▶ Available for either intraocular or extraocular use
- ▶ Extraocular (eyewashes) products should NOT be used intraocularly.

Uses/Indications

Extraocular eyewash preparations are used to irrigate the nasolacrimal duct system, remove debris from the eye, and remove the excess stain after diagnostic staining (eg, fluorescein) of the cornea.

Intraocular irrigation solutions are used during intraocular surgery (eg, cataract removal, vitreal surgery) to maintain the shape of the anterior chamber, cool phacoemulsification handpieces, and lavage emulsified lens and surgical byproducts from the eye.

Pharmacology/Actions

OTC extraocular eyewash preparations are sterile isotonic solutions that contain preservatives and are intended to be used for general external ophthalmic use only.

Intraocular sterile irrigating solutions are sterile, preservative-free, physiologically balanced salt solutions that mimic the composition of the aqueous humor in terms of physiologic pH, osmolality, and ion composition; they contain electrolytes required for normal cellular metabolic functions. Some intraocular irrigating solutions also contain glutathione, which is responsible for stabilizing endothelial cell junctions and intraocular pumping functions.

Contraindications/Precautions/Warnings

Extraocular eyewash preparations are contraindicated for intraocular use, as preservative components are toxic to the corneal endothelium. They are also contraindicated if the patient is hypersensitive to any component of the formulation. Adding drugs or other fluids to intraocular irrigating solutions may damage intraocular tissue.

In humans with diabetic mellitus who are undergoing vitrectomy, caution is recommended with the use of intraocular irrigating solutions, as intraoperative lens changes have been noted in some patients; the same recommendation may apply to diabetic animals. Caution should be used in diabetic animals undergoing vitrectomy for other causes (eg, trauma, neoplasia, endophthalmitis).

Adverse Effects

Intraocular irrigation solutions have caused corneal clouding and edema in some patients due to endothelial damage. In humans, lens changes have been noted in some diabetic patients who are undergoing vitrectomy. A study evaluating cooled versus room temperature irrigating solutions demonstrated no significant adverse effects between the 2 temperatures.¹

Dosages

Extraocular eyewash (extra-label): Flush affected eye(s) as needed; control rate of flow by exerting pressure on the bottle

Intraocular use during surgery (extra-label): Use a sterile, preservative-free balanced salt solution (NOT an OTC eyewash) according to the established practices for each surgical procedure.

Monitoring

- When intraocular solutions are used, monitor for development or worsening of corneal clouding or edema or opacification of the lens.
- In diabetic patients, monitor for lens changes post-vitrectomy.

Client Information

- If you are using an over-the-counter eyewash solution, use proper administration techniques to avoid eye injury or contamination

of the product. Keep the cap tightly closed when the medication is not in use.

- Wait 5 minutes after using the eyewash before applying any medications to the eye.
- Store this product at room temperature away from moisture and sunlight. Do not freeze.
- Discard this product if the color changes or the solution becomes cloudy.

Storage/Stability

Store OTC eyewash products as labeled, usually at controlled room temperature away from moisture and sunlight; do not freeze.

Store solutions for intraocular use at 8°C to 30°C (46°F–86°F); avoid excessive heat, and do not freeze. Discard solution after 6 hours. Store *BSS Plus*® solutions at 2°C to 25°C (36°F–77°F); do not freeze. Discard prepared solution after 6 hours. Do not use unfinished solutions on a different patient.

Compatibility/Compounding Considerations

It is not recommended to add any other solutions or drugs to intraocular or extraocular irrigating solutions; however, extra-label additives in some protocols have included heparin, epinephrine, antibiotics, or local anesthetics.

For reconstituting *BSS Plus*® solutions: Reconstitute solution just prior to use in surgery. Follow the same strict aseptic reconstitution procedures used for IV solution additives. Remove the blue flip-off seal from the part 1 (480 mL) bottle. Remove the blue flip-off seal from the part 2 (20 mL) vial. Clean and disinfect the rubber stoppers on both containers by using sterile alcohol wipes. Transfer the contents of the part 2 vial to the part 1 bottle using the vacuum transfer device (provided). An alternative method of solution transfer may be accomplished by using a syringe to remove precisely 20 mL of the part 2 solution and transfer to the part 1 container through the outer target area of the rubber stopper. **NOTE:** An excess volume of the part 2 solution is provided in each vial; discard unused solution. Gently agitate the contents to mix the solutions. Place a sterile cap on the bottle if the solution is not going to be used immediately. Remove the tear-off portion of the label. Record the time and date of reconstitution and the patient's name on the bottle label.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS:

Several products are marketed as eye washes or rinses for animal patients, including *Clear Eyes*®, *Conquer Hy-Optic*®, *Nutri-Vet*® Dog Eye Rinse, and *Vetericyn*® Plus Eye Wash. Ingredients vary significantly, and some are labeled only as tear stain removers. None of these products appear to be FDA-approved, and they should NOT be used for intraocular procedures.

HUMAN-LABELED PRODUCTS:

Irrigating Solutions, Intraocular

Balanced Salt Solution Irrigating Solution in 15 mL, 18 mL, 30 mL, 250 mL, or 500 mL; preservative-free containing 0.64% NaCl, 0.075% KCl, 0.03% magnesium chloride, 0.048% calcium chloride, 0.39% sodium acetate, 0.17% sodium citrate, and NaOH or HCl; *BSS*®, generic; (Rx)

Balanced Salt Solution Plus; in 2 parts in 10 mL (part 1) and 240 mL (part 2) in a 250 mL bottle; preservative-free containing 0.0154% calcium chloride, 0.714% sodium chloride, 0.038% potassium chloride, 0.02% magnesium chloride, 0.42% sodium phosphate, 0.2% sodium bicarbonate, 0.092% dextrose, 0.0184% glutathione disulfide; *BSS Plus*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Itraconazole Ophthalmic

(it-ruh-**kon**-uh-zohl) *Sporanox*®

Azole Antifungal Agent

Prescriber Highlights

- ▶ Used as a treatment for ocular surface fungal infections
- ▶ Not commercially available as an ophthalmic preparation; must be compounded
- ▶ Compounded forms contain 10% to 30% DMSO, which may be irritating when administered.
- ▶ Warn caregivers to wear nitrile gloves when applying this medication to protect them from DMSO exposure.

Uses/Indications

Itraconazole ophthalmic preparations are primarily used to treat fungal keratitis in horses^{1,2}; however, they have successfully been used in dogs with fungal keratitis.³

Pharmacology/Actions

Itraconazole interacts with 14 alpha-sterol demethylase, a cytochrome P-450 enzyme involved in the conversion of lanosterol to ergosterol. Ergosterol is an essential component of the fungal cell membrane, and preventing its synthesis may increase cellular permeability and cause leakage of cellular contents. Other possible effects of itraconazole include inhibition of endogenous respiration, purine uptake and yeast to mycelial transformation, and impairment of triglyceride and phospholipid biosynthesis. Targeted oxidative enzymes of fungal species, rather than mammalian cells, increase efficacy and decrease host toxicity. Itraconazole is usually effective against *Aspergillus* spp, with variable efficacy versus *Candida* spp and consistently lower efficacy against *Fusarium* spp.^{1,2} Identification of fungal species prior to treatment is critical to the success of treatment.⁴

Contraindications/Precautions/Warnings

Itraconazole is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Compounded itraconazole 1% in DMSO 30% may cause irritation to the cornea. A solution of DMSO 1% is commonly used as a control for corneal irritation in ophthalmic drug studies.⁵

Adverse Effects

Adverse effects reported in humans or animals include stinging and irritation from the DMSO component. Although systemic absorption of topically applied ointments is unlikely, systemic exposure to itraconazole has resulted in hepatotoxicity and hearing loss in humans. Compared with miconazole and natamycin, itraconazole demonstrates lower toxic effects on corneal epithelial cells in vitro.⁶

Dosages

Fungal keratitis (extra-label): 1 – 2 drops, ½ inch strip, or 0.1 mL in the subpalpebral lavage (SPL) catheter to the affected eye(s) every 2 to 3 hours initially, followed by tapering of frequency based on clinical response

Monitoring

- Clinical efficacy: corneal scrapings with cytology and cultures to determine the clearing of fungal organisms
- Resolution of clinical signs of infection
- Toxicity: monitor for ocular irritation (eg, corneal edema, chemosis, blepharospasm) from DMSO

Client Information

- Use this medicine as directed by your veterinarian.
- A garlic-like odor is normal (from the DMSO solvent).
- Wear nitrile gloves when applying to prevent absorption of DMSO solvent found in the medicine.

- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store as directed on package labeling away from moisture and sunlight. Do not freeze. Discard if the color or consistency changes.

Storage/Stability

Store this product as directed on the label from the compounding pharmacy. Keep this product away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Itraconazole ophthalmic preparations should not be mixed directly with any other ophthalmic medications. Itraconazole ophthalmic preparations are not commercially available, but a compounded preparation containing 30% DMSO has been described.^{7,8}

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Itraconazole ophthalmic products are not commercially available and must be prepared (compounded) by a qualified compounding pharmacist. Itraconazole has been compounded into a 1% ophthalmic ointment in a base containing 10% to 30% DMSO or into a 1% ophthalmic solution containing 30% DMSO.

References

For the complete list of references, see wiley.com/go/budde/plumb

Ketorolac Ophthalmic

(kee-**toe**-role-ak) Acular®, Acular LS®, Acuvail®
Nonsteroidal Anti-inflammatory Drug (NSAID)

Prescriber Highlights

- ▶ Used to control anterior segment inflammation, particularly in association with seasonal allergic conjunctivitis
- ▶ May cause transient increases in intraocular pressure (IOP)
- ▶ May delay ocular wound healing and increase risk for intraocular bleeding

Uses/Indications

Ketorolac ophthalmic is indicated for controlling anterior segment inflammation (eg, conjunctivitis, keratitis, anterior uveitis), inducing mydriasis during cataract-removal surgery, and for providing analgesia following surgical removal of cataracts. In diabetic patients, ketorolac may be useful to avoid the systemic effects of topically applied corticosteroids. Ketorolac is also indicated to alleviate the signs of allergic conjunctivitis. Ketorolac did not decrease corneal sensitivity in the eyes of normal dogs,¹ suggesting that its use as an ophthalmic analgesic may be limited; however, analgesic effects might improve in the presence of ocular pain.

Pharmacology/Actions

Ketorolac is an NSAID that blocks prostaglandin synthesis by inhibiting cyclooxygenase 1 and 2. Prostaglandins mediate many of the adverse effects of intraocular inflammation, including disruption of the blood-aqueous humor barrier, vasodilation, increased vascular permeability, leukocytosis, and increased intraocular pressure. Blocking prostaglandin production, therefore, minimizes the adverse short- and long-term effects of intraocular inflammation. Prostaglandins also appear to play a role in the miotic response produced during ocular surgery by constricting the iris sphincter independently of cholinergic mechanisms.

Contraindications/Precautions/Warnings

Ketorolac is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. The potential for cross-sensitivity to acetylsalicylic acid, phenylacetic acid derivatives, and other NSAIDs exists.² In humans that have either a known hypersensitivity to aspirin or NSAIDs or a past medical history of asthma, there have been reports of bronchospasm or exacerbation of asthma associated with the use of ketorolac tromethamine ophthalmic solution. Therefore, caution should be used when treating individuals that have previously exhibited sensitivities to these drugs. Topical NSAIDs may result in corneal epithelial breakdown, delayed corneal wound healing, and corneal thinning, erosion, or ulceration. As such, ketorolac ophthalmic should be used with caution in patients with complicated or repeated ocular surgeries, corneal denervation, corneal epithelial defects, diabetes mellitus, and corneal surface diseases (eg, KCS).² As topical corticosteroids are also known delay healing, concomitant use with topical NSAIDs may increase the potential for delayed or impaired healing. Due to interference with thrombocyte aggregation, ophthalmic NSAIDs may cause increased bleeding of ocular tissues (eg, hyphema) in conjunction with ocular surgery. It is recommended that ketorolac ophthalmic solution be used with caution in patients with known bleeding tendencies (eg, severe liver disease) or in those receiving anticoagulant medications (eg, heparins, rivaroxaban), which may prolong bleeding time.¹

Adverse Effects

Adverse effects reported in humans or animals include transient stinging and burning on instillation, corneal edema, iritis, ocular inflammation, ocular irritation, superficial keratitis, and superficial ocular infections.¹ Ocular bleeding may occur because of interference with thrombocyte aggregation. Other adverse effects that have rarely occurred in humans include corneal infiltrates, corneal ulcers, eye dryness, headaches, and visual disturbance (blurry vision).

Although unlikely, systemic adverse effects may occur with the use of topical NSAIDs, including increased risk for gastric ulceration, decreased platelet aggregation, worsening of asthma and cardiac disease, and renal toxicity. Concurrent use of systemic NSAIDs and corticosteroids increases the risk for systemic adverse effects.

Dosages

Allergic conjunctivitis: 1 drop to the affected eye(s) 2 to 4 times daily

Anterior segment inflammation (extra-label): 1 drop to the affected eye(s) 2 to 4 times daily

Postoperative analgesia: 1 drop to the surgical eye 2 to 4 times daily, beginning one day prior to surgery and continuing for 2 weeks postoperative

Monitoring

- Clinical efficacy (eg, decreased or absent anterior segment inflammation)
- Adverse effects (eg, delayed wound healing)

Client Information

- Use this medicine as directed by your veterinarian.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in solutions.
- Ketorolac ophthalmic may cause burning or stinging on administration. It may also increase the risk for bleeding and may slow down healing in the eye.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.

- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medicine in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Your veterinarian will need to monitor your animal while it is taking this medicine. Do not miss these important follow-up visits.

Storage/Stability

Store as directed on package labeling (15°C-25°C [59°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Ketorolac should not be mixed directly with any other drugs. Ketorolac is not available as an ophthalmic ointment.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Ketorolac Tromethamine 0.5% Ophthalmic Solution (containing ketorolac tromethamine 5 mg/mL; benzalkonium chloride 0.01%; edetate disodium 0.1%; octoxynol 40; purified water; NaCl; HCl and/or NaOH to adjust the pH to 7.4 and osmolality of 290 mOsm/kg), in a 5 mL ophthalmic dropper bottle; *Acular*®, generic; (Rx)

Ketorolac Tromethamine 0.4% Ophthalmic Solution (containing ketorolac tromethamine 0.4%; benzalkonium chloride 0.006%; edetate disodium 0.015%; octoxynol 40; purified water; NaOH; HCl and/or NaOH to adjust pH to ≈7.4), 5 mL in a 10 mL sterile ophthalmic dropper bottle; *Acular LS*®, generic; (Rx)

Ketorolac Tromethamine 0.45% Ophthalmic Solution (containing ketorolac tromethamine 0.45%; carboxymethylcellulose; NaCl; sodium citrate dehydrate; purified water; NaOH and/or HCl to adjust pH), in 30 single-use vials of 0.4 mL each; *Acuvail*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Ketotifen Ophthalmic

(kee-toe-tye-fin) *Alaway*®, *Zaditor*®

Antihistamine Agent

Prescriber Highlights

- ▶ Relatively selective H₁ blocker used to treat allergic conjunctivitis
- ▶ Use up to 2 times daily

Uses/Indications

In veterinary patients, ketotifen is used to provide temporary relief of the clinical signs associated with allergic conjunctivitis.

Pharmacology/Actions

Ketotifen is a relatively selective, noncompetitive histamine antagonist (H₁ receptor) and mast cell stabilizer. Ketotifen inhibits the release of mediators (eg, histamine, leukotrienes C₄ and D₄ [SRS-A], platelet-activating factor) from mast cells, during hypersensitivity reactions; ketotifen also decreases chemotaxis and activation of eosinophils. Evaluation of the effectiveness of the topical ocular application of ketotifen in veterinary species has not been performed.

Contraindications/Precautions/Warnings

Ketotifen is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans include conjunctival injection, rhinitis, ocular allergic reactions, burning or stinging of the eye, con-

junctivitis, eye discharge, dry eye, eye pain, eyelid disorder, itching eye, keratitis, lacrimation disorder, mydriasis, photophobia, rash, and pharyngitis.

Dosages

Allergic conjunctivitis (extra-label): 1 drop in the affected eye(s) twice daily

Monitoring

- Clinical efficacy
- Resolution or relief of signs of allergic conjunctivitis (pruritus, tearing, squinting, erythema)

Client Information

- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 to 10 minutes after applying this medication before applying any other medications to the eye.
- Store this medication in the refrigerator or at room temperature away from moisture and sunlight; do not freeze. Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in solutions.

Storage/Stability

Store this medication in the refrigerator or at controlled room temperature away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Ketotifen should not be mixed directly (in vitro) with any other drugs. Ketotifen is not available as an ophthalmic ointment.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Ketotifen Fumarate 0.025% Ophthalmic Solution (containing: ketotifen fumarate equivalent to 0.25 mg ketotifen per milliliter; benzalkonium chloride 0.01%; glycerol; NaOH/HCl to adjust pH to 4.4 to 5.8; water for injection; with an osmolality of 210 to 300 mOsm/kg), in a 5 mL ophthalmic dropper bottle; *Alaway*®, *Zaditor*®, generic; (OTC)

Ketotifen Fumarate 0.035% Ophthalmic Solution (containing: ketotifen fumarate equivalent to 0.25 mg ketotifen per milliliter; benzalkonium chloride 0.01%; glycerol; NaOH/HCl to adjust pH to 4.4 to 5.8; water for injection; with an osmolality of 210 – 300 mOsm/kg), in a 5 mL ophthalmic dropper bottle; *Alaway*®, generic; (OTC)

Latanoprost Ophthalmic

(la-ta-noe-prost) *Rocklatan*®, *Vyzulta*®, *Xalatan*®, *Xelpros*®

Prostaglandin Analogue

Prescriber Highlights

- ▶ Used to decrease intraocular pressure (IOP) in patients with primary glaucoma or ocular hypertension
- ▶ Not effective for most cases of feline or equine glaucoma
- ▶ Do not use if IOP inflammation is present, as it may worsen inflammation or lead to pupillary block glaucoma.
- ▶ Twice-daily dosing results in less fluctuation of IOP.
- ▶ Preservatives may cause ocular irritation.

Uses/Indications

In dogs, latanoprost is used to reduce intraocular pressure (IOP) to manage primary glaucoma. Up to 60% reduction in IOP can be achieved with either once-daily¹ or twice-daily administration in dogs, with twice-daily administration resulting in lower and less

variable IOP.² When it is administered in combination with a topical NSAID to normotensive canine eyes, latanoprost's ocular hypotensive effects are lessened,³ whereas administration with a topical corticosteroid to normotensive canine eyes does not adversely impact the ocular hypotensive effects.⁴

Latanoprost is generally not indicated for use in cats with glaucoma, as it does not lower IOP in normal feline eyes,¹ and glaucoma in cats is frequently secondary to uveitis (latanoprost is contraindicated in the presence of uveitis). Although encountered uncommonly in clinical practice, primary congenital glaucoma in cats may be effectively managed with latanoprost, with an IOP reduction of up to 60% initially and efficacy decreasing over 3 weeks of treatment.⁵

Use in horses is generally not indicated due to relatively poor IOP-lowering effect and an association with a high incidence of adverse effects (eg, ocular discomfort)⁶; however, concurrent administration of topical latanoprost and diclofenac to normal equine eyes has demonstrated IOP reduction and reduced signs of discomfort.⁷

Pharmacology/Actions

Prostaglandin analogues are chemically modified versions of prostaglandin F₂-alpha, an endogenous inflammatory molecule that mediates ocular hypotensive effects by increasing outflow of aqueous humor through both the trabecular meshwork and uveoscleral outflow routes. In addition to reducing intraocular pressure (IOP), latanoprost leads to miosis.

Contraindications/Precautions/Warnings

Latanoprost is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Latanoprost should be used with caution in the presence of anterior uveitis and secondary glaucoma, particularly in cases already exhibiting significant miosis, as this treatment may exacerbate miosis, promote synechia formation, and worsen glaucoma. Latanoprost should also be used with caution in dogs with primary anterior lens luxation, as the miosis that results may trap the lens and/or vitreous within the pupillary aperture, thus creating pupillary block glaucoma. Latanoprost is not contraindicated in dogs with secondary posterior lens luxation, as the miosis may (favorably) trap the lens in the posterior segment.

Adverse Effects

Adverse effects reported in humans or animals include blurred vision, burning and stinging, conjunctival hyperemia, foreign body sensation, itching, increased pigmentation of the iris, and punctate epithelial keratopathy. Latanoprost may worsen intraocular inflammation. Latanoprost may cause darkening of the iris through direct stimulation of iris melanocytes, and it may also cause eyelash changes (eg, increased length, thickness, pigmentation, and the number of lashes) and eyelid skin darkening. Macular edema, including cystoid macular edema, has been reported after the use of latanoprost in humans. Horses develop conjunctival hyperemia, epiphora, and blepharospasm with latanoprost's administration.⁸

Dosages

Glaucoma (extra-label): 1 drop in the affected eye(s) once to twice daily

Monitoring

- Confirm maintenance of intraocular pressure (IOP) within the normal range for the species being treated

Client Information

- Use this medicine as indicated by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.

- Store unopened bottle(s) under refrigeration. Do not freeze. Once a bottle is opened for use, it may be stored at room temperature (up to 25°C [77°F]), protected from light, for up to for 6 weeks.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in solutions.

Storage/Stability

Protect this product from light. Store unopened bottle(s) under refrigeration at 2°C to 8°C (36°F–46°F)⁹; do not freeze. During shipment to the owner, the bottle may be maintained at temperatures up to 40°C (104°F) for a period not exceeding 8 days. Once a bottle is opened for use, it may be stored at room temperature up to 25°C (77°F) for 6 weeks.

Compatibility/Compounding Considerations

Latanoprost ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Latanoprost 0.005% Ophthalmic Solution (containing 50 µg of latanoprost per milliliter; benzalkonium chloride 0.02%; NaCl; sodium dihydrogen phosphate monohydrate; disodium hydrogen phosphate anhydrous; water for injection, with a pH of ≈6.7 and an osmolality of ≈267 mOsm/kg), in 2.5 mL ophthalmic dropper bottles; *Xalatan*®, generic; (Rx)

Latanoprost 0.005% Ophthalmic Solution (containing 50 µg of latanoprost; potassium sorbate 4.7 mg; castor oil; sodium borate; boric acid; propylene glycol; edetate disodium; polyoxyl 15 hydroxystearate; sodium hydroxide; hydrochloric acid; water for injection), in 2.5 mL ophthalmic dropper bottles; *Xelpros*® (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Netardusil 0.02% and latanoprost 0.005% Ophthalmic Solution (containing netarsudil 0.285 mg; latanoprost 0.050 mg; benzalkonium chloride 0.20 mg; mannitol; boric acid; sodium hydroxide to adjust pH; water for injection), in 2.5 mL ophthalmic dropper bottles; *Rocklatan*® (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb.

Levobunolol Ophthalmic

(lee-voe-**byoon**-uh-lol) Akbeta®, Betagan®

Nonselective Beta-Adrenergic Blocker

Prescriber Highlights

- Decreases aqueous humor production to lower intraocular pressure (IOP) as a treatment for glaucoma
- May delay onset of glaucoma in the normal eye of dogs with unilateral glaucoma
- Use with caution in asthmatic patients due to the potential to induce bronchospasm.

Uses/Indications

Levobunolol is used to reduce intraocular pressure (IOP) in primary glaucoma and as prophylaxis to prevent glaucoma in the normal eye of animals with unilateral glaucoma. In dogs, the magnitude of reduction in IOP is less with levobunolol alone as compared with the combination of levobunolol and dorzolamide,¹ and timolol and dorzolamide.²

Pharmacology/Actions

Levobunolol is a nonselective beta-adrenergic blocking agent with associated intrinsic sympathomimetic activity but without signifi-

cant membrane-stabilizing activity; the significance of which is not fully understood in ophthalmic use. Blockade of ciliary body beta receptors, particularly beta-2 receptors, reduces intraocular pressure (IOP) by decreasing aqueous humor production. The exact mechanism of the ocular hypotensive effect of beta-blockers is not clear. Several mechanisms have been postulated for this effect, including beta-blockade of norepinephrine-induced tonic sympathetic stimulation, leading to decreased activation of cAMP in the ciliary body, inhibition of Na⁺K⁺ATPase activity, or vasoactive mechanisms on the iris and ciliary body. Topically administered beta-blockers also inhibit glutamine-induced increases in intracellular calcium and sodium, thereby providing a potential protective effect against ischemic insult to the retina.³

Contraindications/Precautions/Warnings

Levobunolol is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Levobunolol should be used cautiously, if at all, in patients with heart block, bradycardia, heart failure, asthma, chronic bronchitis, or other cardiopulmonary conditions.¹

Adverse Effects

Adverse effects reported in humans or animals include transient burning, stinging, and irritation upon application to the eye. Topically applied levobunolol led to a significant reduction in heart rate in healthy dogs, thus supporting its limited use in dogs with cardiopulmonary conditions.

Dosages

Glaucoma (extra-label): 1 drop in the affected eye(s) twice daily

Monitoring

- Clinical efficacy
- Confirm maintenance of intraocular pressure (IOP) within the normal range for the species being treated
- Observe asthmatic and diabetic patients for respiratory difficulty or acute hypoglycemia, respectively.
- Heart rate

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 to 10 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in solutions.

Storage/Stability

Store this product at controlled room temperature (15°C–25°C [59°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Levobunolol ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Levobunolol 0.25% Ophthalmic Solution (containing levobunolol HCl 0.5%; benzalkonium chloride 0.004%; polyvinyl alcohol 1.4%; edetate disodium; sodium metabisulfite; sodium phosphate, dibasic; potassium phosphate, monobasic; NaCl; HCl or NaOH to adjust the pH; purified water), in 5, 10, and 15 mL ophthalmic dropper bottles; *Akbeta*®, *Betagan*®, (Rx)

Levobunolol 0.5% Ophthalmic Solution (containing benzalkonium chloride 0.004%; polyvinyl alcohol 1.4%; edetate disodium; sodium metabisulfite; sodium phosphate, dibasic; potassium phosphate, monobasic; NaCl; HCl or NaOH to adjust the pH; purified water), in 5, 10, and 15 mL ophthalmic dropper bottles; *Akbeta*®, *Betagan*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Levofloxacin Ophthalmic

(lev-oh-**flocks**-a sin)

Third-Generation Fluoroquinolone Antibiotic

Prescriber Highlights

- ▶ Used to treat confirmed bacterial conjunctivitis or keratitis based on culture and susceptibility; prophylactic use is not recommended.
- ▶ Physically incompatible with many drugs; therefore, it should not be directly mixed with other ophthalmic medications.
- ▶ Crystalline drug precipitates may be noted in the superficial portion of corneal defects when levofloxacin is used on the eye.

Uses/Indications

Levofloxacin ophthalmic may be used to treat bacterial conjunctivitis or keratitis caused by susceptible strains of *Staphylococcus* spp, *Streptococcus* spp, *Pseudomonas aeruginosa*, *Chlamydia* spp, and *Haemophilus* spp.¹ Levofloxacin is effective against other gram-positive and gram-negative organisms; however, its potency is not considered significantly greater than that of second-generation fluoroquinolones (eg, ciprofloxacin), particularly versus *Staphylococcus* spp.^{1,2} The increased clinical efficacy of levofloxacin may be afforded by increased penetration to ocular tissues.³

The World Health Organization (WHO) has designated levofloxacin as a Critically Important, Highest Priority antimicrobial in human medicine.²

Pharmacology/Actions

Levofloxacin is a third-generation fluoroquinolone antibiotic that acts via the inhibition of bacterial topoisomerase II (DNA gyrase), leading to impaired DNA replication and repair.

Contraindications/Precautions/Warnings

Levofloxacin should not be used prophylactically and is contraindicated in patients that have a history of hypersensitivity to it, other fluoroquinolones, or any component of the formulation. It should only be used as a topical ophthalmic. It should not be injected subconjunctivally or into the anterior chamber of the eye. Fluoroquinolone-induced retinal toxicity in domestic cats has not been demonstrated with this product.

Adverse Effects

Adverse effects reported in humans or animals include blurred vision, tearing, eye pain, redness, itching, and a bad taste in the mouth. Crystalline drug precipitates may be noted in the superficial portion of corneal defects when levofloxacin is used.⁴

Dosages

Conjunctivitis (extra-label): 1 drop in affected eye(s) every 2 to 8 hours

Keratitis (extra-label): 1 drop in affected eye(s) every 2 to 8 hours, or 0.1 mL in the subpalpebral lavage (SPL) catheter every 2 to 8 hours. May precipitate when mixed with other ophthalmic medications; therefore, it is important to flush SPL catheter well with air between medications.

Monitoring

- Corneal cytology and culture to evaluate for the presence of bacteria
- Clinical efficacy: improvement and resolution of clinical signs associated with bacterial infection
- If blepharospasm, uveitis, worsening of ulceration, or no improvement in infection occur within 7 days, reassess microbial susceptibility.

Client Information

- Use this medication as directed by your veterinarian.
- Crystals may appear on the surface of the eye during treatment with levofloxacin ophthalmic.
- Do not mix this medicine with other eye medications.
- Wait 5 minutes after applying this medicine before applying any other medications to the eye.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Your veterinarian will need to monitor your animal while it is receiving this medicine. Do not miss these important follow-up visits.
- Store this medication in the refrigerator or at room temperature. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solutions.

Storage/Stability

Store this product in the at controlled room temperature (15°C–25°C [59°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Fluoroquinolone antibiotics interact with many other drugs, including those containing cations (eg, iron, aluminum, magnesium, calcium). Levofloxacin should not be mixed directly (in vitro) with any other drugs. Levofloxacin is not available as an ophthalmic ointment.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Levofloxacin 0.5% ophthalmic solution (containing levofloxacin 5 mg/mL, benzalkonium chloride 0.005%, NaCl, and water. May also contain HCl and/or NaOH to adjust pH to 6.5 with an osmolality of ≈ 300 mOsm/kg), in a 5 mL ophthalmic dropper bottle; generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Lidocaine Ophthalmic

(līe-doe-kane)

Topical Ocular Anesthetic

For other uses of lidocaine, see *Lidocaine, Systemic*; *Lidocaine, Local Anesthetic*; *Lidocaine, Topical*

Prescriber Highlights

- ▶ Injectable local anesthetic that may be used for analgesia for ocular diagnostic and therapeutic procedures
- ▶ Should never be prescribed for at-home administration
- ▶ Dose- and species-dependent duration, with maximal effect generally between 1 and 5 minutes in dogs, cats, and horses

Uses/Indications

Lidocaine is a local anesthetic available as an injectable solution or an ophthalmic topical gel that can be administered topically to facilitate ophthalmic diagnostic and therapeutic procedures, including

measurement of intraocular pressure (tonometry), corneal and conjunctival sampling for cytology and culture, ocular ultrasonography, and gonioscopy. Lidocaine is indicated as an adjunctive anesthetic for performing therapeutic procedures in the hospital, including removal of foreign bodies and sutures from the cornea, eyelid and conjunctival surgeries, corneal grafting procedures, and cataract extraction. Lidocaine demonstrates maximal effect generally between 1 and 5 minutes, lasting around 20 to 25 minutes in dogs^{1,2} and ≈ 45 to 55 minutes in horses.^{3–5} The degree of anesthesia in normal canine eyes was comparable to that of topically applied bupivacaine and ropivacaine.⁶ Due to the likelihood of developing ocular surface toxicities with prolonged or repeated use, it should never be prescribed for use as an ocular analgesic.

Pharmacology/Actions

Lidocaine induces local anesthesia by inhibiting neuronal depolarization, thus preventing the generation of an action potential. This action occurs due to the binding of the local anesthetic molecule to the intracellular portion of the voltage-gated sodium channel within the neuron, thus inhibiting the intracellular movement of sodium. Variations in the structure of different local anesthetics influence their relative potencies. All local anesthetics have increased efficacy in alkaline environments due to improved intracellular penetration; efficacy is decreased in the acidic environment of inflammation.

Contraindications/Precautions/Warnings

Lidocaine is contraindicated in patients with a history of hypersensitivity to it or any component of the formulation. Prolonged use of a topical ocular anesthetic will likely produce significant corneal epithelial and stromal disease, as well as tear film instability, and may result in loss of vision.⁷

Adverse Effects

Instillation of lidocaine may result in local irritation (eg, blepharospasm, conjunctival hyperemia) several hours after administration. Patients sensitive to other local anesthetics may be less likely to react to proparacaine; however, local or systemic sensitivity may still occur and manifest as a pseudo-allergic (anaphylactoid) reaction mimicking an IgE-mediated vasodilatory event. Cardiovascular, respiratory, and CNS depressant effects that may result from other local anesthetics are extremely rare with the appropriate use of topical anesthetics. Care should be exercised, however, when administering to small veterinary patients (ie, puppies, kittens, “pocket pets”, amphibians) that may be more susceptible to the effects of systemically absorbed drug. Use as a therapeutic analgesic leads to superficial punctate keratitis, which may progress to diffuse necrotizing keratitis, severe infiltrative keratitis, persistent epithelial defects, or damage to Descemet’s membrane. Application of the topical ophthalmic and nonophthalmic gels in normal dog^{1,2} and horse³ eyes induced conjunctival hyperemia and punctate epithelial damage.

Dosages

Ophthalmic procedures requiring local anesthesia in small animal species (extra-label): 1 – 2 drops in the target eye(s) prior to examination or procedure. May repeat 1 drop every 5 to 10 minutes for 2 to 3 additional times. **Additional doses should be avoided in small patients (puppies, kittens, “pocket pets”, amphibians).**

Ophthalmic procedures requiring local anesthesia in horses (extra-label): Using an ophthalmic topical gel, 0.2 mL in the target eye(s) prior to examination or procedure.^{3,5,8} Onset of analgesia is within 1 to 5 minutes and duration is ≈ 45 to 55 minutes.

Monitoring

- Clinical effect. The onset of action indicated by an awake patient that is permitting diagnostic procedures to be performed and by a reduction in anesthetic requirements when administered pre-operatively

Client Information

- Animal owners must **not** administer lidocaine in the home. Its use is limited to diagnostic procedures performed in the veterinary clinical setting.

Storage/Stability

Store at controlled room temperature 15°C to 25°C (59°F–77°F) and protected from sunlight; do not freeze. Discard single-use containers after use.

Compatibility/Compounding Considerations

Lidocaine should not be mixed directly with any other ophthalmic drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Lidocaine HCl 3.5% ophthalmic gel (containing 35 mg lidocaine HCl per mL; hypromellose; NaCl; purified water; HCl, and/or NaOH to adjust pH), in 1 mL tubes and 5 mL in a 10 mL round plastic drop-per bottle; *Akten*® (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Lissamine Green Ophthalmic

(lis-ah-meen) *Green Glo*®

Ocular Diagnostic Agent

Prescriber Highlights

- ▶ Vital dye that demonstrates corneal and conjunctival epithelial cell damage, as well as deficiency in ocular surface mucins
- ▶ Less stinging on administration than is associated with rose bengal
- ▶ Requires broader interpretive experience than fluorescein staining
- ▶ Unapproved drug with periodic market shortages

Uses/Indications

Lissamine green is used as a diagnostic agent when superficial corneal or conjunctival epithelial damage is suspected. Lissamine green does not sting on application like rose bengal, but interpretation of the results requires broader evaluative experience than that for rose bengal and fluorescein. A grading scale that implements slit-lamp biomicroscopy to evaluate the regions of uptake is used in humans, and lissamine green has recently been demonstrated to be an effective adjunctive diagnostic test for the evaluation of qualitative and quantitative tear film deficiencies in dogs.^{1,2}

Pharmacology/Actions

Lissamine green stains the cornea and conjunctiva blue upon instillation, resulting in speckling of the cornea and conjunctiva if ulcerations or dry patches from muco-deficient or damaged epithelial cells are present.

Contraindications/Precautions/Warnings

Lissamine green is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Lissamine green has demonstrated antimicrobial activity against gram-positive and gram-negative organisms; thus, samples for bacterial culture should be collected before administration of lissamine green.³

Adverse Effects

Adverse effects of lissamine green reported in humans or animals include stinging on instillation.

Dosages

Detection of corneal change: Wet the surface of the lissamine green

strip with 1 to 2 drops of normal saline solution. Apply moistened tip to the conjunctiva or fornix as required. Allow the patient to blink several times after application to disperse the dye across the surface of the eye. Wait 60 seconds, then read with a white light source. Blue speckling will be evident for areas of the damaged or muco-deficient cornea.

Monitoring

- Lissamine green ophthalmic is a diagnostic agent. Monitoring is not required.

Client Information

- Lissamine green is not used at home by animal caregivers.

Storage/Stability

Store this product in a well-ventilated place protected from moisture and light; do not freeze.⁴

Compatibility/Compounding Considerations

Lissamine green ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Lissamine Green 1.5 mg Strips (containing 1.5 mg of lissamine green per strip), in individually wrapped strips in containers of 100 strips; *Green Glo*®; (Rx). There are no FDA-approved forms of lissamine green strips.

References

For the complete list of references, see wiley.com/go/budde/plumb

Lodoxamide Ophthalmic

(loe-**doks**-a-mide) *Alomide*®

Mast Cell Stabilizer

Prescriber Highlights

- ▶ Topical mast cell stabilizer used to treat seasonal conjunctivitis

Uses/Indications

Lodoxamide may be used to treat allergic conjunctivitis. No studies describing the efficacy and safety of lodoxamide after topical administration have been performed in veterinary patients.

Pharmacology/Actions

Lodoxamide is a mast cell stabilizer with efficacy in treatment of type I hypersensitivity reactions (IgE-mediated). Mast cell inhibition decreases vascular permeability, thus diminishing the signs associated with conjunctivitis. It does not possess vasoconstrictor, anti-histaminic, cyclooxygenase inhibition, or other anti-inflammatory activity.

Contraindications/Precautions/Warnings

Lodoxamide is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.¹

Adverse Effects

Adverse effects reported in humans include transient burning, stinging, or discomfort on instillation.¹ Additional adverse effects that were infrequently reported include ocular itching, blurred vision, dry eye, tearing, discharge, hyperemic, crystalline deposits, foreign body sensation, corneal erosion/ulcer, lid/lash scales, eye pain, ocular edema/swelling, ocular warming sensation, ocular fatigue, chemosis, anterior chamber cells, keratopathy/keratitis, blepharitis, allergy, stick sensation, and epitheliopathy. Nonocular adverse events that were infrequently reported include headache, heat sensation, dizziness, somnolence, nausea, stomach discomfort, sneezing, dry nose, and rash.

Dosages

Conjunctivitis (extra-label): 1 drop in the affected eye(s) 4 times daily for up to 3 months

Monitoring

- Clinical efficacy: resolution of conjunctival redness, itching, and discomfort

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes or if there are particles in the solution.

Storage/Stability

Store this product at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight.

Compatibility/Compounding Considerations

Lodoxamide ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Lodoxamide Tromethamine Ophthalmic Solution 0.1% (containing lodoxamide tromethamine 1.78 mg, equivalent to 1 mg lodoxamide; benzalkonium chloride 0.007%; mannitol; hypromellose 2910; sodium citrate; tyloxapol; citric acid; edetate disodium; NaOH and/or HCl to adjust pH; purified water), in sterile 10 mL ophthalmic dropper bottle; *Alomide*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Loteprednol Ophthalmic

(lote-uh-**pred**-nole) Lotemax®, Alrex®, Zylet®, Eysuvis®, Inveltys®
Corticosteroid Anti-Inflammatory Agent

Prescriber Highlights

- ▶ A topical corticosteroid used to treat ocular surface and anterior segment inflammatory conditions
- ▶ Do not use in bacterial, fungal, or viral keratitis or if corneal abrasion or ulceration are suspected.
- ▶ Use with caution in avian species due to risk for systemic corticosteroid effects.

Uses/Indications

Loteprednol ophthalmic is used to treat inflammatory conditions of the palpebral and bulbar conjunctiva, cornea, and anterior segment of the globe (eg, blepharitis, conjunctivitis, keratitis, anterior uveitis). In humans with dry eye syndrome, loteprednol has demonstrated benefit in the initial management of the condition before initiation or in conjunction with topically administered cyclosporine.^{1,2} No studies describing the efficacy and safety of loteprednol after topical administration have been performed in veterinary patients.

Pharmacology/Actions

Corticosteroids have broad immunomodulatory and anti-inflammatory actions, which are mediated via direct effects on gene expression,

as well as by nongenomic effects on cell membranes and receptors. Ultimately, corticosteroids decrease pro-inflammatory gene transcription, production of pro-inflammatory prostaglandins and leukotrienes, B- and T-cell number and function, cell-mediated immunity, and humoral immunity.³ The specific mechanisms involved in regulating ocular inflammation are not known. Loteprednol is a “soft” steroid (a metabolically labile moiety that undergoes hydrolysis to its active form within the cornea, thus allowing local anti-inflammatory effects to predominate relative to harmful adverse effects).

Contraindications/Precautions/Warnings

Loteprednol is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Topical corticosteroids are contraindicated in known or suspected corneal ulceration or corneal infection (eg, bacterial, fungal, or viral). Use loteprednol with caution in patients with diabetes mellitus or other endocrine diseases, infectious diseases, chronic renal failure, congestive heart failure, systemic hypertension, or gastric ulceration. When use of loteprednol is necessary in these conditions, the minimal effective dose should be administered for the shortest possible time.

Adverse Effects

Adverse effects reported in humans include development of cataracts (via altered glucose or lenticular metabolic effects), increased intraocular pressure (IOP; via activation of glucocorticoid receptors in the trabecular meshwork), infection (via immunosuppression), decreased wound healing (via increased collagenolytic activity), mydriasis, and calcific keratopathy. Increased IOP and development of cataracts are minimized with loteprednol relative to other topically administered corticosteroids.^{4–6} Systemic effects are rare but include hepatopathy, suppression of endogenous glucocorticoid production, and focal alopecia. Concurrent use of systemic steroids increases the risk for these adverse effects.

Dosages

Conjunctivitis (extra-label): 1 drop or ¼ inch strip in the affected eye(s) every 4 to 8 hours, then gradually decrease the dosage, provided clinical improvement is maintained

Keratitis (extra-label): 1 drop or ¼ inch strip in the affected eye(s) every 4 to 8 hours, then gradually decrease the dosage, provided clinical improvement is maintained

Keratoconjunctivitis sicca (extra-label): 1 drop in the affected eye(s) 4 times daily for 2 weeks

Anterior uveitis (extra-label): 1 drop or ¼ inch strip in the affected eye(s) every 4 to 8 hours, then gradually decrease the dosage, provided clinical improvement is maintained

Perioperative cataract removal (extra-label): 1 drop in the surgical eye(s) 4 times daily, beginning 24 hours after surgery, continuing for 2 weeks

Monitoring

- Clinical efficacy: resolution of inflammation
- Monitor intraocular pressure (IOP) and observe for signs of infection or delayed wound healing
- Monitor for blood and/or urine glucose changes in diabetic animals

Client Information

- Use this medicine as directed by your veterinarian.
- Shake suspensions well before using.
- Use proper administration techniques to avoid contamination of the medicine.
- Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medicine before applying any other medicines to the eye.

- Store this medicine at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes or if there are particles in the solutions.

Storage/Stability

Store this product at room temperature (15°C–25°C [59°F–77°F]) away from moisture and sunlight.

Compatibility/Compounding Considerations

Loteprednol ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Loteprednol Etabonate 0.2% Ophthalmic Suspension (containing loteprednol etabonate 2 mg, edetate disodium, glycerin, povidone, purified water, tyloxapol, HCl and/or NaOH may be added to adjust the pH, and benzalkonium chloride 0.10%, as an isotonic suspension of 250 – 310 mOsm/kg), in 10 mL sterile ophthalmic dropper bottle; *Alrex*®; (Rx)

Loteprednol Etabonate 0.25% Ophthalmic Suspension (containing loteprednol etabonate 2.5 mg, glycerin, sodium citrate dihydrate, sodium chloride, poloxamer 407, edetate disodium dihydrate, citric acid, water for injection, and benzalkonium chloride 0.01%), in 10 mL sterile ophthalmic dropper bottle; *Eysuvis*®; (Rx)

Loteprednol Etabonate 0.38% Ophthalmic Gel (containing loteprednol etabonate 3.8 mg, boric acid, edetate disodium dihydrate, glycerin, hypromellose, poloxamer, polycarbophil, propylene glycol, sodium chloride, water for injection, sodium hydroxide to adjust pH, and benzalkonium chloride 0.003%), in 10 mL ophthalmic dropper bottle; *Lotemax SM*®; (Rx)

Loteprednol Etabonate 0.5% Ophthalmic Gel (containing loteprednol etabonate 5 mg, boric acid, edetate disodium dehydrate, glycerin, polycarbophil, propylene glycol, NaCl, tyloxapol, water for injection, NaOH to adjust pH to 6 to 7, and benzalkonium chloride 0.003%), in sterile 5 g ophthalmic tube; *Lotemax*®; (Rx)

Loteprednol Etabonate Ophthalmic Ointment 0.5% (containing loteprednol etabonate 5 mg, white petrolatum, and mineral oil), in a sterile 3.5 g ophthalmic tube; *Lotemax*®; (Rx)

Loteprednol Etabonate 0.5% Ophthalmic Suspension (containing loteprednol etabonate 5 mg, edetate disodium, glycerin, povidone, purified water, tyloxapol, HCl and/or NaOH to adjust the pH, and benzalkonium chloride 0.01%, as an isotonic suspension of 250 – 310 mOsm/kg), in 5 mL, 10 mL, and 15 mL ophthalmic dropper bottles; *Lotemax*®, generic; (Rx)

Loteprednol Etabonate 1% Ophthalmic Suspension (containing loteprednol etabonate 10 mg, glycerin, sodium citrate dihydrate, poloxamer 407, sodium chloride, edetate disodium dihydrate, citric acid, water for injection, and benzalkonium chloride 0.01%), in 2.8 mL sterile ophthalmic dropper bottle; *Inveltys*®; (Rx)

Loteprednol Etabonate 0.5% and Tobramycin 0.3% Ophthalmic Suspension (containing loteprednol etabonate 5 mg, tobramycin 3 mg, edetate disodium, glycerin, povidone, purified water, tyloxapol, sulfuric acid and/or NaOH to adjust the pH, and benzalkonium chloride, as an isotonic suspension of 260 – 320 mOsmol/kg), in sterile 5 mL and 10 mL ophthalmic dropper bottles; *Zylet*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Miconazole Ophthalmic

(mye-**kon**-uh-zoh)

Azole Antifungal Agent

Prescriber Highlights

- ▶ Used to treat ocular surface fungal infections
- ▶ Not commercially available as an ophthalmic preparation; must be compounded
- ▶ Penetrates the cornea more efficiently than natamycin

Uses/Indications

Miconazole ophthalmic solution is primarily used to treat fungal keratitis in horses; however, it has been successfully used to treat keratomycosis in a dog.¹ Miconazole has a broad antifungal spectrum against most fungi and yeasts of veterinary interest. Sensitive organisms include *Blastomyces dermatitidis*, *Paracoccidioides brasiliensis*, *Histoplasma capsulatum*, *Candida* spp, *Coccidioides immitis*, *Cryptococcus neoformans*, and *Aspergillus fumigatus*. Some *Aspergillus* spp and *Fusarium* spp are only marginally sensitive.

Pharmacology/Actions

Miconazole interacts with 14- α -sterol demethylase, a cytochrome P-450 enzyme involved in the conversion of lanosterol to ergosterol. Ergosterol is an essential component of the fungal cell membrane; preventing its synthesis increases cellular permeability and causes leakage of cellular contents. Other **antifungal** effects of miconazole include inhibition of endogenous respiration, purine uptake, the transformation of yeasts to mycelial forms, and impairment of triglyceride and phospholipid biosynthesis.

Miconazole has a wide antifungal spectrum against most fungi and yeasts of veterinary interest; however, susceptibility of isolates of normal ocular surface microflora and isolates from horses with fungal keratitis indicates variable efficacy. Thus, culture and susceptibility testing should be performed in horses with cytologically confirmed or suspected fungal keratitis.^{1–5}

Contraindications/Precautions/Warnings

Miconazole is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans or animals include transient burning or stinging on application. Miconazole may cause local irritation if it is injected subconjunctivally. Although systemic absorption of topically applied solutions is unlikely, systemic exposure to miconazole has resulted in hepatotoxicity and hearing loss in humans. Miconazole had more cytotoxic effects on cultured corneal epithelial cells than itraconazole, but miconazole had less cytotoxic effects than natamycin.⁶

Dosages

Equine fungal keratitis (extra-label): 1 – 2 drops to the affected eye(s) or 0.1 mL in the subpalpebral lavage (SPL) catheter every 2 to 3 hours initially, then taper the frequency based on clinical response

Monitoring

- Clinical efficacy: improvement and resolution of clinical signs associated with the fungal infection
- Corneal scrapings with cytology and cultures to determine the clearing of fungal organisms

Client Information

- Use this medication as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.

- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Your veterinarian will need to monitor your animal while it is receiving this medicine. Do not miss these important follow-up visits.
- Store this medication as directed on the product label. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in solutions.

Storage/Stability

Store this product in the refrigerator (2°C-8°C [36°F-46°F]) or at controlled room temperature (20°C- 25°C [68°F-77°F]); check individual labels for details, as recommendations vary by product. Do not freeze. Keep product away from moisture and sunlight.

Compatibility/Compounding Considerations

Miconazole should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Miconazole has been compounded into a 1% ophthalmic solution that mimics the formerly commercially available *Monistat IV*®. Miconazole topical creams and ointments for dermatological use should **NOT** be used in the eye.

References

For the complete list of references, see wiley.com/go/budde/plumb

Mitomycin Ophthalmic

(*mye-toe-mye-sin*) Mitosol®, Mitomycin C
Cytotoxic Agent

Prescriber Highlights

- ▶ Antifibrotic agent used adjunctively in periocular surgeries (most commonly used in glaucoma implant procedures)
- ▶ Antineoplastic agent used for susceptible periocular and ocular surface tumors
- ▶ Product is normally blue in color.
- ▶ May cause periocular alopecia following treatment
- ▶ The National Institute for Occupational Safety and Health (NIOSH) classifies mitomycin as a hazardous drug; use appropriate precautions when handling.

Uses/Indications

Because of its ability to reduce fibroblastic proliferation and prevent subsequent scarring, mitomycin is used adjunctively in ocular surgeries, particularly to treat glaucoma by placement of drainage shunts or alternation of the natural aqueous humor outflow tracts; however, efficacy is unknown.¹⁻³ Mitomycin is also used as an adjunct to surgical removal for treatment of cutaneous and ocular surface neoplastic conditions (eg, squamous cell carcinoma, hemangiosarcoma) to prevent recurrence.⁴⁻⁸

Pharmacology/Actions

Mitomycin binds DNA, which leads to cross-linking of DNA and subsequent inhibition of DNA synthesis and function. Evidence suggests that at high concentrations of the drug, cellular RNA and protein synthesis are also suppressed.

Contraindications/Precautions/Warnings

Mitomycin is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Mitomycin is contraindicated in patients with thrombocytopenia, coagulation

disorder, or an increase in bleeding tendency due to other causes. Pregnant women or women attempting to become pregnant should avoid contact with this drug.

The National Institute for Occupational Safety and Health (NIOSH) classifies mitomycin as a hazardous drug; personal protective equipment (PPE) should be used accordingly to minimize the risk for exposure. Refer to the Occupational Safety and Health Association (OSHA) website for more information.

Adverse Effects

Adverse effects reported in humans or animals include burning and stinging on application, corneal irritation, and edema formation.⁴ Periocular alopecia has been reported in horses after topical application.⁵ Topical administration in the presence of corneal epithelial defects has been associated with stromal ulceration and bullous keratopathy.⁵ The systemic use of mitomycin has been associated with myelosuppression, particularly thrombocytopenia and leukopenia, nephrotoxicity, and pulmonary toxicity, although these effects are unlikely with topical therapy.

Dosages

Equine periocular squamous cell carcinoma (extra-label): ½ inch strip of a 0.04% ointment to the affected site 3 times daily for 21 days in a cycle of 7 days of drug application, followed by a 7-day drug vacation.^{5,6} If using a subpalpebral lavage (SPL) system, instill 0.1 mL of a 0.04% solution 3 times daily for 21 days in a cycle of 7 days of drug application, followed by a 7-day drug vacation; maintain catheter patency by flushing the line with air after each dose.

Adjunctive antifibrotic agent for glaucoma implant surgery (extra-label): A single, 5-minute intraoperative application of 0.5 mg/mL solution at the filtration site following surgical placement of the shunt.¹ **NOTE:** Multiple protocols exist for use during surgical procedures for correcting glaucoma, varying with the type of procedure and other patient parameters. As these procedures should be performed by board-certified veterinary ophthalmologists, the protocols are not detailed here.

Monitoring

- Clinical efficacy depending on indication:
 - Maintenance of patency of the glaucoma implant, leading to control of intraocular pressure (IOP)
 - Cell or tissue samples for evaluation to confirm remission of neoplasia

Client Information

- This medicine is a chemotherapy drug. Pregnant women and sick or immunocompromised individuals should avoid contact with this medicine.
- Personal protective equipment (eg, gloves, mask) should be worn when giving this medicine to your animal. Dispose of any waste in a doubled plastic bag, then place the bag in a secure disposal location that children and animals cannot access.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine in the refrigerator. Do not freeze.
- Do not use this medicine if the color changes (it is normally blue), if it becomes cloudy, or if there are particles in the solution.
- This medication is considered to be a hazardous drug as defined by the National Institute for Occupational Safety and Health (NIOSH). Talk with your veterinarian or pharmacist about the use of personal protective equipment when handling this medicine.

Storage/Stability

Store this product in the refrigerator (2°C–8°C [36°F–46°F]), away from moisture and sunlight. Store compounded products as stated on product label.

Compatibility/Compounding Considerations

Mitomycin ophthalmic preparations should not be mixed directly with any other ophthalmic medications. Mitomycin ophthalmic solution is not commercially available, but mitomycin lyophilized powder for injection can be compounded into appropriate ophthalmic solutions as follows:

- If using the 5 mg vial, reconstitute vial with 12.5 mL sterile 0.9% sodium chloride injection to make a 0.4 mg/mL (0.04%) solution.
- If using the 20 mg vial, reconstitute with 50 mL sterile 0.9% sodium chloride injection to make a 0.4 mg/mL (0.04%) solution.

NOTE: Commercially available mitomycin 20 mg vials may not contain 50 mL volume. Use caution when reconstituting, and when volume restrictions require, aseptically transfer contents into a 50 mL empty sterile vial.

Mitomycin ophthalmic ointments are not commercially available but may be prepared by qualified compounding pharmacists.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Mitomycin ophthalmic topical solution for multi-use is not commercially available, but mitomycin lyophilized powder for injection (5 mg or 20 mg vial) can be compounded into a suitable ophthalmic. A commercially available kit for mitomycin ophthalmic topical (single use) as an adjunct to ab externo glaucoma surgery is available; *Mitosol*®, generic; (Rx).

References

For the complete list of references, see wiley.com/go/budde/plumb

Morphine Ophthalmic

(*mor*-feen)

Opioid

Prescriber Highlights

- ▶ Opioid analgesic that may be beneficial in the treatment of ocular pain
- ▶ Must be compounded by a qualified compounding pharmacist, as no ophthalmic dosage forms are commercially available
- ▶ Do not use in animals receiving naloxone, nalbuphine, or nalorphine, as these drugs have opioid antagonist properties.
- ▶ DEA Schedule II (C-II) controlled substance

Uses/Indications

Morphine ophthalmic solution may be used to manage ocular pain. Dogs with experimentally induced corneal ulcers that received morphine solution 3 times daily experienced a reduction in signs of ocular pain without prolonged healing times¹; however, in a clinical trial of dogs and cats with naturally occurring corneal ulcers, a single drop of morphine solution did not improve clinical signs of pain.² Combined topical administration and subconjunctival injection did not appear to provide ocular pain relief (as evaluated by changes in blepharospasm, conjunctival hyperemia, blinking rates, tearing) in dogs following phacoemulsification surgery.³ Topical morphine did not affect corneal sensitivity when it was administered to normal equine eyes.^{4,5}

Pharmacology/Actions

Morphine is primarily an opioid mu receptor agonist that interacts with mu binding sites located in the brain, as well as within the spinal

cord and on peripheral sensory nerves. They are present in the cornea in smaller numbers than delta receptors in the canine cornea.¹ The analgesic effects on the cornea can be variable. It is possible that receptor specificity and localization, frequency of administration, and location of pain stimulus relative to drug penetration all impact clinical efficacy. Topical ocular morphine administration to eyes of normal horses results in measurable intraocular and systemic morphine concentrations, without local or systemic adverse effects.⁴

Contraindications/Precautions/Warnings

Morphine ophthalmic is contraindicated in patients with a history of hypersensitivity to it or any component of the formulation. Adverse effects may occur following concomitant use with monoamine oxidase inhibitors (eg, amitraz, linezolid, selegiline) or within 14 days of such treatment. Because of potential systemic absorption, morphine ophthalmic should be used with caution in patients with respiratory insufficiency or depression; severe CNS depression; bronchoconstrictive disorders (eg, asthma); heart failure secondary to pulmonary hypertension; arrhythmias; increased intracranial or cerebrospinal pressure (eg, head injury, brain tumor); or seizures.

Adverse Effects

Adverse effects reported in humans or animals include stinging and burning on application. Although systemic adverse effects are unlikely with topical administration, they could occur. See *Morphine*.

Dosages

Corneal ulcers: 1 drop of a morphine 1% solution in the affected eye(s) every 8 hours

Monitoring

- Clinical efficacy: control of signs of ocular discomfort (eg, blepharospasm, conjunctival hyperemia, tearing)

Client Information

- Use this medicine as directed by your veterinarian.
- This medicine is a controlled substance and may not be refilled without a new prescription.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine as directed on package label. Keep product away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product as directed on the package labeling away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Morphine ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Morphine sulfate ophthalmic solution is not commercially available; it must be compounded. Preservative-free morphine sulfate injection may be aseptically diluted with sterile 0.9% sodium chloride injection to achieve a 1% (10 mg/mL) aseptic ophthalmic solution.

References

For the complete list of references, see wiley.com/go/budde/plumb

Moxifloxacin Ophthalmic

(mox-ih-**flox**-uh-sin) Vigamox®, Moxeza®
Fourth-Generation Fluoroquinolone Antimicrobial

Prescriber Highlights

- ▶ Used to treat confirmed bacterial conjunctivitis or keratitis based on culture and susceptibility; prophylactic use is not recommended
- ▶ Physically incompatible with many drugs; therefore, it should not be directly mixed with other ophthalmic medications
- ▶ Crystalline drug precipitates may be noted in the superficial portion of corneal defects when moxifloxacin is used on the eye.

Uses/Indications

Moxifloxacin ophthalmic may be used to treat bacterial conjunctivitis or keratitis caused by susceptible strains of *Staphylococcus* spp, *Streptococcus* spp, *Pseudomonas aeruginosa*, *Chlamydomydia* spp, and *Haemophilus* spp. Moxifloxacin is effective against other gram-positive and gram-negative organisms. *Pseudomonas aeruginosa* isolates from dogs with bacterial keratitis showed slightly greater resistance in vitro to moxifloxacin than to other fluoroquinolones,¹ whereas *Staphylococcus pseudintermedius* isolates from dogs with bacterial keratitis demonstrated lower rates of resistance in vitro to moxifloxacin than to other fluoroquinolones.² Penetration of topical ophthalmic moxifloxacin into the tears, cornea, and aqueous humor of horses with normal eyes was significantly greater than with the second-generation fluoroquinolone ciprofloxacin.^{3,4}

The World Health Organization (WHO) has designated moxifloxacin as a Critically Important, Highest Priority antimicrobial in human medicine.²

Pharmacology/Actions

Moxifloxacin is a fourth-generation fluoroquinolone antibiotic that acts via inhibition of bacterial topoisomerase II (DNA gyrase) and topoisomerase IV, leading to impaired DNA replication and repair. The inhibition of both enzymes decreases the rates of resistance as compared to earlier generations of fluoroquinolones.

Contraindications/Precautions/Warnings

Moxifloxacin is contraindicated in patients that have a history of hypersensitivity to moxifloxacin, to other fluoroquinolones, or to any of the components of this medication. This medication should only be used as a topical ophthalmic; it should not be injected subconjunctivally or into the anterior chamber of the eye. Moxifloxacin should not be used prophylactically. Fluoroquinolone-induced retinal toxicity in domestic cats has not been demonstrated with this product.

Adverse Effects

Adverse effects reported in humans or animals include blurred vision, tearing, eye pain, redness, itching, and a bad taste in the mouth. Crystalline drug precipitates may be noted in the superficial portion of corneal defects when moxifloxacin is used.

Dosages

Conjunctivitis (extra-label): 1 drop in the affected eye(s) every 2 to 8 hours

Keratitis (extra-label): 1 drop in the affected eye(s), or 0.1 mL in the subpalpebral lavage (SPL) catheter, every 2 to 8 hours. May precipitate when mixed with other ophthalmic medications; therefore, it is important to flush SPL catheter well with air between medications

Monitoring

- Corneal cytology and culture to evaluate for the persistence of bacteria
- Clinical efficacy: improvement and resolution of clinical signs associated with bacterial infection

- If blepharospasm, uveitis, worsening of ulceration, or no improvement in infection occur within 7 days, reassess microbial susceptibility.

Client Information

- Use this medication as directed by your veterinarian.
- Crystals may appear on the surface of the eye during treatment with moxifloxacin ophthalmic.
- Do not mix this medicine with other eye medications.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Your veterinarian will need to monitor your animal while it is receiving this medicine. Do not miss these important follow-up visits.
- Store this medication in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in solutions.
- Dispose of unused medication in regular trash; do not put it into the sewer system.

Storage/Stability

Store this product in the refrigerator (2°C-8°C [36°F-46°F]) or at controlled room temperature (20°C-25°C [68°F-77°F]); check individual labels for details as recommendations vary by product. Do not freeze. Keep product away from moisture and sunlight.

Compatibility/Compounding Considerations

Fluoroquinolone antibiotics interact with many other drugs, including those containing cations (eg, iron, aluminum, magnesium, or calcium). Moxifloxacin should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Moxifloxacin 0.5% Ophthalmic Solution (containing moxifloxacin HCl 5.45 mg, boric acid, NaCl, purified water, and HCl and/or NaOH to adjust pH), 3 mL in sterile ophthalmic dropper bottle; Vigamox®, generic; (Rx)

Moxifloxacin 0.5% Ophthalmic Solution (containing NaCl, xanthan gum, boric acid, sorbitol, tyloxapol, purified water, and HCl and/or NaOH to adjust pH), 3 mL in sterile ophthalmic dropper bottle; Moxeza®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Nalbuphine Ophthalmic

(nal-byoo-feen) Nubain®
Opioid Agonist/Antagonist Analgesic Agent

Prescriber Highlights

- ▶ Used to provide ocular analgesia in patients
- ▶ Do not use in combination with other opioid ocular analgesics (eg, morphine sulfate).

Uses/Indications

Nalbuphine may be compounded as an ophthalmic preparation to provide ocular surface analgesia. In dogs, nalbuphine 1% administered 3 times daily after the creation of a corneal epithelial wound

was not effective as an analgesic,¹ whereas nalbuphine 0.8% administered 3 times daily following phacoemulsification reduced corneal sensitivity and signs of ocular discomfort.² Corneal sensitivity was not decreased in normal horses following a single administration of 1.2% nalbuphine.³ Additionally, 1.2% nalbuphine did not decrease corneal sensation in rabbits.⁴

Pharmacology/Actions

Nalbuphine is a synthetic opioid agonist/antagonist analgesic of the phenanthrene series. Nalbuphine's analgesic potency is essentially equivalent to that of morphine on a milligram basis. The opioid antagonist activity of nalbuphine is one-fourth as potent as nalorphine and 10 times as potent as pentazocine. Nalbuphine, by itself, has potent opioid antagonist activity at doses equal to or lower than its analgesic dose. When nalbuphine is administered following or concurrently with mu-agonist opioid analgesics (eg, morphine, oxymorphone, fentanyl), it may partially reverse or block opioid-induced respiratory depression from the mu-agonist analgesic.⁵

Contraindications/Precautions/Warnings

Nalbuphine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Nalbuphine is a partial opioid antagonist and should not be used in patients receiving other opioid agents as analgesia may be reversed by nalbuphine. When nalbuphine is used systemically, it has potential to depress respiratory and cardiac systems; however, these effects are unlikely when the drug is used topically.

Adverse Effects

Adverse effects reported in humans or animals include burning, stinging, and irritation on application. Systemic adverse effects, although unlikely following topical use, include respiratory and cardiovascular depression, somnolence, and ataxia.

Dosages

Corneal analgesia (extra-label): 1 drop in the affected eye(s) every 8 hours

Monitoring

- Clinical efficacy as demonstrated by adequate ocular analgesia

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wear gloves when administering the medicine to avoid inadvertent contact with it.
- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine as directed on the package labeling, away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solutions.

Storage/Stability

Because this medication must be compounded, storage recommendations can vary; follow storage requirements listed on the packaging, away from moisture and sunlight. Do not freeze.

Compatibility/Compounding Considerations

Nalbuphine ophthalmic preparations should not be mixed directly with any other ophthalmic medications. Nalbuphine is not commercially available and must be compounded by a qualified compounding pharmacist.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Nalbuphine is not commercially available as an ophthalmic preparation but may be compounded by qualified compounding pharmacists into 0.8% to 1.2% ophthalmic solution.

References

For the complete list of references, see wiley.com/go/budde/plumb

Natamycin Ophthalmic

(nat-uh-**mye**-sin) Natacyn®

Polyene Antifungal

Prescriber Highlights

- ▶ Commercially available ophthalmic suspension used to treat fungal keratitis, primarily associated with *Fusarium* spp
- ▶ Poorly soluble and does not efficiently penetrate corneal epithelium following topical administration
- ▶ Well tolerated following topical administration
- ▶ Natamycin suspensions may block subpalpebral lavage (SPL) catheters; therefore, ensure catheter patency by flushing the line with air after each dose.
- ▶ Expensive

Uses/Indications

Natamycin is primarily used to treat equine fungal keratitis caused by *Fusarium* spp. As natamycin is not as effective for deep stromal lesions, efficacy may be enhanced by creation of superficial ulceration (ie, therapeutic penetrating keratoplasty) to increase transcorneal penetration.¹

Pharmacology/Actions

Natamycin is a tetraene polyene antibiotic derived from *Streptomyces natalensis*. It possesses in vitro activity against various yeasts and filamentous fungi, including *Candida* spp, *Aspergillus* spp, *Cephalosporium* spp, *Fusarium* spp, and *Penicillium* spp; culture and susceptibility testing is recommended to ensure appropriate therapy.²⁻⁴ The mechanism of action appears to be through binding to the sterol moiety of the fungal cell membrane, thus altering membrane permeability and allowing leakage of essential fungal intracellular constituents. Although the activity against fungi is dose-related, natamycin is predominantly fungicidal.

Contraindications/Precautions/Warnings

Natamycin is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.⁵

Adverse Effects

Adverse effects reported in humans or animals include corneal opacity, eye discomfort, eye edema, eye hyperemia, foreign body sensation, paresthesia, and tearing.⁵ Allergic reactions, dyspnea, and chest pain have been reported in humans that were administered natamycin.

Dosages

Equine fungal keratitis (extra-label): 0.1 mL to the affected eye(s) every 2 to 6 hours, depending upon the severity of infection (can be used more frequently initially). If administering via subpalpebral lavage (SPL) catheter, maintain catheter patency by flushing the line with air after each dose.

Monitoring

- Corneal scrapings with cytology and culture and susceptibility testing to determine the clearance of fungal organisms
- Resolution of clinical signs of infection

Client Information

- Give this medicine to your animal as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.

- Shake the medication well before using.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication at room temperature away from moisture and sunlight; do not freeze.
- Avoid exposure to light and excessive heat.
- Do not use this medication if the color changes.

Storage/Stability

Store this medication at 2°C to 24°C (36°F–75°F) away from moisture and sunlight (dispensed in a brown bottle to minimize exposure to light); do not freeze.⁵

Compatibility/Compounding Considerations

Natamycin should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Natamycin 5% Ophthalmic Suspension (containing natamycin 50 mg/mL, benzalkonium chloride 0.02%, NaOH and/or HCl to adjust the pH to 5 to 7.5, and purified water), in a 15 mL amber glass dropper bottle; *Natascyn*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Nedocromil Ophthalmic

(ne-doe-**krow**-mill) *Alocril*®

Mast Cell Stabilizing Ophthalmic Agent

Prescriber Highlights

- Topical mast cell stabilizer used to treat allergic conjunctivitis

Uses/Indications

Nedocromil may be used for the treatment of ocular itching associated with allergic conjunctivitis. The efficacy of nedocromil in animals following topical ocular administration has not been evaluated.

Pharmacology/Actions

Nedocromil is a mast cell stabilizer and inhibits histamine release from mast cells. Decreased chemotaxis and decreased activation of eosinophils (anti-inflammatory effects) have also been demonstrated following administration of nedocromil.¹

Contraindications/Precautions/Warnings

Nedocromil is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans include ocular burning, irritation and stinging, unpleasant taste, and nasal congestion. Approximately 40% of humans report headaches and a bad taste in their mouths after use.² Other less common adverse effects associated with nedocromil administration include asthma, conjunctivitis, eye redness, photophobia, and rhinitis.

Dosages

Allergic conjunctivitis (extra-label): 1 drop in the affected eye(s) twice daily. Treatment should be continued throughout exposure (ie, until exposure to the offending allergen is terminated), even when clinical signs are absent.

Monitoring

- Clinical efficacy: resolution of signs of allergic conjunctivitis

Client Information

- Use this medicine as directed by your veterinarian.

- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product in the refrigerator (2°C–8°C [35.6°F–46.4°F]) or at controlled room temperature (20°C–25°C [68°F–77°F])² away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Nedocromil ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Nedocromil 2% Ophthalmic Solution (containing nedocromil 20 mg/mL; benzalkonium chloride 0.01%; edetate disodium 0.05%; purified water; NaCl 0.5%, buffered to a pH of 4 to 5.54 with an osmolality range of 270 – 330 mOsm/kg), in a 5 mL ophthalmic dropper bottle; *Alocril*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Nepafenac Ophthalmic

(ne-**paf**-en-ak) *Nevanac*®, *Ilevro*®

Nonsteroidal Anti-Inflammatory Drug (NSAID)

Prescriber Highlights

- Used to control anterior segment inflammation, particularly in association with surgical removal of cataracts
- Prodrug to amfenac, which is bioavailable to the posterior segment
- May increase risk for intraocular bleeding, inhibit corneal wound healing, and promote corneal melting

Uses/Indications

Nepafenac is used to decrease inflammation following cataract surgery. It may also be used to control other anterior segment inflammatory conditions (eg, conjunctivitis, keratitis, anterior uveitis).¹ Studies evaluating efficacy in veterinary patients have not been performed.

Pharmacology/Actions

Nepafenac is an NSAID and analgesic prodrug. After topical ocular dosing, nepafenac penetrates the cornea, where it is converted by ocular tissue hydrolases to amfenac, which inhibits the action of prostaglandin H synthase (cyclooxygenase), an enzyme required for prostaglandin production. Prostaglandins mediate many of the adverse effects of intraocular inflammation, including disruption of the blood-aqueous humor barrier, vasodilation, increased vascular permeability, leukocytosis, and alterations in intraocular pressure (IOP). Blocking prostaglandin production, therefore, minimizes the adverse short- and long-term effects of intraocular inflammation.

Contraindications/Precautions/Warnings

Nepafenac is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation or other NSAIDs. Caution should be employed when using nepafenac in patients that have previously exhibited sensitivity to other NSAID drugs, as there

is potential for cross-sensitivity. Nepafenac has been associated with increased bleeding of ocular tissues (including hyphema) in conjunction with ocular surgery due to interference with platelet aggregation. All topical NSAIDs may slow or delay epithelial wound healing. Concomitant use with topical steroidal agents may increase the potential for delayed healing. Nepafenac should be used cautiously in patients receiving systemic anticoagulants (eg, aspirin, clopidogrel, warfarin, and heparin), as concomitant use may increase the risk for intraocular bleeding. Use of topical NSAIDs may result in keratitis due to epithelial breakdown, corneal thinning, corneal erosion, corneal ulceration, or corneal perforation. It should be discontinued immediately in patients exhibiting evidence of corneal epithelial breakdown.

Postmarketing experience with topical NSAIDs suggests that use more than 24 hours before surgery or beyond 14 days after surgery may increase patient risk for the occurrence of corneal adverse effects.²

Adverse Effects

Adverse effects reported in humans or animals include capsular opacity, decreased visual acuity, foreign body sensation, increased intraocular pressure (IOP), and sticky sensation.² Other ocular adverse effects include conjunctival edema, corneal edema, dry eye, lid margin crusting, ocular discomfort, ocular hyperemia, ocular pain, ocular pruritus, photophobia, tearing, and vitreous detachment. Some of these effects may be the consequence of the cataract surgical procedure.³ Nonocular adverse effects reported by humans include headache, hypertension, nausea, vomiting, and sinusitis.² Increased bleeding time, delayed corneal wound healing, keratitis, and corneal weakening have also been reported.

Although unlikely, systemic adverse effects may occur with the use of topical NSAIDs, including increased risk for gastric ulceration, decreased platelet aggregation, worsening of asthma and cardiac disease, and renal toxicity. Concurrent use of systemic NSAIDs and corticosteroids increases the risk for systemic adverse effects.

Dosages

Perioperative for cataract surgery (extra-label): 1 drop in the affected eye(s) 1 to 3 times daily for up to 2 weeks following cataract surgery

Anterior segment inflammation (extra-label): 1 drop in the affected eye(s) every 8 hours

Monitoring

- Clinical efficacy: decreased or absent inflammation. Monitor for delayed wound healing

Client Information

- Use this medicine as directed by your veterinarian.
- Shake this medicine well before using.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes.

Storage/Stability

Store in the refrigerator (2°C–8°C [35.6°F–46.4°F]) or controlled room temperature (20°C–25°C [68°F–77°F])² away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Nepafenac ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Nepafenac 0.1% Ophthalmic Solution (containing nepafenac 1 mg/mL; mannitol; carbomer 974P; NaCl; tyloxapol; edetate disodium; benzalkonium chloride 0.005%; NaOH and/or HCl to adjust pH to 7.4; purified water, USP, with an osmolality of 305 mOsm/kg), 3 mL in a sterile 4 mL ophthalmic dropper bottle; *Nevanac*®; (Rx)

Nepafenac 0.3% Ophthalmic Suspension (containing nepafenac 0.3%; boric acid, propylene glycol; carbomer 974P; NaCl; guar gum; carboxymethylcellulose sodium; edetate disodium; benzalkonium chloride 0.005%; NaOH and/or HCl to adjust pH; purified water, USP), 1.7 mL in a sterile 4 mL ophthalmic dropper bottle, or 3 mL in a sterile 4 mL ophthalmic dropper bottle; *Ilevro*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Ocular Lubricants

Tear Replacement and Ocular Lubricating Agents

Prescriber Highlights

- Used in the temporary enhancement of the quality and quantity of precorneal tear film
- Hyaluronate/hyaluronic acid-containing preparations may exert a beneficial effect on corneal ulcer healing.
- Most artificial tears and ocular lubricants are labeled for humans and are available OTC.
- Preservative-free solutions are less likely to cause stinging or corneal damage with long-term use.
- Petroleum-based ocular products should not be used in procedures where combustion may occur (eg, electrocautery, laser) or when contact with the intraocular environment is possible (eg, penetrating corneal wound).
- Basic components of an artificial tear/ocular lubricant include demulcent (soothing agent), emollient (a fat-based agent that softens and protects tissues), inactive ingredients (impact viscosity, tonicity, and pH), and preservatives (to prevent bacterial growth).

Uses/Indications

Artificial tear solutions and ocular lubricants are used when there is a decrease in tear production, an increase in tear loss through evaporation, and a decrease in the tear film's break up time, such as in the presence of qualitative and quantitative keratoconjunctivitis sicca (KCS)¹, buphthalmos (enlarged globe that adversely affects the ability to blink), or lagophthalmos (inability to blink due to the conformational or functional abnormality of the eyelid). Ocular lubricants are also beneficial during sedation and anesthesia to prevent corneal desiccation² and as a vehicle for compounding water-soluble medications for ocular application. Although the evidence is variable, preparations containing hyaluronic acid/sodium hyaluronate may accelerate the healing of corneal ulcers.^{3,4}

Pharmacology/Actions

Improving the quality (eg, viscosity, osmolality, retention time) of the precorneal tear film results in improvement in comfort for patients with tear film deficiencies, as well as reduction in ocular surface inflammation and damage. Preparations include a demulcent that acts as a soothing and anti-inflammatory agent, an emollient to protect tissues and prevent drying, inactive ingredients to stabilize the preparation for ocular administration, and preservatives to prevent contamination.⁵

Demulcents may include polyethylene glycol (PEG), propylene glycol (PG), glycerin, cellulose derivatives, hydroxymethylcellulose (HMC), hydroxypropylcellulose (HPC), hydroxypropylmethylcellulose (HPMC), and polyvinyl alcohol (PVA). Although each demul-

cent varies slightly in a specific function, broadly, they are all responsible for increasing viscosity of the preparation, decreasing viscosity and osmolarity of the tear film, protecting mucous membranes, relieving inflammation, and improving the stability of the tear film.⁶

Emollients may include lanolin, mineral oil, petrolatum, paraffin, and wax. Although each emollient varies slightly in a specific function, broadly, they are all responsible for lubrication and soothing, stabilizing the lipid layer of the tear film, and sealing in moisture.⁶

Inactive ingredients may include sorbitol, hyaluronic acid (HA), sodium hyaluronate, L-carnitine and erythritol, hydroxypropyl guar, polyacrylic acid, tyloxapol, and tromethamine. Although each inactive ingredient varies slightly in a specific function, they are responsible for lowering the osmolarity of the tear film, stabilizing the viscosity of the tear film, prolonging contact time of active ingredients, promoting appropriate corneal epithelial metabolism, and maintaining hydration.⁶

Preservatives may include benzalkonium chloride, polyquad, sodium perborate, stabilized oxychloro complex, edetate disodium, and chlorobutanol. Although each preservative varies in a specific function, broadly, they are responsible for decreasing bacterial contamination of the preparation, increasing drug penetration, and increasing shelf life.⁶ Some preservatives (eg, benzalkonium chloride) are known ocular irritants and cause destabilization of the tear film, which is beneficial in some conditions to allow for improved penetration of active ingredients. Although this action may be beneficial for some ocular conditions (eg, glaucoma), the resulting epithelial cell damage can be harmful in ocular surface diseases (eg, corneal ulceration). Thus, preservatives have been developed that are stable in solution but degrade upon contact to minimize ocular surface damage.

The combination of demulcent and emollient, as well as other inactive ingredients and preservatives, decreases friction on the ocular surface and increases tear retention and tear film moisture while minimally impacting ocular clarity.

Contraindications/Precautions/Warnings

Artificial tear solutions are contraindicated in patients that have a history of hypersensitivity to any component of the formulation. Interactions between topical ocular lubricants and other ocular preparations may occur, leading to alterations in viscosity of the lubricant, decreased efficacy in stabilizing the precorneal tear film, precipitate formation, or patient irritation or discomfort.

Adverse Effects

Adverse effects reported in animals include eye pain, burning or stinging on instillation, and redness. Preservative-free solutions may decrease these adverse effects. Blurred vision is commonly reported in humans after the application of ocular ointments; thus, many OTC preparations are formulated as solutions with varied demulcent

and emollient proportions. Sticky eyelashes and increased sensitivity to light are commonly reported by humans using solutions containing hydroxypropyl cellulose.

Dosages

Tear replacement (extra-label): 1 drop or a ¼ – ½ inch strip to the affected eye(s) every 2 to 3 hours or as often as possible

Ocular lubrication during anesthetic procedures (extra-label): ½ – 1 inch strip to each eye every 90 minutes. Do not use in procedures where the combustion of petroleum-based ocular lubricants or contact of the lubricant with the intraocular environment may occur.

Monitoring

- Clinical efficacy
- Monitor corneal surface for signs of damage, as chronic use of preserved solutions may cause corneal damage.

Client Information

- Place these drops in your animal's eye(s) as often as possible during the day. Administer longer-acting ointments at bedtime.
- If you notice any increase in squinting, eye discharge, or ocular redness, discontinue use and contact your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. To prevent contamination, do not touch the dropper tip (or ointment) or allow it to touch your animal's eye or any other surface.
- Keep the cap tightly closed when the medicine is not in use.
- If applying multiple medications to your animal's eye(s), apply lubricating medications last, at least 5 minutes following administration of other eye medications.
- Store this medicine at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if the solution becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight.

Compatibility/Compounding Considerations

Artificial tears and ocular lubricants are often used as vehicles when compounding topical ocular medications. Check to ensure compatibility of all ingredients and preservatives before compounding.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS:

None are FDA-approved.

Hyaluronic Acid-Containing Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
water for injection; sodium chloride; 0.25% viscoadaptive hyaluronan; glycerin; sodium phosphate; potassium sorbate; EDTA	OTC; 10 mL in a sterile ophthalmic dropper bottle	<i>I-Drop® Vet Plus Lubricant</i> NOTE: There is also a 0.3% gel, <i>I-Drop® Vet Gel</i> .
sterile water; sorbitol; carbomer; sodium hydroxide; disodium EDTA; cetrimide	OTC; 0.7 oz tube	<i>OptixCare® Eye Lube</i>
sterile water; sorbitol; carbomer; hyaluron; sodium hydroxide; disodium EDTA; cetrimide	OTC; 0.7 oz tube	<i>OptixCare® Eye Lube Plus</i>
water; 0.4% Hyasent-S (modified, cross-linked HA); sodium chloride; disodium; phosphate; potassium chloride; potassium phosphate	Rx; 10 mL in a sterile ophthalmic dropper bottle	<i>Remend® Eye Lubricating Drops</i> NOTE: There is also a 0.75% gel that is marketed as a corneal repair gel for superficial corneal ulcers: <i>Remend® Corneal Repair Gel</i> .

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HUMAN-LABELED PRODUCTS:

There are many products marketed, and this list is not comprehensive:

Polyvinyl Alcohol-Containing Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
polyvinyl alcohol 1%; polyethylene glycol 400 1%; benzalkonium chloride 0.1 mg/mL; dextrose; edetate disodium; purified water; HCl and/or NaOH to adjust pH	OTC; 15 mL in a sterile ophthalmic dropper bottle	<i>HypoTears</i> ®
polyvinyl alcohol 0.5%; povidone 0.6%; benzalkonium chloride; dextrose; edetate disodium; potassium chloride; purified water; sodium bicarbonate; sodium chloride; sodium citrate; sodium phosphate (mono- and dibasic)	OTC; 15 mL in a sterile ophthalmic dropper bottle	<i>Murine Tears</i> ®

Polyvinylpyrrolidone-Containing Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
polyvinyl pyrrolidone 2%; polyvinyl alcohol 2.7%; boric acid; disodium edetate dihydrate; ethanol; glycerin; lecithin; polixetonium; polysorbate-80; potassium chloride; purified water; sodium chloride	OTC; 15 mL in a sterile ophthalmic dropper bottle	<i>Freshkote</i> ®

Carboxymethylcellulose-Containing Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
sodium carboxymethylcellulose 1%; borate buffers; calcium chloride; magnesium chloride; potassium chloride; purified water; sodium bicarbonate; NaCl; sodium phosphate	OTC; 0.6 mL single-use vials	<i>TheraTears</i> ® Liquid Gel
sodium carboxymethylcellulose 0.25%; borate buffers; calcium chloride; <i>Dequest</i> ®; potassium chloride; purified water; sodium bicarbonate; sodium borate; sodium chloride; magnesium chloride; sodium perborate; sodium phosphate; trehalose	OTC; 15 mL in a sterile ophthalmic dropper bottle	<i>TheraTears</i> ® Lubricant Eye Drops

Hydroxypropyl Methylcellulose-Containing Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
dextran 70 0.1%; hydroxypropyl methylcellulose 2910 0.3%; calcium chloride; magnesium chloride; potassium chloride; purified water; sodium bicarbonate; NaCl; zinc chloride; HCl and/or NaOH and/or carbon dioxide to adjust pH	OTC; 0.4 mL single-use containers in cartons of 28	<i>Bion Tears</i> ®
dextran 70 0.1%; glycerin 0.2%; hypromellose 0.3%; boric acid; calcium chloride; glycine; HCl and/or NaOH to adjust pH; magnesium chloride; POLYQUAD (polyquaternium-1) 0.002% preservative; polysorbate 80; potassium chloride; purified water; sodium chloride; zinc chloride	OTC; 15 mL in a sterile ophthalmic dropper bottle	<i>Genteal</i> ® Moderate Lubricating Liquid NOTE: Many other <i>Genteal</i> ® formulations exist.
glycerin 0.2%; hydroxypropyl methylcellulose 0.2%; polyethylene glycol 400 1%; ascorbic acid; benzalkonium chloride; boric acid; dextrose; disodium phosphate; glycine; magnesium chloride; potassium chloride; purified water; sodium borate; sodium chloride; sodium citrate; sodium lactate	OTC; in a sterile 15 mL or 30 mL ophthalmic dropper bottle	<i>Visine Tears</i> ® Dry Eye Relief NOTE: Many other <i>Visine</i> ® formulations exist.

Hydroxypropyl Guar-Containing Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
hydroxypropyl methylcellulose 0.3%; carbopol 980; phosphonic acid; purified water; sodium perborate; sorbitol	OTC; 10 g in sterile ophthalmic dropper bottles	<i>Systane</i> ® Lubricant Eye Gel NOTE: Many other <i>Systane</i> ® formulations exist.

Polyethylene Glycol/Hyaluronate-Containing Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
polyethylene glycol 400 0.25%; boric acid; calcium chloride; magnesium chloride; potassium chloride; purified water; sodium borate; sodium chloride; sodium chlorite (<i>Ocupure</i> ® brand); sodium hyaluronate	OTC; 15 mL in a sterile ophthalmic dropper bottle	<i>Blink</i> ® Gel Tears NOTE: Many other <i>Blink</i> ® formulations exist.

Petroleum-Based Products

Active ingredients (formulation)	Label status; size(s)	Trade names/additional information
mineral oil 15%; white petrolatum 85%	OTC; 3.5 g ophthalmic ointment tube	<i>Gentleal</i> ® Night-Time Ointment
light mineral oil 1%; mineral oil 4.5%	OTC; 15 mL ophthalmic dropper bottle	<i>Soothe XP</i> ®
mineral oil 20%; white petrolatum 80%	OTC; 3.5 g ophthalmic ointment tube	<i>Tears Again</i> ®

References

For the complete list of references, see wiley.com/go/budde/plumb

Ofloxacin Ophthalmic

(o-**flocks**-uh-sin) *Ocuflox*®

Second-Generation Fluoroquinolone Antimicrobial

Prescriber Highlights

- ▶ Used to treat bacterial conjunctivitis or keratitis
- ▶ Physically incompatible with many drugs; therefore, it should not be directly mixed with other ophthalmic medications.
- ▶ Crystalline drug precipitates may be noted in the superficial portion of corneal defects during use.

Uses/Indications

Ofloxacin ophthalmic may be used for the treatment of bacterial conjunctivitis or keratitis caused by susceptible strains of *Staphylococcus* spp, *Streptococcus* spp, *Pseudomonas aeruginosa*, *Chlamydomphila* spp, and *Haemophilus* spp.¹ Ofloxacin achieves greater aqueous humor concentrations than ciprofloxacin following topical administration to dogs.² In the UK, horses with bacterial keratitis did not demonstrate increased resistance to ofloxacin.³

The World Health Organization (WHO) has designated fluoroquinolones as a Critically Important, Highest Priority antimicrobial in human medicine.⁴

Pharmacology/Actions

Ofloxacin is a bactericidal and concentration-dependent fluoroquinolone antibiotic that is believed to act by inhibiting bacterial DNA-gyrase (a type-II topoisomerase), thereby preventing DNA supercoiling and DNA synthesis. The activity of ofloxacin is greater against gram-negative organisms than gram-positive organisms. *Staphylococcus* spp demonstrates lower susceptibility to second-generation fluoroquinolones than most other gram-positive agents.¹

Contraindications/Precautions/Warnings

Ofloxacin is contraindicated in patients that have a history of hypersensitivity to ofloxacin, to other fluoroquinolones, or to any of the components of this medication. Ofloxacin should only be used as a topical ophthalmic. Do not inject ofloxacin subconjunctivally or into the anterior chamber of the eye. Fluoroquinolone-induced retinal toxicity in domestic cats has not been demonstrated with this product.

Adverse Effects

Adverse effects reported in humans or animals include blurred vision, tearing, eye pain, redness, and itching.⁵ Crystalline drug precipitates may be noted in the superficial portion of corneal defects during use. The arthropathy that occurs in immature animals associated with oral administration of quinolones has not been noted following topical ophthalmic administration.⁵ Alterations in ocular surface microflora, with the increased population of resistant organisms, have been documented in dogs during 6 weeks of topical ofloxacin therapy following cataract surgery.⁶

Dosages

Conjunctivitis (extra-label): 1 drop in the affected eye(s) every 2 to 8 hours

Keratitis (extra-label): 1 drop in the affected eye(s), or 0.1 mL in the subpalpebral lavage (SPL) catheter, every 2 to 8 hours. It may precipitate when mixed with other ophthalmic medications; therefore, it is important to flush the SPL catheter well with air between medications.

Monitoring

- Corneal cytology and culture to evaluate for the persistence of bacteria
- Evaluation for clinical signs suggestive of continued infection

Client Information

- Use this medication as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Do not touch the dropper tip or allow it to touch your animal's eye or any other surface to prevent contamination.
- Crystals may appear on the ocular surface after initiation of treatment with ofloxacin.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Keep the cap tightly closed when the medication is not in use. Store this medication in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this medication in the refrigerator (2°C-8°C [35.6°F-46.4°F]) or at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight; do not freeze.⁵

Compatibility/Compounding Considerations

Fluoroquinolone antibiotics interact with many other drugs, including those containing cations (eg, iron, aluminum, magnesium, or calcium). Ofloxacin should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Ofloxacin 0.3% Ophthalmic Solution (containing ofloxacin 3 mg/mL; benzalkonium chloride 0.005%; NaCl; HCl and/or NaOH to adjust pH to a range of 6 to 6.8; purified water; with an osmolality of 300 mOsm/kg), in a 5 mL and 10 mL ophthalmic dropper bottle; *Ocuflox*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Olopatadine Ophthalmic

(o-la-**pat**-a-deen) *Pataday*®, *Patanol*®, *Pazeo*®
H₁ and H₂ Receptor Antagonist Antihistamine Agent

Prescriber Highlights

- ▶ Used to relieve pruritus caused by allergic conjunctivitis
- ▶ Longer duration of action allows once- to twice-daily dosing

Uses/Indications

Olopatadine ophthalmic is used to relieve pruritus caused by allergic conjunctivitis. Evaluation of olopatadine's clinical efficacy in veterinary species has not been performed.

Pharmacology/Actions

Olopatadine is a selective, noncompetitive H₁ and H₂ receptor antagonist and a mast cell stabilizer. It is also an inhibitor of proinflammatory mediators such as leukotrienes and thromboxane.¹

Contraindications/Precautions/Warnings

Olopatadine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans include transient burning or stinging on instillation in the eye, although instillation was more comfortable compared with other ophthalmic antihistamines.¹

Dosages

Allergic conjunctivitis (extra-label): 1 drop in the affected eye(s) once daily (*Pataday*®, *Pazeo*®) or twice daily (*Pataday*®, *Patanol*®)

Monitoring

- Clinical efficacy: resolution of pruritus and conjunctivitis

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine in refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solutions.

Storage/Stability

Store this product in the refrigerator (2°C-8°C [35.6°F-46.4°F]) or at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Olopatadine ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Olopatadine 0.1% Ophthalmic Solution (containing 1.11 mg olopatadine HCl equivalent to 1 mg olopatadine per mL; povidone; dibasic sodium phosphate; NaCl; edentate disodium; benzalkonium chloride 0.01%; HCl/NaOH to adjust pH to ≈7; purified water, with an osmolality of ≈300 mOsm/kg), in 2.5 mL ophthalmic dropper bottles; *Pataday*®, *Patanol*®; generic; (Rx)

Olopatadine 0.2% Ophthalmic Solution (containing 2.22 mg olopatadine HCl equivalent to 2 mg olopatadine per mL; povidone; dibasic sodium phosphate; NaCl; edentate disodium; benzalkonium chloride 0.01%; HCl/NaOH to adjust pH to ≈7; purified water, with an osmolality of ≈300 mOsm/kg), in 2.5 mL ophthalmic dropper bottles; *Pataday*®; generic; (Rx)

Olopatadine 0.7% Ophthalmic Solution (containing 7.76 mg olopatadine HCl equivalent to 7 mg olopatadine per mL; povidone; hydroxypropyl-gamma-cyclodextrin; polyethylene glycol 400; hydroxypropyl methylcellulose; boric acid; mannitol; benzalkonium chloride 0.015%; HCl/NaOH to adjust pH; purified water), in 2.5 mL ophthalmic dropper bottles; *Pataday*®, *Pazeo*®; generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Oxytetracycline/Polymyxin B Ophthalmic

(ocks-ih-**tet**-rih-sye-kleen/pol-lee-**mix**-in) *Terramycin*®
Combination Antibiotic

For information related to systemic use, see *Oxytetracycline and Polymyxin B*

Prescriber Highlights

- ▶ Used to treat superficial ocular bacterial infections in dogs, cats, horses, cattle, and sheep
- ▶ Frequently effective for treating *Chlamydia* spp and *Mycoplasma* spp conjunctivitis in cats
- ▶ *Chlamydia* spp infections should be treated for 3 to 4 weeks beyond resolution of clinical signs to disrupt *Chlamydia* spp reproductive cycle.
- ▶ Some cats experience local irritation or more severe anaphylactic reactions that necessitate discontinuation of treatment.
- ▶ Used to decrease keratomalacia induced by matrix metalloproteinases in response to corneal infections
- ▶ May accelerate corneal epithelial wound healing in dogs

Uses/Indications

Oxytetracycline/Polymyxin B is used to prevent or treat superficial ocular infections involving the conjunctiva and/or cornea caused by gram-positive, gram-negative, and intracellular bacteria. It is commonly used to treat known or suspected *Chlamydia* spp and *Mycoplasma* conjunctivitis in cats.

Following a single dose of topical oxytetracycline/polymyxin B administered to normal canine eyes, tear film concentrations surpass MICs for common ocular pathogens.¹ Topical tetracyclines have variable effects on the rate of corneal epithelial wound healing in dogs without bacterial ocular surface infections.^{2,3} Topical tetracyclines are useful in decreasing keratomalacia occurring due to bacte-

rial infections of corneal ulcers⁴; however, the twice-daily dosing advised by the manufacturer may not be sufficient to achieve sustained collagenolytic activity within the tear film of normal canine eyes.¹

Pharmacology/Actions

The reversible binding of tetracyclines to the 30S ribosomal subunits of susceptible organisms inhibits the binding of the tRNA and its associated amino acid to the mRNA-ribosome complex, which disrupts protein synthesis. Tetracyclines demonstrate good activity against *Rickettsia* spp, *Borrelia* spp, *Chlamydia* spp, *Mycoplasma* spp, *Moraxella* spp, and *Brucella* spp, with variable activity against *Staphylococcus* spp and *Streptococcus* spp. Tetracyclines are not effective against *Pseudomonas* spp.

The increased healing rate of uninfected corneal ulcers in dogs treated with topical tetracyclines is thought to be due to upregulation of transforming growth factor-beta.² Oxytetracycline is also a potent inhibitor of matrix metalloproteinase-9 and a moderate inhibitor of matrix metalloproteinase-2. The reduction of keratomalacia occurring in infected corneal ulcers is due to inhibition of matrix metalloproteinases (MMP), which is induced by tetracyclines binding calcium and zinc (cofactors necessary for MMP activity).⁵

Polymyxin B sulfate, one of a group of related antibiotics derived from *Bacillus polymyxa*, is rapidly bactericidal. This action is exclusively effective against gram-negative organisms, particularly *Pseudomonas aeruginosa*.⁶ Polymyxin B sulfate interacts with the lipopolysaccharide of the cytoplasmic outer membrane of gram-negative bacteria, altering membrane permeability and causing cell death; it does not need to enter the cell. Polymyxin B sulfate has a bactericidal action against almost all gram-negative bacilli except the Proteus group. It is ineffective against gram-positive bacteria, gram-negative cocci, or obligate anaerobes. It lacks activity against viral, fungal, or parasitic pathogens. Pus and cellular debris reduce effectiveness.

Contraindications/Precautions/Warnings

Oxytetracycline and polymyxin are contraindicated in patients that have a history of hypersensitivity to either drug, drugs in the same class, or any component of the formulation.

Adverse Effects

Adverse effects reported in humans or animals are rare. Oxytetracycline with polymyxin B sulfate is well tolerated by the epithelial membranes and other tissues of the eye. Allergic or inflammatory reactions due to individual hypersensitivity are rare; however, some cats may have an anaphylactic reaction to the polymyxin B component.^{6,7}

Dosages

Conjunctivitis, keratitis (label dosage; FDA approved): ¼ inch strip in the affected eye(s) 2 to 4 times daily⁶

Monitoring

- Clinical efficacy
- Resolution of conjunctivitis or keratitis

Client Information

- Use this medication as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Do not touch the dropper tip or allow it to touch your animal's eye or any other surface to prevent contamination.
- Wait 5 minutes after applying this medicine before applying any other medicine to your animal's eye(s).
- Crystals may appear on the surface of your animal's eye during treatment. This change is not harmful and will go away once treatment is complete.
- Keep the cap tightly closed when the medication is not in use. Store this medication at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes.

Storage/Stability

Store at controlled room temperature (20°C-25°C [68°F-77°F]), with excursions between 15°C to 30°C (59°F-86°F), away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Oxytetracycline should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS:

Oxytetracycline HCl and Polymyxin B Ophthalmic Ointment (containing oxytetracycline 5 mg/g; polymyxin B sulfate 10,000 units/g; white petrolatum; liquid petrolatum), in 3.5 g ointment tubes; *Teramycin*®; (Rx)

HUMAN-LABELED PRODUCTS: NONE

References

For the complete list of references, see wiley.com/go/budde/plumb

Phenylephrine Ophthalmic

(fen-il-**ef**-rin) *Cyclomydril*®

Alpha Agonist Vasoconstrictor

For systemic use, refer to **Phenylephrine**

Prescriber Highlights

- ▶ Direct-acting sympathomimetic agent (alpha-1 agonist)
- ▶ Used to augment mydriasis, most commonly during intraocular surgical procedures and to limit bleeding during minor ocular surface procedures
- ▶ Not an effective mydriatic in cats or horses when used alone but may increase mydriatic effect when combined with atropine
- ▶ Used to differentiate second- from third-order Horner syndrome
- ▶ Local discomfort may occur after application.
- ▶ Cardiovascular adverse effects are more likely in small dogs and cats and following use of 10% concentrations.

Uses/Indications

Phenylephrine is most commonly used to induce mydriasis prior to phacoemulsification for cataract removal. It is also used prior to minor ocular surface procedures (eg, conjunctival biopsy) to limit bleeding. Phenylephrine is also used to distinguish second- from third-order lesions in Horner syndrome because of the phenomenon of denervation hypersensitivity. In dogs, maximal pupillary dilation occurs in 55 minutes¹; however, phenylephrine alone does not lead to mydriasis in cats² and horses.³

Pharmacology/Actions

Phenylephrine is a direct alpha-1-adrenergic receptor agonist with potent local vasoconstrictor effects. Mydriatic effects are variable and are achieved through direct stimulation of the radial iris dilator muscle.

Contraindications/Precautions/Warnings

Phenylephrine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Phenylephrine should be used cautiously in patients with thyrotoxicosis or hypertension, as adverse cardiovascular effects may occur.

Adverse Effects

Adverse effects reported in humans or animals include eye pain and stinging on instillation, temporary blurred vision and photophobia, and conjunctival sensitization. Cardiovascular effects following topical administration have been reported in dogs³ and cats⁴ due to

systemic absorption, which may result in adverse cardiovascular effects (eg, marked increase in blood pressure, syncope, myocardial infarction, bradycardia, and ventricular and supraventricular arrhythmias) that could require therapeutic intervention.⁴⁻⁶ The systemic effects can be severe and may require intervention. The effects of phenylephrine will be increased when it is administered after topical local anesthetic agents because of epithelial damage induced by topical local anesthetic agents. In normal dogs, phenylephrine leads to a statistically (but not necessarily clinically) significant increase in intraocular pressure (IOP) following topical administration.¹

Dosages

Diagnosis of Horner syndrome (extra-label): 1 drop of phenylephrine 0.25% solution in each eye. Patients suffering from third-order Horner syndrome will experience mydriasis from this low concentration, whereas healthy patients will not. If there is no response in 20 to 30 minutes, apply 1 drop of a 2.5% solution to each eye. If mydriasis occurs, second-order Horner syndrome is likely.

Preoperative cataract removal (extra-label): 1 drop of phenylephrine 2.5% or 10% in the affected eye(s) as a single dose or every 15 minutes for 2 hours. Severe cardiovascular adverse effects are more likely with small dogs and cats and following use of the 10% solution, thus caution should be exercised both with the concentration and the frequency of administration. Effects are augmented when it is administered following topical anesthetic agents.

Monitoring

- Clinical efficacy
- Monitor degree of pupillary dilation
- Monitor bleeding after minor ocular surface surgical procedures
- Monitor heart rate and blood pressure

Client Information

- This medicine is rarely (if ever) indicated for administration to animals at home.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine as directed on package labeling away from moisture and sunlight; do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product as directed on package labeling either in the refrigerator (2°C-8°C [35.6°F-46.4°F]) for 10% concentration or at controlled room temperature (20°C-25°C [68°F-77°F]) for 2.5% concentration; do not freeze.

Compatibility/Compounding Considerations

Phenylephrine ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Phenylephrine HCl 2.5% Ophthalmic Solution (containing phenylephrine hydrochloride 25 mg (2.5%); sodium phosphate monobasic; sodium phosphate dibasic; boric acid; water for injection; hydrochloric acid and/or sodium hydroxide (to adjust pH); benzalkonium chloride), in a sterile 15 mL ophthalmic dropper bottle; generic; (Rx)

Phenylephrine HCl 10% Ophthalmic Solution (containing phenylephrine hydrochloride 100 mg (10%); sodium phosphate monoba-

sic; sodium phosphate dibasic; boric acid; water for injection; hydrochloric acid and/or sodium hydroxide (to adjust pH); benzalkonium chloride), in a sterile 10 mL ophthalmic dropper bottle; generic; (Rx)

Phenylephrine HCl 1% and Cyclopentolate HCl 0.2% Ophthalmic Solution (containing cyclopentolate hydrochloride 0.2%; phenylephrine hydrochloride 1%; benzalkonium chloride 0.01%; edetate disodium; boric acid; hydrochloric acid and/or sodium carbonate (to adjust pH); purified water), in sterile 5 mL ophthalmic dropper bottle; *Cyclomydril*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Pilocarpine Ophthalmic

(pye-loe-kar-peen) *Isopto Carpine*®

Cholinergic Miotic Agent

Prescriber Highlights

- ▶ Primarily used to treat neurogenic keratoconjunctivitis sicca (KCS). It may also be used to treat primary glaucoma in dogs but has largely been replaced by other treatments.
- ▶ Contraindicated in cases of secondary glaucoma (eg, caused by uveitis or lens luxation) because of the risk for pupillary block
- ▶ Ophthalmic solution may be administered to the affected eye or be given orally on food as a treatment for neurogenic KCS.
- ▶ Monitor for ocular and systemic toxicity.

Uses/Indications

The primary use of pilocarpine in veterinary patients is for the treatment of neurogenic keratoconjunctivitis sicca (KCS), which is caused by interrupted neurologic signaling to the otherwise normal lacrimal glands. In dogs with coinciding xeromyetria (dryness of the nasal mucous membranes), pilocarpine also increases nasal secretions. Oral administration and topical ocular administration, alone or in combination, increases tear production in dogs.¹⁻³ Pilocarpine, administered topically to the eye, may also be used to diagnose parasympathetic denervation of the iris sphincter caused by lesions of or trauma to the oculomotor nerve (cranial nerve III). Pilocarpine may also be used topically to treat primary glaucoma in dogs,⁴ but it has largely been replaced by beta-blockers, carbonic anhydrase inhibitors, and prostaglandin agents. Pilocarpine does not reduce intraocular pressure (IOP) in horses and is therefore not used to treat equine glaucoma.⁵

Pharmacology/Actions

Pilocarpine mimics acetylcholine and acts as a direct-acting cholinergic agonist at the neuromuscular junctions. In the eye, this cholinergic stimulation leads to lacrimal secretion, and can therefore be used to treat neurogenic keratoconjunctivitis sicca (KCS; ie, lacrimal gland tissue is normal, but innervation is interrupted). Cholinergic stimulation also leads to ciliary body muscle contraction, which places posteriorly directed tension on the base of the iris to mechanically pull open the iridocorneal angle, increasing aqueous humor outflow in dogs with elevated intraocular pressure (IOP).

Contraindications/Precautions/Warnings

Pilocarpine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Avoid its use in dogs with glaucoma secondary to uveitis or lens luxation, as the induction of miosis may lead to blockage of aqueous humor flow through the pupil, thus worsening glaucoma (pupillary block glaucoma).

Adverse Effects

Adverse effects reported in humans or animals include significant

irritation (due to the low pH) upon application, which may result in inflammation of the uveal tract.^{1,6} Local irritation has been reported in dogs within the first 2 days of treatment, but it appears to improve and cause minimal problems thereafter.⁷ Long-term use may result in irreversible miosis due to dilator muscle atrophy and sphincter muscle fibrosis. Systemic absorption of pilocarpine, which occurs when the medication is administered orally, may result in vomiting, diarrhea, increased salivation, increased urination, bronchiolar spasm, and pulmonary edema.

Dosages

Diagnosis of ocular parasympathetic denervation (oculomotor nerve [cranial nerve III] lesions) (extra-label): 1 drop of diluted pilocarpine (0.1–0.2%) solution in the affected eye(s)⁸

Neurogenic keratoconjunctivitis sicca (KCS; extra-label): 1 drop of pilocarpine 0.25% solution in the affected eye(s) 4 times daily, alone or in combination with 2 drops/9.1 kg (20 lb) of a pilocarpine 2% solution in food to be consumed orally twice daily.⁹ The dose is increased weekly until signs of toxicity occur or until clinical signs are controlled. If signs of toxicity develop, the dose must be decreased.

Primary glaucoma (extra-label): 1 drop of pilocarpine 1% or 2% solution in the affected eye(s) 3 times daily. **NOTE:** Topical pilocarpine is not recommended for first-line or chronic therapy.

Monitoring

- Clinical efficacy (neurogenic keratoconjunctivitis sicca [KCS]): improved tear production as assessed by Schirmer tear test, patient comfort, and decreased ocular redness and irritation
- Clinical efficacy (primary glaucoma): decrease in/control of intraocular pressure (IOP)
- Observe for signs of systemic toxicosis, including vomiting, diarrhea, increased salivation, increased urination, bronchiolar spasm (coughing), and pulmonary edema, which may occur with both oral and topical administration.
- Monitor for signs of ocular irritation (eg, redness, squinting, cloudiness) when the drug is administered topically.

Client Information

- Give this medication as directed by your veterinarian.
- Observe your animal for signs of toxicity, including vomiting, diarrhea, increased salivation, increased urination, and coughing. Contact your veterinarian if any of these signs occur.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medicine at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solutions.

Storage/Stability

Store this medication at controlled room temperature (15°C–25°C [59°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Pilocarpine should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Pilocarpine HCl 1%, 2%, or 4% Ophthalmic Solution (containing pilocarpine 10 mg/mL, 20 mg/mL, or 40 mg/mL, respectively; benzal-

konium chloride 0.01%; monobasic sodium phosphate; hypromellose; edetate disodium; dibasic sodium phosphate, purified water; NaOH and/or HCl may be added to adjust pH to 3.5 to 5.5), in a 15 mL ophthalmic dropper bottle; *Isopto Carpine*®, generic; (Rx).

References

For the complete list of references, see wiley.com/go/budde/plumb

Pimecrolimus Ophthalmic

(pi-mek-*kroe*-li-mus)

T-cell Immunosuppressive Lacromimetic Agent

Prescriber Highlights

- Used to treat dogs with keratoconjunctivitis sicca (KCS), chronic superficial keratitis (CSK), and other immune-mediated ocular surface inflammatory conditions
- Useful for cyclosporine-resistant cases of KCS
- Not commercially available, must be compounded. *Elidel*® topical (skin) pimecrolimus ointment should NOT be used in the eye.
- Advise caregivers to wear gloves when handling pimecrolimus.

Uses/Indications

Pimecrolimus is used to stimulate tear production in dogs with keratoconjunctivitis sicca (KCS) that do not respond to topical cyclosporine. Pimecrolimus 1% administered 3 times daily to patients with KCS and chronic superficial keratitis (CSK) produced favorable improvement in clinical signs.¹ A prospective study in dogs comparing pimecrolimus 1% with cyclosporine 0.2% over an 8-week period found significantly greater improvement in clinical signs that were treated with pimecrolimus rather than cyclosporine.² Topical ophthalmic pimecrolimus has not been evaluated in cats or horses.

Pharmacology/Actions

Pimecrolimus is a calcineurin inhibitor (like cyclosporine) that reversibly inhibits T-cell proliferation and prevents the release of pro-inflammatory cytokines. Pimecrolimus binds to intracellular macropilin, forming a complex that subsequently binds to and inhibits calcineurin. In blocking calcineurin, translocation of the cytoplasmic component of the nuclear factor of activated T cells to the nucleus is prevented. This prevention of translocation impairs transcription of the genes encoding IL-2 and other cytokines, thereby suppressing T-cell proliferation and normal immune function.

Contraindications/Precautions/Warnings

Pimecrolimus is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Caregivers should be advised to avoid contact with pimecrolimus, as skin contact with pimecrolimus has been associated with the development of cancer in humans.

Adverse Effects

Adverse effects to ocular use of pimecrolimus are rare but include orbital alopecia, which may be related to oil vehicles used to compound pimecrolimus solutions.

Dosages

Keratoconjunctivitis sicca (KCS) (extra-label): 1 drop of a pimecrolimus 1% solution in affected eye(s) 2 to 3 times daily

Chronic superficial keratitis (CSK) (extra-label): 1 drop of a pimecrolimus 1% solution in affected eye(s) 2 to 3 times daily

Monitoring

- Baseline and periodic Schirmer tear test, as well as reduction in mucoid ocular discharge and conjunctival hyperemia. Evaluate for lack

of progression (and possibly regression) of chronic inflammatory changes, such as corneal pigmentation, vascularization, and fibrosis

Client Information

- Use this medicine as directed by your veterinarian.
- This is a lifelong medicine and must be given daily. Missed doses may cause return of keratoconjunctivitis sicca (KCS) or chronic superficial keratitis (CSK).
- Wear gloves when handling pimecrolimus. Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine as directed on the package labeling away from moisture and sunlight; do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if particles are seen in solutions.

Storage/Stability

Store this product as directed on the package labeling away from moisture and sunlight.

Compatibility/Compounding Considerations

Pimecrolimus ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Pimecrolimus ophthalmic preparations are not commercially available but have been prepared (compounded) by qualified compounding pharmacists in 1% solutions in oil vehicles including olive oil, corn oil, and medium-chain triglyceride oil.

References

For the complete list of references, see wiley.com/go/budde/plumb

Povidone-Iodine Ophthalmic

(*poe*-veh-done *eye*-oh-dyne) *Betadine*®

Ocular Disinfectant

Prescriber Highlights

- ▶ Primarily used as a component of preoperative periocular surface preparation
- ▶ An inexpensive alternative to traditional antifungal, antibacterial, or antiviral ophthalmic agents
- ▶ May be used to debride nonvital corneal epithelium from indolent ulcers
- ▶ Solutions greater than 1% must be thoroughly lavaged from the eye within 5 minutes of administration to avoid corneal toxicity.
- ▶ Topical administration causes discomfort.

Uses/Indications

Povidone-iodine ophthalmic solutions (1% to 5%) are used as a component of periocular surface preparation for all surgical procedures. Additionally, they may be used to chemically debride loose epithelium associated with superficial chronic corneal epithelial defects (SCCED; also called indolent ulcer), or as treatment for bacterial, fungal, and viral infections of the ocular surface. Contrary to what may be expected, lower concentrations (less than 1%) of povidone-iodine demonstrate greater bactericidal activity than higher (10%) concentrations.¹ Povidone-iodine 1.25% led to comparable clinical improvement relative to topical neomycin/polymyxin B/gramicidin and ciprofloxacin in human patients with bacterial keratitis caused by various organisms.²

Pharmacology/Actions

Povidone-iodine is an iodophor that damages bacterial cell walls and organelle membranes through interactions with fatty acid double bonds. Additionally, povidone-iodine induces oxidative damage to amino acids and nucleotides. These actions lead to formation of pores in membranes, producing bactericidal and virucidal, and, potentially, fungicidal and sporicidal activity, depending on organism and contact time.

Contraindications/Precautions/Warnings

Povidone-iodine is contraindicated in patients with a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans or animals include local burning and stinging on application, as well as chemosis and conjunctival hyperemia; however, concentrations less than 1% are better tolerated than more concentrated preparations. Some patients may experience a local hypersensitivity reaction. Solutions of 0.25% to 10% demonstrate varying degrees of toxicity to ocular surface cells in vitro.³

Dosages

Periocular preoperative preparation (extra-label): Saturate sterile swabs or ocular gauze with povidone-iodine solution and apply to surgical area as disinfectant component of presurgical preparation; irrigate the ocular surface with sterile saline solution, leaving it in contact with the surgical area for 2 minutes. Residual solution should be removed via irrigation with sterile saline solution.

Fungal, bacterial, or herpesvirus keratitis (extra-label): 1 drop of a 5% solution to the affected eye(s) once every 24 hours. Treated eye(s) must be thoroughly lavaged with sterile saline solution not more than 5 minutes after administration to avoid corneal damage.

Debridement of nonvital epithelium in superficial chronic corneal epithelial defects (SCCED) ulcers (extra-label): 1 – 2 drops of a 5% solution to the affected eye(s) once. Irrigate thoroughly with sterile saline solution to remove all povidone-iodine not more than 5 minutes after application to avoid damage to the healthy corneal epithelium.

Monitoring

- Clinical efficacy

Client Information

- Use this medicine as recommended by your veterinarian.
- Rinse the animal's eyes thoroughly with sterile eyewash solution within 5 minutes of application.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Store at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if particles are seen in the solution.

Storage/Stability

Store at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Povidone-iodine preparations should not be mixed directly with any other medications. Sterile povidone-iodine ophthalmic preparation is available as a 5% solution intended for application to the eyes, but solutions of 0.5% to 1% may be prepared by diluting this solution with sterile normal saline. Nonsterile povidone-iodine solutions should not be administered to the eye.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Povidone-Iodine 5% Ophthalmic Prep (containing povidone-iodine 5%, with 0.5% available iodine; purified water; citric acid; glycerin; nonoxynol-9; NaCl; NaOH; and dibasic sodium phosphate) in a sterile 30 mL bottle; *Betadine Sterile Ophthalmic Prep*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Prednisolone Ophthalmic

(pred-*niss*-oh-lone) *Pred Forte*®, *Omnipred*®, *Blephamide*®,
Pred G®

Corticosteroid

For systemic use, refer to *Prednisolone/Prednisone*

Prescriber Highlights

- ▶ Topical corticosteroid used to treat ocular surface and intraocular inflammatory conditions
- ▶ Superior penetration into the anterior segment compared with other topical corticosteroids, particularly in acetate formulation (versus phosphate)
- ▶ Use cautiously in small diabetic patients to avoid blood glucose dysregulation.
- ▶ Contraindications include a hypersensitivity to prednis(ol)one; an existing or suspected bacterial, viral, or fungal ocular infection; and the presence of a corneal abrasion or ulceration.

Uses/Indications

Prednisolone ophthalmic is used to treat inflammatory conditions of the palpebral and bulbar conjunctiva, the cornea, and the anterior segment of the globe (eg, blepharitis, conjunctivitis, keratitis, anterior uveitis). Greater intraocular penetration (and therefore potential clinical efficacy) is achieved with acetate formulations relative to phosphate. Administration of prednisolone acetate to the eyes of cats reduced experimentally induced uveitis more significantly than dexamethasone, flurbiprofen, and diclofenac.¹

Prednisolone is available in combination with antimicrobial agents for topical ophthalmic use.

Pharmacology/Actions

Corticosteroids have broad immunomodulatory and anti-inflammatory actions, which are mediated via direct effects on gene expression, as well as by nongenomic effects on cell membranes and receptors. Ultimately, corticosteroids decrease pro-inflammatory gene transcription, production of pro-inflammatory prostaglandins and leukotrienes, B- and T-cell number and function, cell-mediated immunity, and humoral immunity.² Specific mechanisms involved in controlling ocular inflammatory conditions are not known. Prednisolone has a relative glucocorticoid potency of 4 (versus 1 for hydrocortisone and 25 for dexamethasone); however, its superior intraocular penetration when in the acetate formulation leads to increased clinical efficacy relative to dexamethasone ophthalmic formulations.

Contraindications/Precautions/Warnings

Prednisolone is contraindicated in patients with a history of hypersensitivity to it or any component of the formulation. Due to inhibition of epithelial wound healing and potentiation of infection, prednisolone ophthalmic is contraindicated when corneal ulceration is suspected or present or in the presence of corneal infection (bacterial, fungal, or viral),³ infectious diseases, chronic renal failure, congestive heart failure, systemic hypertension, or gastric ulceration. When use of prednisolone is necessary in these conditions, the minimal effective dose should be administered for the shortest possible time.

Adverse Effects

Adverse effects reported in humans include development of cataracts

(via altered glucose or lenticular metabolic effects), increased intraocular pressure (IOP; via activation of glucocorticoid receptors in the trabecular meshwork), infection (via immunosuppression), decreased wound healing (via increased collagenolytic activity), mydriasis, and calcific keratopathy. Increased IOP may occur in some animals with long-term use of prednisolone; however, most reports suggest the change is reversible.^{4,5} Systemic effects are rare but include hepatopathy, suppression of endogenous glucocorticoid production, and focal alopecia. Concurrent use of systemic steroids increases risk for these adverse effects.

Dosages

Conjunctivitis (extra-label): 1 drop or ¼ inch strip in the affected eye(s) every 6 hours until clinical improvement occurs, then gradually decrease, provided clinical improvement is maintained

Keratitis (extra-label): 1 drop or ¼ inch strip in the affected eye(s) every 6 hours until clinical improvement occurs, then gradually decrease, provided clinical improvement is maintained

Anterior uveitis (extra-label): 1 drop or ¼ inch strip in the affected eye(s) every 6 hours until clinical improvement occurs, then gradually decrease, provided clinical improvement is maintained

Post-cataract removal (extra-label): 1 drop in the affected eye(s) every 6 hours

Monitoring

- Clinical efficacy
- Resolution or development of signs of inflammation
- Intraocular pressure (IOP; especially in cats) and ocular surface integrity (eg, fluorescein stain) as needed based on clinical presentation
- Blood or urine glucose changes based on development of clinical signs (eg, polyuria, polydipsia) in diabetic animals

Client Information

- This medicine can be given for various lengths of time. Be sure you understand how often and for how long your animal should be given this medicine.
- Shake suspensions well before using.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Contact your veterinarian as soon as possible if your animal's eye condition worsens or does not improve while using this medicine.
- Your veterinarian will need to monitor your animal's eye condition while taking this medicine. Do not miss these important follow-up visits.
- Store this medicine as directed on the product label. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are visible particles in the solution.

Storage/Stability

Storage recommendations can vary; follow storage requirements listed on packaging. Most commonly, this product is recommended to be stored at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Prednisolone ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Prednisolone Acetate 0.12% Ophthalmic Suspension (containing

prednisolone acetate [microfine suspension] 0.12%, benzalkonium chloride, boric acid, edetate disodium, hypromellose, polysorbate 80, purified water, sodium bisulfite, sodium chloride, and sodium citrate) in 10 mL sterile ophthalmic dropper bottle; *Pred Mild*®; (Rx)

Prednisolone Acetate 1% Ophthalmic Suspension (containing prednisolone acetate [microfine suspension] 1%; benzalkonium chloride; boric acid; edetate disodium; hypromellose; polysorbate 80; purified water; sodium bisulfite; NaCl; and sodium citrate, with a pH of 5 to 6) in 5 mL and 10 mL ophthalmic dropper bottles; *Pred Forte* 1%, *Omnipred*® generic; (Rx)

Prednisolone Sodium Phosphate 1% Ophthalmic Solution (containing prednisolone sodium phosphate 10 mg/mL [equivalent to 9.1 mg/mL prednisolone phosphate] in a buffered isotonic solution; hypromellose; monobasic and dibasic sodium phosphate; NaCl, edetate disodium and purified water; NaOH and/or HCl may be added to adjust the pH to 6.2 to 8.2) in 5 mL and 10 mL ophthalmic dropper bottles; generic; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Gentamicin and Prednisolone Acetate Ophthalmic Ointment 0.3%/0.6% (containing gentamicin sulfate equivalent to 0.3% gentamicin base, prednisolone acetate 0.6%, chlorobutanol [chloral derivative] 0.5%, mineral oil, petrolatum [and] lanolin alcohol, purified water, and white petrolatum) in a sterile 3.5 g ophthalmic tube; *Pred-G*®; (Rx)

Gentamicin and Prednisolone Acetate Ophthalmic Suspension 0.3%/1% (containing gentamicin sulfate equivalent to 0.3% gentamicin base, prednisolone acetate [microfine suspension] 1%, benzalkonium chloride 0.005%, edetate disodium, hypromellose, polyvinyl alcohol 1.4%, polysorbate 80, purified water, sodium chloride, sodium citrate dehydrate, NaOH and/or HCl to adjust pH [5.4 to 6.6]) in a sterile 5 mL ophthalmic dropper bottle; *Pred-G*®; (Rx)

Sulfacetamide Sodium and Prednisolone Acetate Ophthalmic Ointment (10%/0.2%) (containing sulfacetamide sodium 10%, prednisolone acetate 0.2%, phenylmercuric acetate 0.0008%, white petrolatum, mineral oil, and petrolatum and lanolin alcohol) in a sterile 3.5 g ophthalmic tube; *Blephamide S.O.P*®; (Rx)

Sulfacetamide Sodium and Prednisolone Acetate Ophthalmic Suspension (10%/0.2%) (containing sulfacetamide sodium 10%, prednisolone acetate 0.2%, benzalkonium chloride 0.004%, edetate disodium, polysorbate 80, polyvinyl alcohol 1.4%, potassium phosphate monobasic, purified water, sodium phosphate dibasic, sodium thiosulfate, and HCl and/or NaOH to adjust the pH [6.6 to 7.2]); *Blephamide*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Proparacaine Ophthalmic

(proe-**pare**-a-kane) *Alcaine*®, *Fluaine*®

Topical Ocular Anesthetic

Prescriber Highlights

- ▶ Rapid-acting topical anesthetic used for ocular diagnostic procedures
- ▶ Should never be prescribed for therapeutic administration
- ▶ Dose- and species-dependent duration, with maximal effect generally between 5 and 15 minutes in dogs, cats, and horses
- ▶ Long-term use results in tear film instability and globe-threatening corneal epithelial and stromal damage.
- ▶ Some products must be stored in the refrigerator; consult product label.

Uses/Indications

Proparacaine is indicated for use in ophthalmic diagnostic proce-

dures, including measurement of intraocular pressure (tonometry), corneal and conjunctival sampling for cytology and culture, ocular ultrasonography, and gonioscopy. Proparacaine is indicated as an adjunctive anesthetic for performing therapeutic procedures in the hospital, including removal of foreign bodies and sutures from the cornea; eyelid and conjunctival surgeries; corneal grafting procedures; and cataract extraction. Proparacaine has dose- and species-dependent duration of action, with maximal effect between ≈5 and 15 minutes in dogs, cats, and horses, and duration of effect lasting around 30 minutes.^{1–5} The degree of maximal effect of proparacaine is less than that of viscous tetracaine in dogs.² Favorably, administration of topical proparacaine has not been associated with decreased positive bacterial or fungal cultures in dogs, cats, or horses.^{6,7} Because of the likelihood of developing ocular surface toxicities with prolonged or repeated use, proparacaine should never be prescribed for use as an ocular analgesic.

Pharmacology/Actions

Proparacaine induces local anesthesia by inhibiting neuronal depolarization, which prevents generation of an action potential. This action occurs due to binding of the local anesthetic molecule to the intracellular portion of the voltage-gated sodium channel within the neuron, thus inhibiting intracellular movement of sodium. Variations in the structure of different local anesthetics influence their relative potencies. All local anesthetics have increased efficacy in alkaline environments due to improved intracellular penetration; efficacy of local anesthetics is decreased in acidic environments, such as when there is inflammation.

Contraindications/Precautions/Warnings

Proparacaine is contraindicated in patients with a history of hypersensitivity to it or any component of the formulation. Prolonged use of a topical ocular anesthetic will likely produce significant corneal epithelial and stromal disease, as well as tear film instability, and may result in loss of vision.⁸

Adverse Effects

Instillation of proparacaine may result in local irritation several hours after administration. Although patients that are sensitive to other local anesthetics may be less likely to react to proparacaine, local or systemic sensitivity may manifest as a pseudo-allergic (anaphylactoid) reaction mimicking an IgE-mediated vasodilatory event. Cardiovascular, respiratory, and CNS depressant effects that may result from other local anesthetics are extremely rare with appropriate use of topical anesthetics. Care should be exercised, however, when proparacaine is administered to small veterinary patients (ie, puppies, kittens, “pocket pets”), or those more susceptible to systemic absorption (ie, amphibians). Use of proparacaine as a therapeutic analgesic leads to superficial punctate keratitis, which may progress to diffuse necrotizing keratitis, severe infiltrative keratitis, persistent epithelial defects, or damage to the Descemet membrane.

Dosages

Ophthalmic procedures requiring local anesthesia (extra-label): 1 – 2 drops in the target eye(s) prior to examination or procedure. May repeat 1 drop every 5 to 10 minutes 2 to 3 additional times. With additional applications, the duration of corneal anesthesia can be extended to 55 minutes in dogs. Additional doses should be avoided in small patients (ie, puppies, kittens, “pocket pets”).

Monitoring

- Clinical effect: Onset of action is indicated by awake patient permitting diagnostic procedures to be performed, and by a reduction in anesthetic requirements when administered preoperatively.

Client Information

- Proparacaine is **not** administered by pet owners in the home. It is reserved for diagnostic procedures in the veterinary practice.

Storage/Stability

Store this product in a refrigerator or at controlled room temperature (depending on the product label) away from moisture and sunlight; do not freeze.⁹

Discard this product if the solution changes color, if it becomes cloudy, or if it develops particles.

Compatibility/Compounding Considerations

Proparacaine should not be mixed directly (in vitro) with any other drugs.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Proparacaine HCl 0.5% ophthalmic solution (containing 5 mg/mL proparacaine; benzalkonium chloride 0.01%; glycerin; HCl and/or NaOH to adjust pH; purified water) in 15 mL ophthalmic dropper bottles; *Alcaine*®, generic; (Rx)

Proparacaine HCl 0.5% with Fluorescein Sodium Ophthalmic Solution (containing proparacaine 5 mg/mL; fluorescein sodium 2.5 mg/mL; povidone; boric acid; water for injection; NaOH and/or HCl to adjust pH; methylparaben), 5 mL, in sterile plastic or glass ophthalmic dropper bottles; *Fluaine*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Rose Bengal Stain, Ophthalmic

(rose *ben*-gall) *Glostrips*®

Ocular Diagnostic Agent

Prescriber Highlights

- ▶ Vital dye that demonstrates corneal and conjunctival epithelial cell damage, as well as deficiency in ocular surface mucins
- ▶ Full-thickness loss of corneal epithelium is not necessary to cause rose bengal uptake by epithelial cells (in contrast to fluorescein).
- ▶ Very toxic to cornea; irrigate eye(s) well after use
- ▶ Stings upon application

Uses/Indications

Rose bengal stain is used to detect epithelial cell damage prior to full-thickness epithelial cell loss (ulceration), as most commonly occurs with viral infections. Rose bengal staining may also be used as an adjunctive diagnostic procedure for nonulcerative epithelial and subepithelial keratomycosis in horses.¹ Rose bengal uptake also occurs in patients with qualitative (mucin) tear deficiency syndromes. Rose bengal stain has recently been demonstrated to be an effective adjunctive diagnostic test for evaluation of quantitative tear film deficiencies in dogs.²

Pharmacology/Actions

Rose bengal stains both the nuclei and cell walls of dead or degenerated epithelial cells of the cornea and conjunctiva. In contrast to fluorescein stain, rose bengal does not require full-thickness loss of corneal epithelium to indicate damage to the cornea. Rose bengal uptake is decreased in cells that are covered appropriately by mucin, a component of the tear film.

Contraindications/Precautions/Warnings

Rose bengal is contraindicated in patients with a history of hypersensitivity to it or any component of the formulation. Topical anesthetic agents may be indicated to prevent stinging on application. Rose bengal is very toxic to the cornea, and treated eye(s) should be thoroughly irrigated after exposure. Rose bengal is virucidal and may confound results of diagnostic tests for viral keratitis; therefore,

culture samples should be obtained prior to application of the stain.

Adverse Effects

Adverse effects reported in humans or animals include burning and stinging upon application.

Dosages

Detection of corneal and conjunctival cell damage (extra-label; NOT FDA-approved): Wet the surface of the rose bengal strip with 1 – 2 drops of normal saline solution. Apply moistened tip of strip or instill 1 drop of rose bengal 1% solution to conjunctival fornix. Examine the eye with white light or with a red-free absorption filter. Degenerated or dead epithelial cells will appear red in color. Flush eye(s) after exposure.

Monitoring

- Evaluate corneas for signs of toxicity (eg, ulceration)

Client Information

- Rose bengal strips should not be used by pet owners at home.

Storage/Stability

Store this product in the refrigerator or at controlled room temperature (below 30°C) away from moisture and sunlight as directed on packaging or by compounding pharmacist; do not freeze. Unpreserved solutions of rose bengal carry a high risk of contamination with bacteria. Store this product as directed and observe all assigned beyond-use dates.

Compatibility/Compounding Considerations

Rose bengal ophthalmic preparations should not be mixed directly with any other ophthalmic medications. Rose bengal solution is no longer commercially available but may be compounded as a 1% solution either with or without preservatives; however, preservation systems are strongly recommended to prevent microbial and fungal contamination.³

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Rose Bengal 0.13% Strips (containing rose bengal 1.3 mg/strip) as individually wrapped strips in containers of 100 sterile strips; *Glostrips*®, generic; (Rx). No FDA-approved dosage forms of rose bengal are currently available.

Rose bengal may be prepared (compounded) by qualified compounding pharmacists into a 1% sterile solution in sterile water for injection.

References

For the complete list of references, see wiley.com/go/budde/plumb

Silver Sulfadiazine (SSD) Ophthalmic

(sil-ver sul-fa-*dye*-ahy-zeen) *Silvadene*®, *SSD*®, *Thermazene*®

Broad-Spectrum Antibacterial; Antifungal Agent

Prescriber Highlights

- ▶ Antibacterial and antifungal dermatologic topical cream used as treatment for fungal keratitis in horses
- ▶ Long-term use may result in discoloration of the sclera due to argyrisms.
- ▶ Do not use this medication in subpalpebral lavage (SPL) catheters, as occlusion will result.
- ▶ Best dispensed in 1 mL sterile tuberculin syringes with sterile tip caps for administration to the ocular surface
- ▶ Warn caregivers with sulfonamide allergies to avoid contact with this medication.

Uses/Indications

Silver sulfadiazine is used as a broad-spectrum antibacterial and antifungal ophthalmic agent. It is used most commonly as a less expensive alternative to conventional therapy for fungal keratitis in horses. Silver sulfadiazine was found to be superior to natamycin in sensitivity testing for fungal isolates in horses with fungal keratitis.¹

Pharmacology/Actions

A specific mechanism of action has not been determined for silver sulfadiazine's antibacterial and antifungal activity; however, its effectiveness may be from either synergistic interactions between the silver and sulfadiazine components, or actions of each component individually. Silver is a biocide that binds to a broad range of targets, thus exerting a variety of effects, such as enzyme inhibition, protein denaturation, or membrane leakage, depending upon the target bound. Sulfadiazine interferes with microbial folic acid synthesis via competitive inhibition of bacterial dihydropteroate synthetase.

Contraindications/Precautions/Warnings

Silver sulfadiazine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. It should not be administered through subpalpebral lavage (SPL) catheters as occlusion will result. The package labeling of the topical cream clearly states that it is **NOT** for use in the eye. As such, it is recommended to obtain informed consent from owners before prescribing.

Adverse Effects

Adverse effects reported in humans or animals include argyrisms (bluish discoloration of the skin and sclera due to silver accumulation), pain, burning, and itching. Although rare, systemic effects that may occur in association with the sulfonamide component include transient leukopenia, skin rashes and inflammation, blood dyscrasias, GI reactions, hepatitis, and toxic nephrosis.

Dosages

NOTE: All dosages are extra-label, and the package insert of the topical cream clearly states that it is **NOT** for use in the eye. As such, it is recommended to obtain informed consent from owners before prescribing.

Bacterial or fungal keratitis: ¼ – ½ inch strip or 0.1 – 0.2 mL in the conjunctival sac of the affected eye(s) every 1 to 2 hours initially, then tapered to every 4 hours and then per clinical response. Avoid long-term use.

Monitoring

- Clinical efficacy: resolution of keratitis as evidenced by cytology and cultures
- Sclera: discontinue use if sclera becomes blue or gray in color (argyrisms)

Client Information

- Use this medicine as directed by your veterinarian.
- Do not use this medicine in subpalpebral lavage (SPL) catheters.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this product at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes.

Storage/Stability

Store this product at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Silver sulfadiazine should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Silver sulfadiazine is **NOT** approved for ophthalmic use. The package insert of all brands of silver sulfadiazine topical creams warns against use in the eye. As such, it is recommended to obtain informed consent from owners before prescribing. It is available as a topical cream, silver sulfadiazine 10 mg/g in a water miscible base in 20 g, 50 g, 400 g, and 1000 g containers. It is optimally dispensed in sterile, single-use tuberculin syringes tipped with a sterile tip cap (needle removed) for administration in the conjunctival sac.

References

For the complete list of references, see wiley.com/go/budde/plumb

Sodium Chloride (Hypertonic) Ophthalmic

(soe-dee-um *klor*-ide [hye-per-ton-ik]) Muro 128®, Altachlore®, Sochlor®

Hyperosmotic Agent

Prescriber Highlights

- ▶ Used for temporary reduction of corneal edema
- ▶ Ointment forms have longer contact time and are preferred relative to solution forms.
- ▶ Available without prescription
- ▶ Compromise of the corneal surface (eg, corneal ulcer) may diminish the benefits of hypertonic sodium chloride due to loss of a functional epithelial barrier.

Uses/Indications

Hypertonic sodium chloride ophthalmic is used for the temporary reduction of corneal edema and associated bullous keratopathy, superficial corneal erosions, and chronic corneal edema that develop due to endothelial dysfunction induced by a variety of causes. Reduction of edema is short-lived; however, the reduction may be sufficient to favorably impact corneal clarity. Topical administration of 5% sodium chloride ophthalmic ointment to normal canine corneas resulted in decreased corneal thickness.¹ Although topical hypertonic saline has been administered to dogs with spontaneous chronic corneal epithelial defects (SCCEDs) undergoing various debridement techniques,² use of 5% sodium chloride ophthalmic did not significantly change the rate of corneal healing, and even prolonged healing time in some cases.³

Pharmacology/Actions

Tear film and aqueous humor are continuously absorbed by the hydrophilic corneal stroma; fluid balance is controlled by a normally functioning corneal epithelium and endothelium. When this balance is disrupted because of endothelial dysfunction, fluid accumulates in the corneal stroma and migrates to the surface of the cornea, causing opacification, bullae formation, ulceration, and visual impairment. The use of topical hypertonic saline may reduce the fluid accumulation in the corneal stroma.

Contraindications/Precautions/Warnings

Hypertonic sodium chloride is contraindicated in patients with a history of hypersensitivity to any component of the formulation. It may have lesser efficacy in the presence of corneal ulceration, as intact corneal epithelium is required to create the concentration gradient by which fluid is drawn from the bullae. Application of hypertonic sodium chloride may further increase discomfort associated with ulcerative keratitis.

Adverse Effects

Adverse effects reported in humans or animals include temporary burning and stinging upon application. Short-term application of hypertonic sodium chloride to normal canine eyes did not induce signs of toxicity.¹

Dosages

Corneal Edema (extra-label): 1 – 2 drops or a ¼ – ½ inch strip in the affected eye(s) every 3 to 4 hours

Monitoring

- Clinical efficacy
- Observe for resolution of corneal edema. Treatment should be discontinued if discomfort or corneal ulceration develop.

Client Information

- Use this medication as directed by your veterinarian.
- This medication is available without a prescription. Only use products recommended by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait 5 to 10 minutes after applying this medication before applying any other medications to the eye.
- Store this medication as directed on the product label. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are visible particles in the solution.

Storage/Stability

Store at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight; do not freeze. Discard this medication if the color changes, if it becomes cloudy, or if particles are observed in the solution.

Compatibility/Compounding Considerations

Hypertonic sodium chloride should not be mixed directly (in vitro) with any other drugs to avoid dilution of the active ingredient.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Sodium Chloride 5% Ophthalmic Ointment (containing sodium chloride 50 mg/mL, lanolin oil, mineral oil, water for injection and white petrolatum) in a 3.5 g ophthalmic ointment tube; *Muro 128*®, *Altachlore*®, *Sochlor*®; (OTC)

Sodium Chloride 2% or 5% Ophthalmic Solution (containing sodium chloride 20 mg/mL or 50 mg/mL, methylparaben 0.023%, propylparaben 0.01% boric acid, hypromellose, propylene glycol, purified water, sodium borate, NaOH and/or HCl to adjust pH) in a 15 mL ophthalmic dropper bottle; *Muro 128*®, *Altachlore*®, *Sochlor*®; (OTC)

References

For the complete list of references, see wiley.com/go/budde/plumb

Sulfacetamide Ophthalmic

(sul-fa-seet-uh-mide) *Bleph-10*®, *Blephamide*®
Sulfonamide Antimicrobial Agent

Prescriber Highlights

- ▶ Topical antimicrobial used to treat ocular surface bacterial infections
- ▶ Warn caregivers with sulfonamide allergies to avoid contact with this medication.
- ▶ Some sulfacetamide ophthalmic products are combined with prednisolone; do not use these formulations in patients with bacterial, viral, or fungal keratitis.

Uses/Indications

Sulfacetamide is used to treat conjunctivitis and other superficial ocular infections due to susceptible microorganisms. Sulfacetamide is considered active against *Escherichia coli*, *Staphylococcus aureus*, *Streptococcus pneumoniae* and *Streptococcus viridans*, *Haemophilus*, *Klebsiella*, and *Enterobacter*. Sulfacetamide has no activity when it is applied topically for *Neisseria* spp, *Serratia marcescens*, or *Pseudomonas aeruginosa*. A significant percentage of staphylococcal isolates are completely resistant to sulfa drugs. Sulfacetamide is also found in combination products with prednisolone (eg, *Blephamide*®).

Pharmacology/Actions

All sulfonamides are bacteriostatic agents with a similar spectrum of activity. Sulfonamides inhibit bacterial synthesis of dihydrofolic acid via competitive inhibition of dihydropteroate synthetase. Resistant strains have altered dihydropteroate synthetase with reduced affinity for sulfonamides or produce increased quantities of aminobenzoic acid.¹

Contraindications/Precautions/Warnings

Sulfacetamide is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Sulfacetamide may induce serious allergic reactions and blood dyscrasias, but this is rare following topical ophthalmic administration.¹ Patients with corneal infections due to bacterial, viral, or fungal organisms should not receive corticosteroid-containing sulfacetamide products.

Adverse Effects

Adverse effects reported in humans or animals following ocular administration include local irritation, stinging, and burning.¹ Although rare, systemic effects that may occur in association with the sulfonamide component include transient leukopenia, skin rashes and inflammation, toxic epidermal necrolysis, blood dyscrasias, GI reactions, fulminant hepatic necrosis, and toxic nephrosis.

In dogs, systemic sulfonamides are known to cause keratoconjunctivitis sicca (KCS) due to a direct toxicity to the lacrimal acinar cells; however, there have been no reports of topical sulfacetamide causing KCS. This toxic effect to the lacrimal acinar cells is caused by the nitrogen-containing pyridine and pyrimidine rings in sulfonamides and is dose-related. The estimated incidence of KCS is 15% to 25% in dogs treated with sulfonamides and may be latent in onset.² Dogs exhibiting blepharospasm or excess mucus ocular discharge should be evaluated for KCS. This condition may be reversible if sulfonamide use is discontinued in time. See **Monitoring**.

Dosages

Susceptible bacterial conjunctivitis and ocular infections (extra-label): 1 drop or ¼ – ½ inch strip on the affected eye(s) every 2 to 3 hours initially and then tapered according to clinical response. Duration of therapy is usually 7 to 10 days.

Monitoring

- Clinical efficacy: resolution of bacterial conjunctivitis as evidenced by cytology and cultures
- Baseline and periodic Schirmer tear testing, especially in predisposed breeds. Dogs should be observed for signs of developing KCS (eg, mucus ocular discharge, blepharospasm).

Client Information

- Use this medicine as directed by your veterinarian.
- Caregivers with sulfa allergies should avoid contact with this medicine.
- Observe dogs for signs of dry eye (eg, squinting, excessive mucus eye discharge, redness of the whites of the eyes) and consult your veterinarian if this occurs.
- Use proper administration techniques to avoid contamination of

the medicine. Keep the cap tightly closed when the medicine is not in use.

- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if it becomes cloudy, or if particles are seen in the solutions. Sulfacetamide solutions will darken over time. The medicine should be thrown away if this occurs.

Storage/Stability

Store this product at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Sulfacetamide ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Sulfacetamide Sodium 10% Ophthalmic Solution (containing sulfacetamide sodium 10%; benzalkonium chloride 0.005%; edetate disodium; polysorbate 80; polyvinyl alcohol 1.4%; purified water; sodium phosphate dibasic; sodium phosphate monobasic; sodium thiosulfate; and HCl and/or NaOH to adjust pH to 6.8 to 7.5), 5 mL in sterile 10 mL ophthalmic dropper bottles; *Bleph-10*®, generic; (Rx)

Sulfacetamide Sodium 10% Ophthalmic Ointment (containing sulfacetamide sodium USP 100 mg; white petrolatum; mineral oil) in 3.5 g tubes; generic; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Sulfacetamide Sodium 10% and Prednisolone Acetate 0.2% Ophthalmic Suspension (containing sulfacetamide sodium 10%; prednisolone acetate 0.2%; benzalkonium chloride 0.004%; edetate disodium; polysorbate 80; polyvinyl alcohol 1.4%; potassium phosphate monobasic; purified water; sodium phosphate dibasic; sodium thiosulfate; and HCl and/or NaOH to adjust the pH to 6.6 to 7.2), 5 mL in 10 mL bottle, and 10 mL in 15 mL bottle; *Blephamide*®, (Rx)

Sulfacetamide Sodium 10% and Prednisolone Sodium Phosphate 0.23% Ophthalmic Suspension (containing sulfacetamide sodium 100 mg; prednisolone sodium phosphate 2.5 mg, equivalent to prednisolone phosphate 2.3 mg; poloxamer 407; boric acid; edetate disodium; purified water; NaOH and/or HCl to adjust pH to 6.5 to 7.5) in 5 mL and 10 mL; generic; (Rx)

Sulfacetamide Sodium 10% and Prednisolone Acetate 0.2% Ophthalmic Ointment (containing sulfacetamide sodium 10%, prednisolone acetate 0.2%, phenylmercuric acetate 0.0008%, white petrolatum, mineral oil, petrolatum (and) lanolin alcohol) in sterile 3.5 g ophthalmic tube; *Blephamide*®, (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Tacrolimus Ophthalmic

(ta-*kroe*-li-mus)

T-cell Inhibitor Immunosuppressive Agent

Prescriber Highlights

- ▶ Used to treat dogs with keratoconjunctivitis sicca (KCS), chronic superficial keratitis (CSK), and other immune-mediated ocular surface inflammatory conditions
- ▶ Useful for cyclosporine-resistant cases of KCS
- ▶ It is not commercially available; it must be compounded (*Protopic*® topical dermal tacrolimus ointment should **NOT** be used in the eye).
- ▶ The National Institute for Occupational Safety and Health (NIOSH) classifies tacrolimus as a hazardous drug; use appropriate precautions when handling.

Uses/Indications

Tacrolimus is used to stimulate tear production in dogs with quantitative and qualitative keratoconjunctivitis sicca (KCS). Tacrolimus 0.03% has been demonstrated to be equally efficacious as cyclosporine 2% for canine KCS.¹ Concentrations of 0.01% to 0.03% have been reported to be successful in controlling canine KCS.^{1–4} Tacrolimus can also manage other ocular surface inflammatory conditions, such as chronic superficial keratitis (CSK). Tacrolimus topical ophthalmic use has not been evaluated in cats or horses.

Pharmacology/Actions

Tacrolimus is a macrolide antibiotic that has a T-cell suppressant activity similar to that of cyclosporine, but the potency is ≈100 times that of cyclosporine. Reversible T-cell suppression results from inhibition of calcineurin by tacrolimus. Within T cells, the binding of tacrolimus with the intracellular protein (FK-506) forms a complex that subsequently binds to and inhibits intracellular calcineurin. Inhibition of calcineurin prevents transcription of pro-inflammatory cytokines and T-cell signal transduction, resulting in immunosuppressive effects.

Contraindications/Precautions/Warnings

Tacrolimus is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. The degree of systemic absorption from ophthalmic preparations is not known; use with caution in small-sized animals.

The National Institute for Occupational Safety and Health (NIOSH) classifies tacrolimus as a hazardous drug; personal protective equipment (PPE) should be used accordingly to minimize the risk for exposure.

Adverse Effects

Adverse effects associated with the ocular use of tacrolimus are rare but include orbital alopecia, which may be related to oil vehicles used to compound tacrolimus solutions.

Dosages

Keratoconjunctivitis sicca (KCS) (extra-label): 1 drop in each eye 2 times daily

Chronic superficial keratitis (CSK) (extra-label): 1 drop in each eye 2 times daily

Monitoring

- Clinical efficacy: baseline and periodic Schirmer tear test, reduction in mucoid ocular discharge and conjunctival hyperemia, and lack of progression (and possibly regression) of chronic inflammatory changes, such as corneal pigmentation, vascularization, and fibrosis

Client Information

- Use this medicine as directed by your veterinarian.

- This is a lifelong medication and must be given daily to be effective. Missed doses may cause rapid (within days) return of keratoconjunctivitis sicca (KCS).
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 minutes after applying this medication before applying other medications to the eye.
- Store this medicine as directed on the package labeling away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solutions.
- This medication is considered to be a hazardous drug as defined by the National Institute for Occupational Safety and Health (NIOSH). Talk with your veterinarian or pharmacist about the use of personal protective equipment when handling this medicine.

Storage/Stability

Store this product as directed on the package labeling away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Tacrolimus ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

It is not commercially available and must be prepared (compounded) by a qualified compounding pharmacist.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Tacrolimus ophthalmic preparations are not commercially available but may be compounded by qualified compounding pharmacists in concentrations varying from 0.01% to 0.03% in oil vehicles including olive oil, corn oil, and medium-chain triglyceride oil.

References

For the complete list of references, see wiley.com/go/budde/plumb

Tafluprost Ophthalmic

(*ta*-floo-prost) Zioptan®

Prostaglandin Analogue

Prescriber Highlights

- ▶ Used as a treatment for primary glaucoma in dogs
- ▶ Unlikely to be effective for feline or equine glaucoma
- ▶ Twice-daily dosing of tafluprost may result in less fluctuation of intraocular pressure (IOP).
- ▶ Preservative-free and may be better tolerated than preservative-containing prostaglandin analogues
- ▶ May worsen intraocular inflammation through prostaglandin-mediated mechanisms

Uses/Indications

Tafluprost is indicated to reduce intraocular pressure (IOP) in canine glaucoma. It has not been evaluated as a treatment for feline or equine glaucoma; however, other prostaglandin analogues are minimally effective in those species.

Pharmacology/Actions

Prostaglandin analogues are chemically modified versions of prostaglandin F₂-alpha. Prostaglandin F₂-alpha, an endogenous inflammatory mediator, mediates ocular hypotensive effects, presumably by increasing the outflow of aqueous humor through the trabecular meshwork and uveoscleral routes. In addition to lowering intraocular pressure (IOP), tafluprost induces miosis by activating prosta-

glandin receptors. In normal canine eyes, tafluprost significantly decreased IOP and induced miosis following a single administration.¹ When combined with betaxolol or timolol, a single administration of tafluprost to normal canine eyes produced a greater IOP reduction than tafluprost alone.²

Contraindications/Precautions/Warnings

Tafluprost is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Tafluprost should be used with caution in the presence of anterior uveitis and secondary glaucoma, particularly in cases already exhibiting significant miosis, as this treatment may exacerbate miosis, promote synechiae formation, and worsen glaucoma. Also, in dogs with primary anterior lens luxation, tafluprost should be used with caution, as the resulting miosis may trap the lens and/or vitreous within the pupillary aperture, thus creating pupillary block glaucoma. Tafluprost is not contraindicated in dogs with secondary posterior lens luxation, as the miosis may (favorably) trap the lens in the posterior segment.

Adverse Effects

Adverse effects associated with tafluprost reported in humans or animals include blurred vision, burning and stinging, conjunctival hyperemia, foreign body sensation, itching, increased pigmentation of the iris, and punctate epithelial keratopathy. Tafluprost may cause darkening of the iris through direct stimulation of iris melanocytes, and it may cause increased eyelash growth. Tafluprost may worsen intraocular inflammation. Macular edema, including cystoid macular edema, has also been reported after the use of tafluprost in humans.

Dosages

Glaucoma (extra-label): 1 drop in the affected eye(s) 1 to 2 times daily

Monitoring

- Clinical efficacy: maintenance of normal intraocular pressure (IOP)

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- The solution from one individual unit is to be used immediately in one or both eyes. Because sterility cannot be maintained after the individual unit is opened, the remaining contents should be discarded immediately after administration.
- Store this medicine in the original pouch in the refrigerator. After opening the pouch, the unopened single-use containers may be stored in the opened foil pouch for up to 28 days at room temperature. Protect this medicine from light and moisture. Discard any unused containers 28 days after first opening the pouch.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solutions.

Storage/Stability

Store this product in the original pouch in the refrigerator (2°C-8°C [36°F-46°F]). After opening the pouch, the unopened single-use containers may be stored in the opened foil pouch for up to 28 days at room temperature (20°C-25°C [68°F-77°F]). Protect this product from light and moisture. Discard any unused containers 28 days after first opening the pouch.

Compatibility/Compounding Considerations

Tafluprost ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status**VETERINARY-LABELED PRODUCTS:** NONE**HUMAN-LABELED PRODUCTS:**

Tafluprost 0.0015% Ophthalmic Solution (containing tafluprost 0.015 mg/mL; glycerol; sodium dihydrogen phosphate dihydrate; disodium edetate; polysorbate 80; water for injection; HCl and/or NaOH to adjust pH to 5.5 to 6.75 and an osmolality of 260 – 300 mOsm/kg), in 0.3 mL single-use containers in boxes of 30 or 90 pouches; *Zioptan*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Tetracaine Ophthalmic

(*teh*-trah-kane) *Altacaine*®, *Tetravisc*®

Topical Ocular Anesthetic

Prescriber Highlights

- ▶ Rapid-acting topical anesthetic for ocular diagnostic procedures; it should not be prescribed for therapeutic administration, as it will cause ocular surface toxicity
- ▶ Anesthetic duration of the solution is 20 to 30 minutes; viscous forms have a longer duration of action.
- ▶ Comparable efficacy to oxybuprocaine, as evaluated in dogs; increased efficacy relative to proparacaine, as evaluated in horses
- ▶ Increased ocular surface irritation as compared with proparacaine and oxybuprocaine
- ▶ Long-term use results in tear film instability and corneal epithelial and stromal damage, and it may progress to vision loss.

Uses/Indications

Tetracaine is indicated for use in ophthalmic diagnostic procedures, including measurement of intraocular pressure (tonometry), corneal and conjunctival sampling for cytology and culture, ocular ultrasonography, and gonioscopy. Tetracaine is indicated as an adjunctive anesthetic for therapeutic procedures in the hospital, including removal of foreign bodies and sutures from the cornea; eyelid and conjunctival surgeries; corneal grafting procedures; and cataract extraction. In veterinary patients, tetracaine may produce a more rapid and deeper plane of local anesthesia relative to proparacaine.¹ A single application of 1% solution in dogs induces anesthesia for a duration of 30 minutes,² whereas viscous tetracaine has a significantly longer duration of action than proparacaine solution and lidocaine gel.³ With a single application of 0.5% in horses, the anesthesia is 20 to 30 minutes long; additional applications of 1% solution prolongs anesthesia up to 1 hour.⁴ A formulation of viscous tetracaine decreases corneal sensitivity and has a longer duration of action (1 hour) in horses compared with proparacaine or nonviscous tetracaine.¹

Because of the likelihood of developing ocular surface toxicities with prolonged, repeated use, tetracaine should never be prescribed for use as an ocular analgesic. See *Adverse Effects*.

Pharmacology/Actions

Tetracaine induces local anesthesia by inhibiting neuronal depolarization, which prevents the generation of an action potential. This action occurs due to the binding of tetracaine to the intracellular portion of the voltage-gated sodium channel within the neuron, thus inhibiting the intracellular movement of sodium. Variations in the structure of different local anesthetics influence their relative potencies; however, all local anesthetics have increased efficacy in alkaline environments due to improved intracellular penetration (efficacy is decreased in the acidic environment of inflammation).

Contraindications/Precautions/Warnings

Tetracaine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Prolonged use of a topical ocular anesthetic leads to significant corneal epithelial and stromal disease, as well as tear film instability, and may result in loss of vision.⁵

Adverse Effects

Upon initial administration, instillation of tetracaine in the eye, at recommended concentration and dosage, may produce mild irritation, conjunctival redness, lacrimation, or squinting; however, some local irritation may occur several hours after the instillation, and this irritation may be greater than that seen with proparacaine.² Tetracaine appears safe for use in patients that are sensitive to other local anesthetics, but local or systemic sensitivity occasionally occurs and rarely may manifest as a pseudo-allergic (anaphylactoid) reaction mimicking an IgE-mediated vasodilatory event. Cardiovascular, respiratory, and CNS depressant effects that may result from other local anesthetics are rare with the appropriate use of topical anesthetics. Care should be exercised when administering tetracaine to small patients (ie, puppies, kittens, “pocket pets”) or animals known to have increased absorption of the drug (ie, amphibians). The use of tetracaine as a therapeutic analgesic leads to superficial punctate keratitis, which may progress to diffuse necrotizing keratitis, severe infiltrative keratitis, persistent epithelial defects, or damage to the Descemet membrane. In horses, reduction in the tear film breakup time has been reported, which is indicative of decreased stability of the precorneal tear film.⁶

Dosages

Ophthalmic procedures requiring local anesthesia (extra-label): 1 – 2 drops in the target eye(s) before examination or procedure. May repeat 1 drop every 5 to 10 minutes for 2 to 3 additional times. With a single application of tetracaine 0.5% in horses, the duration of anesthesia is 20 to 30 minutes; additional applications, application of 1% solution, or application of viscous formulation may prolong anesthesia for as long as 1 hour in horses. Additional doses should be avoided in small patients (ie, puppies, kittens, and “pocket pets”).

Monitoring

- Clinical effect. The onset of action is indicated by awake patients permitting diagnostic procedures to be performed and by a reduction in anesthetic requirements when tetracaine is administered preoperatively.

Client Information

- This medicine is only given in the veterinary clinic for use on your animal's eye to allow for examination.

Storage/Stability

Storage recommendations can vary; follow storage requirements listed on packaging. Most commonly, this product is recommended to be stored at controlled room temperature (20°C–25°C [68°F–77°F]) away from moisture and sunlight; do not freeze. Discard the medication if the solution changes color, if it becomes cloudy, or if it develops particles.

Compatibility/Compounding Considerations

Tetracaine ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status**VETERINARY-LABELED PRODUCTS:** NONE**HUMAN-LABELED PRODUCTS:**

Tetracaine HCl 0.5% Viscous Ophthalmic Solution (containing tetracaine HCl 0.5%; benzalkonium chloride; boric acid; edetate disodium; hypromellose; potassium chloride; sodium borate; NaCl; water

for injection) in 0.6 mL single unit dose plastic containers and 5 mL ophthalmic dropper bottles; *Tetravisc*®; (Rx) (not US FDA-approved) Tetracaine HCl 0.5% Ophthalmic Solution (containing tetracaine HCl 5 mg; boric acid; potassium chloride; edetate disodium; purified water; NaOH and/or HCl to adjust pH to 3.7 to 6; chlorobutanol 0.4%) in 15 mL ophthalmic dropper bottles; generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Timolol Ophthalmic

(*tim*-oh-lol) *Timoptic*®, *Betimol*®, *Istalol*®, *Combigan*®, *Cosopt*®
Nonselective Beta-Blocker

Prescriber Highlights

- ▶ Beta-blocker that decreases aqueous humor production to decrease intraocular pressure (IOP) in patients with glaucoma
- ▶ Although less effective than 0.5% concentrations, 0.25% concentrations of timolol are associated with fewer systemic adverse effects.
- ▶ Delays onset of glaucoma in the normal eye of dogs with unilateral glaucoma
- ▶ Use with caution in asthmatic patients due to the potential to induce bronchospasm.

Uses/Indications

Timolol is used to reduce intraocular pressure (IOP) in primary glaucoma and to prevent glaucoma in the normal eye of animals with unilateral glaucoma. The 0.25% concentration has minimal ocular hypotensive effects in animals and the 0.5% concentration is preferentially employed; however, the potential for systemic adverse effects is greater with the 0.5% solution.^{1,2} The maximal IOP-lowering effect is ≈20% to 25% in normotensive dogs^{3,4} and horses,⁵ and 10% in cats.⁶ Gel-forming 0.5% solutions of timolol (meant to increase ocular surface residence time by utilizing the interaction of xanthum gum with tears) reduce IOP by a mean of 5.3 mm Hg in dogs.⁷ However, IOP reduction in glaucomatous cats is less consistent.⁸ Reduction of IOP following the administration of timolol 0.5% in combination with dorzolamide 2% twice daily in normal eyes of horses is variable but has been reported to be as high as 17%.^{5,9,10} Although timolol may also be combined with alpha-adrenergic agonists (eg, brimonidine) to potentiate ocular hypotensive effects through the increased outflow of aqueous humor and decreased aqueous humor production, the adverse effects of brimonidine decrease its utility in small animal patients. This combination does not produce a clinically significant IOP reduction in normal horses.¹¹

Pharmacology/Actions

Timolol is a nonselective beta-adrenergic receptor blocking agent with intrinsic sympathomimetic activity, but it lacks membrane-stabilizing activity. Blockade of ciliary body beta-receptors, particularly beta-2 receptors, reduces intraocular pressure (IOP) by decreasing aqueous humor production. The exact mechanism of the ocular hypotensive effect of beta-blockers has not been definitively demonstrated; however, several mechanisms have been postulated for this effect including beta blockade of norepinephrine-induced tonic sympathetic stimulation, leading to decreased cAMP activation in the ciliary body, inhibition of Na⁺/K⁺-ATPase activity, or vasoactive activity on the iris and ciliary body. Topically administered beta-blockers also inhibit glutamine-induced increases in intracellular calcium and sodium thereby providing a potential protective effect against ischemic insult to the retina.¹²

Contraindications/Precautions/Warnings

Timolol is contraindicated in patients that have a history of hy-

persensitivity to it or any component of the formulation. Timolol should be used cautiously, if at all, in patients with atrioventricular block, bradycardia, heart failure, asthma, chronic bronchitis, or other cardiopulmonary conditions. Use this product cautiously in diabetic patients, as timolol can mask clinical signs associated with hypoglycemia.

Adverse Effects

Adverse effects reported in humans or animals include transient burning, stinging, and irritation upon application to the eye. In dogs, cats, and horses, the administration of timolol may cause slight miosis. It may trigger attacks in cats with asthma. Timolol produces significant reductions in heart rate in healthy cats, thus supporting its limited use in patients with known cardiopulmonary disease.¹

Dosages

Glaucoma or prophylaxis in the normal eye of unilateral glaucoma (extra-label): 1 drop in the affected eye(s) twice daily

Monitoring

- Clinical efficacy
- Intraocular pressure (IOP) and disease progression
- Respiratory rate in asthmatic patients; blood glucose in diabetic patients; heart rate and resting respiratory rate in cardiac patients (especially cats)

Client Information

- Use this medicine as directed by your veterinarian.
- This medication may delay the onset of glaucoma in the normal eye of dogs with glaucoma in their other eye.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Tell your veterinarian about any conditions or diseases your animal may have now or has had in the past, especially asthma or diabetes.
- Use proper administration techniques to avoid contamination of the medication. Do not touch the dropper (or ointment) tip or allow it to touch your animal's eye or any other surface to prevent contamination.
- Keep the cap tightly closed when the medication is not in use.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in solutions.
- Your veterinarian will need to monitor your animal's eye condition while taking this medicine. Do not miss these important follow-up visits.

Storage/Stability

Store this medication at controlled room temperature (15°C-25°C [59°F-77°F]) away from moisture and sunlight; do not freeze. Discard this medication if the solution color change, if it becomes cloudy, or if there are particles in the solution.

Compatibility/Compounding Considerations

Timolol maleate ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Timolol 0.25% Ophthalmic Solution (containing 2.56 mg timolol hemihydrate equivalent to 2.5 mg timolol; monosodium and disodium phosphate dehydrate to adjust pH; water for injection; benzalkonium chloride 0.01%), in 5 mL, 10 mL, and 15 mL sterile ophthalmic dropper bottles; *Betimol*®; (Rx)

Timolol 0.5% Ophthalmic Solution (containing 5.12 mg timolol hemihydrate equivalent to 5 mg timolol; monosodium and disodium

phosphate dehydrate to adjust pH; water for injection; benzalkonium chloride 0.01%), in 5 mL, 10 mL, and 15 mL sterile ophthalmic dropper bottles; *Betimol*®, *Istalol*®; (Rx)

Timolol 0.25% Ophthalmic Solution (containing 3.4 mg timolol maleate equivalent to 2.5 mg timolol; monobasic and dibasic sodium phosphate; NaOH to adjust pH; purified water; benzalkonium chloride 0.01%), in sterile 5 mL and 10 mL ophthalmic dropper bottles; *Timoptic*®, generic; (Rx)

Timolol 0.5% Ophthalmic Solution (containing 6.8 mg timolol maleate equivalent to 5 mg timolol; monobasic and dibasic sodium phosphate; NaOH to adjust pH; purified water; benzalkonium chloride 0.01%), in sterile 5 mL and 10 mL ophthalmic dropper bottles; *Timoptic*®, generic; (Rx)

Timolol 0.25% Gel-Forming Solution (containing 3.4 mg of timolol maleate equivalent to 2.5 mg timolol; gellan gum; tromethamine; mannitol; water for injection; benzododecinium bromide 0.012%), in sterile 5 mL ophthalmic dropper bottles; *Timoptic-XE*®, generic; (Rx)

Timolol 0.5% Gel-Forming Solution (containing 6.8 mg of timolol maleate equivalent to 5 mg timolol; gellan gum; tromethamine; mannitol; water for injection; benzododecinium bromide 0.012%), in sterile 5 mL ophthalmic dropper bottles; *Timoptic-XE*®, generic; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Timolol 0.5% and Dorzolamide 2% Ophthalmic Solution (containing 20 mg/mL dorzolamide (22.26 mg of dorzolamide HCl) and 5 mg/mL timolol (6.83 mg timolol maleate); benzalkonium chloride 0.0075%; sodium citrate; hydroxyethyl cellulose; NaOH; mannitol; water for injection, with a pH of ≈ 5.65 and an osmolarity of 242 – 323 mOsm/kg), in 10 mL ophthalmic dropper bottles; *Cosopt*®; (Rx)

Timolol 0.5% and Dorzolamide 2% Ophthalmic Solution (containing 20 mg/mL dorzolamide (22.3 mg of dorzolamide HCl) and 5 mg/mL timolol (6.8 mg timolol maleate); hydroxyethyl cellulose; mannitol; sodium citrate; NaOH (to adjust pH); water for injection), supplied in a foil pouch containing 15 low-density polyethylene 0.2 mL single-use containers; *Cosopt PF*®; (Rx)

Brimonidine Tartrate 0.2% and Timolol Maleate 0.5% Ophthalmic Solution (containing brimonidine tartrate 0.2%; timolol maleate 0.5%; benzalkonium chloride 0.005%; citric acid; polyvinyl alcohol; NaCl; sodium citrate; purified water; HCl and/or NaOH added to adjust pH to 5.6 to 6.6, with an osmolality of 280 – 330 mOsm/kg), in a 10 mL dropper bottle; *Combigan*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Tobramycin Ophthalmic

(tobe-ra-mye-sin) *Tobrex*®, *AkTob*®, *Tobradex*®, *Tobradex ST*®, *Zylet*®

Aminoglycoside Antimicrobial Agent

For systemic use, see *Tobramycin*

Prescriber Highlights

- Used as a treatment for susceptible bacterial infections of the ocular surface
- Do not confuse pure tobramycin ophthalmic preparations with preparations containing glucocorticoids.
- Fortified solutions of tobramycin greater than 13.5 mg/mL may be compounded for use in susceptible infections where commercially available products are not concentrated enough to achieve inhibitory concentrations.

Uses/Indications

Tobramycin ophthalmic is primarily used to treat ocular surface infections caused by *Pseudomonas aeruginosa*; however, susceptibility of *Pseudomonas* spp isolates may be decreasing.¹ Preparations of tobramycin in combination with a glucocorticoid are used primarily in the management of inflammatory conditions of the conjunctiva that are associated with or at risk for developing a superficial bacterial infection²; it may also be used postoperatively following ocular surface or intraocular surgical procedures.

The World Health Organization (WHO) has designated tobramycin as a Critically Important, High Priority antimicrobial for human medicine.³ The Office International des Epizooties (OIE) has designated tobramycin as a Veterinary Critically Important Antimicrobial (VCIA) Agent in equine species.⁴

Pharmacology/Actions

Tobramycin is actively transported into bacteria and binds to a specific aminoglycoside receptor on the 30S subunit of bacterial ribosomes and interferes with an initiation complex between mRNA and the 30S subunit to inhibit protein synthesis. Aminoglycosides are bactericidal, whereas most other antibiotics that interfere with protein synthesis are bacteriostatic. Tobramycin's spectrum of activity includes coverage against many aerobic gram-negative bacteria, including most species of *Escherichia coli*, *Klebsiella* spp, *Proteus* spp, *Pseudomonas* spp, *Salmonella* spp, *Enterobacter* spp, *Serratia* spp, and *Shigella* spp. Efficacy against *Staphylococcus* spp is limited, and efficacy against *Streptococcus* spp is negligible.^{5,6}

The antimicrobial activity of the aminoglycosides is enhanced in an alkaline environment; the acidic nature of pus or cellular debris can reduce efficacy.

The aminoglycoside antibiotics are inactive against fungi, viruses, and anaerobic bacteria.

Contraindications/Precautions/Warnings

Tobramycin is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Cross sensitivity of tobramycin with other aminoglycosides has been known to occur.⁷

Commercially available tobramycin solutions and ointments may not achieve inhibitory concentrations in the eye and may be either compounded in fortified concentrations (greater than or equal to 13.5 mg/mL) or injected subconjunctivally.^{8,9}

Adverse Effects

Adverse effects associated with tobramycin reported in humans or animals include ocular burning and irritation upon drug instillation, nonspecific conjunctivitis, conjunctival epithelial defects, and conjunctival hyperemia.

When tobramycin is used systemically, it can cause nephrotoxicity and neurotoxicity (ototoxicity and vestibular toxicity).⁷ Because cats are more susceptible to these toxic effects of aminoglycosides, it is recommended that all cats receiving topical tobramycin be monitored for systemic toxicity.

Dosages

Susceptible ocular surface infections (extra-label): 1 drop or ¼ – ½ inch strip in the affected eye(s) 4 to 6 times daily. The frequency may be increased to every 30 to 60 minutes for treating significant infections.

Management of inflammatory conditions of the conjunctiva associated with or at risk for developing a superficial bacterial infection | Postoperative ocular surface or intraocular procedures (extra-label): Using a tobramycin/glucocorticoid combination preparation, 1 drop or ¼ – ½ inch strip in the affected eye(s) 3 to 4 times daily. The frequency is decreased following control of ocular surface inflammation, with a corresponding reduced risk for infection.

Monitoring

- Clinical efficacy: corneal cytology with culture and susceptibility testing to evaluate for the persistence of bacteria. Evaluate clinical signs suggestive of continued infection.
- Baseline and periodic urinalysis with sediment evaluation to check for casts, which may be an early indicator for renal toxicity

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine in the refrigerator or at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product in the refrigerator or at controlled room temperature (2°C–25°C [46°F–77°F]) as directed in product labeling, away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Tobramycin ophthalmic preparations should not be mixed directly with any other ophthalmic medications. Tobramycin may be compounded into a fortified ophthalmic solution in concentrations of greater than or equal to 13.5 mg/mL by aseptically adding tobramycin injection in appropriate amounts to commercially available tobramycin ophthalmic solution.⁸

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Tobramycin 0.3% Ophthalmic Solution; *Tobrex*®; *Aktob*®; generic; (Rx)

Tobramycin 0.3% Ophthalmic Ointment; *Tobrex*®; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Tobramycin and Dexamethasone Ophthalmic Ointment (containing tobramycin 0.3% (3 mg), dexamethasone 0.1% (1 mg), chlorobutanol 0.5% (5 mg), mineral oil, white petrolatum), in a sterile 3.5 g tube; *Tobradex*®; (Rx)

Tobramycin and Dexamethasone Ophthalmic Suspension (containing tobramycin 0.3% (3 mg), dexamethasone 0.1% (1 mg), benzalkonium chloride 0.01%, tyloxapol, edetate disodium, NaCl, hydroxyethyl cellulose, sodium sulfate, sulfuric acid and/or NaOH to adjust pH, purified water), in a sterile 2.5 mL ophthalmic dropper; *Tobradex*®, generic; (Rx)

Tobramycin and Dexamethasone Ophthalmic Suspension (containing tobramycin 0.3% (3 mg), dexamethasone 0.05% (0.5 mg), xanthan gum, tyloxapol, disodium edetate, NaCl, propylene glycol, sodium sulfate, benzalkonium chloride 0.01%, NaOH and/or HCl to adjust pH, purified water), in a sterile 5 mL ophthalmic dropper; *Tobradex ST*®; (Rx)

Tobramycin 0.3% and Loteprednol 0.5% Ophthalmic Suspension in 5 mL and 10 mL; *Zylet*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Travoprost Ophthalmic

(trav-oh-prost) Travatan Z®

Prostaglandin Analogue

Prescriber Highlights

- ▶ Used to decrease intraocular pressure (IOP) in dogs with primary glaucoma or ocular hypertension
- ▶ Based on receptor differences, it is not effective in feline glaucoma and unlikely to be effective in equine glaucoma.
- ▶ Do not use it if intraocular inflammation is present, as it may worsen inflammation or lead to pupillary block glaucoma.
- ▶ Twice-daily dosing of travoprost results in less fluctuation of IOP.
- ▶ Preservative systems may cause ocular irritation and discomfort.

Uses/Indications

Travoprost is used in dogs to reduce intraocular pressure (IOP) to manage primary glaucoma. It has not been evaluated for efficacy in either cats or horses. In normotensive canine eyes, once-daily administration of travoprost reduced IOP comparably to latanoprost. It also resulted in miosis and conjunctival hyperemia.¹ In glaucomatous canine eyes, once- and twice-daily administration of travoprost 0.004% significantly reduced IOP;² whereas concentrations above 0.0001% were necessary to mediate ocular hypotension and miosis in a dose-response study.³

Pharmacology/Actions

Prostaglandin analogues are chemically modified versions of prostaglandin F₂-alpha, an endogenous inflammatory mediator that mediates ocular hypotensive effects by increasing outflow of aqueous humor through both the trabecular meshwork and uveoscleral routes. In addition to reducing intraocular pressure (IOP), travoprost leads to miosis.

Contraindications/Precautions/Warnings

Travoprost is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. Travoprost should be used with caution in the presence of anterior uveitis and secondary glaucoma, particularly in cases already exhibiting significant miosis, as this treatment may exacerbate miosis, promote synechiae formation, and worsen glaucoma. In addition, in dogs with primary anterior lens luxation, travoprost should be used with caution, as the resulting miosis may trap the lens and/or vitreous within the pupillary aperture, thus creating pupillary block glaucoma. Travoprost is not contraindicated in dogs with secondary posterior lens luxation, as the miosis may (favorably) trap the lens in the posterior segment.

Adverse Effects

Adverse effects associated with travoprost reported in humans or animals include blurred vision, burning and stinging, conjunctival hyperemia, foreign body sensation, itching, increased pigmentation of the iris, increased eyelash growth, and punctate epithelial keratopathy. Travoprost may cause darkening of the iris through direct stimulation of iris melanocytes, and it may worsen intraocular inflammation. Macular edema, including cystoid macular edema, has been reported after the use of travoprost in humans.

Dosages

Glaucoma (extra-label): 1 drop in the affected eye(s) once to twice daily

Monitoring

- Clinical efficacy: sustained reduction of intraocular pressure (IOP)

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait 5 minutes after applying this medicine before applying any other medicines to the eye.
- Store this medicine in the refrigerator or at room temperature. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solution.

Storage/Stability

Store this product in the in the refrigerator or at room temperature (2°C-25°C [36°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Travoprost preparations should not be mixed directly with any other medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Travoprost 0.004% Ophthalmic Solution (containing travoprost 0.04 mg; benzalkonium chloride 0.015%; polyoxyl 40 hydrogenated castor oil; tromethamine; boric acid; mannitol; edetate disodium; NaOH and/or HCl to adjust pH; water for injection), in sterile 2.5 mL and 5 mL ophthalmic dropper bottles; generic; (Rx)

Travoprost 0.004% Ophthalmic Solution (containing travoprost 0.04 mg; polyoxyl 40 hydrogenated castor oil; sofzia® [boric acid, propylene glycol; sorbitol; zinc chloride]; NaOH and/or HCl to adjust pH; purified water), in a sterile 2.5 mL ophthalmic dropper bottle; *Travatan Z*®; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Trifluridine Ophthalmic

Trifluorothymidine (TFT)

(try-**floor**-ih-deen) *Viroptic*®

Antiviral (Nucleoside Analogue)

Prescriber Highlights

- ▶ Topical antiviral agent used in cats with feline herpesvirus-1 (FHV-1)
- ▶ Effective as treatment of dogs with experimentally induced canine herpesvirus-1 (CHV-1) infection
- ▶ Must be used 4 to 5 times daily
- ▶ In vitro efficacy of trifluridine is greater than that of acyclovir and idoxuridine.
- ▶ May cause more irritation and pain upon administration as compared with other topical antiviral agents
- ▶ Trifluridine is a corneal toxin and should not be used for more than 3 weeks at a time.
- ▶ The National Institute for Occupational Safety and Health (NIOSH) classifies trifluridine ophthalmic as a hazardous drug; use appropriate precautions when handling.

Uses/Indications

Trifluridine is used for topical treatment of feline herpesvirus-1 (FHV-1) keratitis. It may also be used for canine herpesvirus-1 (CHV-1) keratitis.^{1,2} Evaluation of trifluridine's efficacy for equine herpesvirus-2 (EHV-2) keratitis has not been performed.

Pharmacology/Actions

Trifluridine interferes with DNA synthesis; however, its antiviral mechanism of action is not completely known. It is believed that trifluridine incorporates into viral DNA, as well as reversibly inhibits thymidylate synthetase, an essential enzyme in DNA synthesis. Trifluridine's in vitro efficacy is consistently greater than that of acyclovir and idoxuridine.^{3,4} Trifluridine is virustatic.

Contraindications/Precautions/Warnings

Trifluridine is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

The National Institute for Occupational Safety and Health (NIOSH) classifies trifluridine ophthalmic as a hazardous drug; personal protective equipment (PPE) should be used accordingly to minimize the risk for exposure.

Adverse Effects

Adverse effects reported in humans or animals include acute ocular irritation, including burning, corneal stippling, vascularization, and clouding. The irritation associated with administration of trifluridine may be more intense than the inflammation caused by FHV-1. Following prolonged use of trifluridine, ocular irritation characterized by follicular conjunctivitis, blepharitis with punctal swelling, bulbar conjunctival hyperemia, and corneal epithelial staining have also been reported. In any of these cases, the drug should be discontinued.

Dosages

Keratitis caused by feline herpesvirus-1 (FHV-1) or canine herpesvirus-1 (CHV-1) (extra-label): 1 drop in the affected eye(s) every 2 to 3 hours for 48 hours and then 4 to 5 times daily for a week beyond the resolution of clinical signs. Trifluridine is a corneal toxin and should not be used for greater than 3 weeks at a time.

Monitoring

- Clinical efficacy: resolution of clinical signs (eg, corneal ulceration, corneal vascularization, conjunctivitis)

Client Information

- Give this medicine to your animal as directed by your veterinarian.
- Do not use this medication for more than 3 consecutive weeks.
- Trifluridine may cause stinging upon administration.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait for 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication in the refrigerator away from moisture and sunlight. Do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solutions.
- This medication is considered to be a hazardous drug as defined by the National Institute for Occupational Safety and Health (NIOSH). Talk with your veterinarian or pharmacist about the use of personal protective equipment when handling this medicine.

Storage/Stability

Store in the refrigerator (2°C-8°C [36°F-46°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Trifluridine should not be mixed directly with any other ophthalmic medications. Trifluridine is not available as an ophthalmic ointment.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS:

Trifluridine aqueous 1% Ophthalmic Solution (containing trifluri-

dine 10 mg in an aqueous solution with glacial acetic acid and sodium acetate (buffers); NaCl; thimerosal 0.001%, in a 7.5 mL ophthalmic dropper bottle; *Viropic*®; generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Triple Antibiotic Ophthalmic (Neomycin/Polymyxin B/Bacitracin; Neomycin/Polymyxin B/Gramicidin) Double Antibiotic Ophthalmic (Bacitracin/Polymyxin B)

Broad-Spectrum Antimicrobial Combination Agent

Prescriber Highlights

- ▶ First choice for prophylactic antimicrobial coverage in dogs and horses with ocular surface inflammatory conditions, including ulcerative keratitis
- ▶ First choice for therapeutic antimicrobial coverage in dogs and horses with ulcerative keratitis until culture and sensitivity results are available
- ▶ Available as an ointment (neomycin/polymyxin B/bacitracin; polymyxin B/bacitracin) or solution (neomycin/polymyxin B/gramicidin)
- ▶ Some animals, particularly cats, may be hypersensitive to neomycin or polymyxin components in ointments.
- ▶ Do not confuse hydrocortisone-containing triple antibiotic ophthalmic ointment with those that lack corticosteroids.
- ▶ Prolonged use of antibiotic preparations may result in overgrowth of nonsusceptible organisms, including fungi.

Uses/Indications

Triple antibiotic ophthalmic products are used for topical treatment of superficial infections of the external eye and its adnexa caused by susceptible bacteria. Triple antibiotic ophthalmic is often used as prophylactic antimicrobial treatment prior to and following ophthalmic surgery. Triple antibiotic ophthalmic is also used for antimicrobial prophylaxis of corneal injuries and wounds. Triple antibiotic ophthalmic ointments may also contain a corticosteroid; thus, it is important to ensure that the appropriate formulation is selected. Specifically, corticosteroid-containing preparations should not be administered in patients with ulcerative keratitis. Triple antibiotic ophthalmic solutions substitute gramicidin for bacitracin. Double antibiotic ophthalmic preparations do not contain neomycin.

Pharmacology/Actions

Neomycin sulfate, an aminoglycoside, is actively transported into the bacterial cell and binds a specific receptor protein on the 30S subunit of bacterial ribosomes. It consequently interferes with the initiation complex between messenger RNA (mRNA) and the 30S subunit, inhibiting protein synthesis by inducing misreading of the mRNA. Aminoglycosides are bactericidal, whereas most other antibiotics that interfere with protein synthesis are bacteriostatic. Aminoglycosides are used primarily in infections involving aerobic, gram-negative bacteria, such as *Pseudomonas* spp, *Acinetobacter* spp, and *Enterobacter* spp. Infections caused by gram-positive bacteria, such as *Staphylococcus* spp and *Streptococcus* spp, are significantly less susceptible to neomycin.¹⁻⁴ Aminoglycosides are mostly ineffective against anaerobic bacteria, fungi, and viruses.

Polymyxin B increases the permeability of the bacterial cell membrane by interacting with the phospholipid components of the membrane for a variety of gram-negative organisms, with lesser efficacy

versus gram-positive organisms such as *Streptococcus* spp.¹⁻⁴ It is bactericidal.

Bacitracin, a polypeptide antibiotic, inhibits bacterial cell wall synthesis by preventing transfer of mucopeptides into the growing cell wall. It has bactericidal activity in vitro against a variety of gram-positive organisms and a few gram-negative organisms. In practice however, topical bacitracin is most useful for gram-positive bacillus infections. Importantly, bacitracin is effective against beta-hemolytic *Streptococcus equi* ocular infections, whereas efficacy against *Pseudomonas aeruginosa* is negligible.²⁻⁴ Bacitracin poorly penetrates the cornea and therefore is of limited value in deep corneal or intraocular infections.

In combination, neomycin sulfate, polymyxin B sulfate, and bacitracin zinc are considered active against *Staphylococcus aureus*; streptococci, including *Streptococcus pneumoniae*; *Escherichia coli*; *Haemophilus influenzae*; *Klebsiella* spp; *Enterobacter* spp; *Neisseria* spp; and *Pseudomonas aeruginosa*.

Gramicidin has activity against gram-positive organisms (not gram-positive bacilli) and some gram-negative organisms. Its mechanism of action is to bacterial cell wall permeability to inorganic cations.

Contraindications/Precautions/Warnings

Triple antibiotic combinations are contraindicated in patients that have a history of hypersensitivity to any component of the formulation. Some cats are hypersensitive to the neomycin or polymyxin component of triple antibiotic ointments.⁵ This hypersensitivity has not been associated with steroid-containing triple antibiotic products. Although there is concern that prolonged use of antibiotic preparations may result in overgrowth of nonsusceptible organisms, including fungi, minimal impact on the ocular surface microbiome of dogs or horses was noted after 7 days of administration to normal eyes.^{6,7}

Adverse Effects

Adverse effects reported in humans or animals include irritation and swelling upon instillation. Ophthalmic antibiotics, especially those containing neomycin sulfate and polymyxin B, may cause cutaneous sensitization and anaphylaxis, particularly in cats.⁵ Clinical signs associated with sensitization to topical antibiotics can include itching, reddening, and edema of the conjunctiva and eyelid but may progress to anaphylaxis. Clinical signs usually subside quickly on withdrawing the medication, and death has been rarely reported.⁸

Dosages

Ulcerative conjunctivitis or keratitis: 1 drop of solution or ¼ inch strip of ointment in the affected eye(s) 3 to 4 times daily until resolution of ulceration and infection or until microbial susceptibilities necessitate a change to another antimicrobial agent

Nonulcerative conjunctivitis or keratitis: 1 drop of solution or ¼ inch strip of ointment in the affected eye(s) 3 to 4 times daily until resolution of infection and inflammation

Monitoring

- Clinical efficacy. Resolution of conjunctivitis or keratitis as evidenced by cytology, cultures, or remission of clinical signs

Client Information

- Use this medication as directed by your veterinarian.
- Cats may have a severe allergic reaction to one of the components of this medication. Contact your veterinarian immediately if your cat has difficulty breathing or has facial swelling after receiving this medication.
- Use proper administration techniques to avoid contamination of the medication. Do not touch the dropper/ointment tip or allow it to touch animal's eye or any other surface to prevent contamination.

- Wait for 5 to 10 minutes after applying any other medicine to your animal's eye.
- Keep the cap tightly closed when the medication is not in use. Store this medicine at room temperature away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes or becomes cloudy.

Storage/Stability

Store at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Triple antibiotic ophthalmic preparations should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS:

Neomycin/Polymyxin B/Bacitracin Ophthalmic Ointment (containing neomycin sulfate equivalent to 3.5 mg neomycin base, polymyxin B sulfate equivalent to 10,000 polymyxin B units, bacitracin zinc equivalent to 400 bacitracin units, and white petrolatum and mineral oil, q.s.), in a 3.5 g ophthalmic ointment tube; *Vetropolycin*®; (Rx)

Neomycin/Polymyxin B/Bacitracin with Hydrocortisone Ophthalmic Ointment; each gram contains bacitracin zinc USP 400 units, neomycin sulfate 0.5% (equivalent to 3.5 mg neomycin base), polymyxin B sulfate USP 10,000 units, and hydrocortisone acetate USP 1% in a base of white petrolatum and mineral oil; *Vetropolycin HC*®; (Rx)

HUMAN-LABELED PRODUCTS:

Neomycin/Polymyxin B/Bacitracin Ointment (containing neomycin sulfate equivalent to 3.5 mg neomycin base, polymyxin B sulfate equivalent to 5,000 or 10,000 polymyxin B units, bacitracin zinc equivalent to 400 bacitracin units, and white petrolatum, q.s.), in a 3.5 g ophthalmic ointment tube; *Neosporin*®, *Lumi-Sporyn*®, generic; (Rx)

Neomycin/Polymyxin B/Gramicidin Ophthalmic Solution (containing polymyxin B equivalent to 1.75 mg polymyxin B base, polymyxin B sulfate equivalent to 10,000 polymyxin B units, gramicidin 0.025 mg/mL, alcohol 0.5%, thimerosal 0.001%, propylene glycol, polyoxyethylene polyoxypropylene compound, NaCl, and water for injection), in a 10 mL ophthalmic dropper bottle; *Neosporin*®, generic; (Rx)

Bacitracin and Polymyxin B Sulfate Ophthalmic Ointment (containing polymyxin B sulfate equivalent to 10,000 polymyxin B units, bacitracin zinc equivalent to 500 bacitracin units, and white petrolatum, q.s.), *AK-Poly-Bac*®, generic; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Neomycin and Polymyxin B Sulfates and Dexamethasone Ophthalmic Suspension (containing neomycin sulfate equivalent to neomycin 3.5 mg, polymyxin B sulfate 10,000 units, dexamethasone 0.1%, benzalkonium chloride 0.004%, hypromellose 2910 0.5%, sodium chloride, polysorbate 20, HCl and/or NaOH to adjust pH, and purified water), in a 5 mL sterile ophthalmic dropper bottle; *Maxitrol*®, *Dexasporin*®; (Rx)

Neomycin and Polymyxin B Sulfates and Dexamethasone Ophthalmic ointment (containing neomycin sulfate equivalent to neomycin 3.5 mg, polymyxin B sulfate 10,000 units, dexamethasone 0.1%, methylparaben 0.05%, propylparaben 0.01%, white petrolatum, and anhydrous liquid lanolin), in a sterile 3.5 g tube; *Maxitrol*®, generic; (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Tropicamide Ophthalmic

(troe-**pik**-uh-myde) Mydracil®, Paremyd®, Tropicacyl®
Anticholinergic Mydriatic; Cycloplegic Agent

Prescriber Highlights

- ▶ Mydriatic agent with mild cycloplegic action
- ▶ Used to produce short-term dilation for funduscopic examination or to prevent synechiae formation following cataract surgery
- ▶ More rapid onset and shorter duration of mydriasis compared with atropine
- ▶ Poor cycloplegic ability compared with atropine, resulting in minimal control of ciliary body muscle spasm
- ▶ Good corneal penetration due to lack of ionization at physiological pH
- ▶ May increase intraocular pressure (IOP); use with caution in patients with glaucoma

Uses/Indications

Tropicamide is used to produce rapid- and short-acting mydriasis for funduscopic evaluation. Tropicamide has also been used to break down synechiae in uveitis and to prevent synechiae formation following cataract surgery. It is not as effective as atropine in inducing cycloplegia and is consequently a poor choice for controlling ocular pain caused by ciliary spasm. Tropicamide has been demonstrated to overcome the miosis induced by topical ophthalmic latanoprost¹ and parenteral butorphanol² administration in dogs with normal eyes, without leading to elevated intraocular pressure, and thus can be used to facilitate posterior segment examination.

Pharmacology/Actions

Tropicamide causes anticholinergic effects by blocking the muscarinic receptors of the sphincter muscle of the iris and the ciliary body muscle, which produces pupillary dilation (mydriasis) and, to a lesser degree, paralysis of accommodation (cycloplegia). Tropicamide induces mydriasis in about 15 to 30 minutes, and the pupil returns to normal in 6 to 12 hours.

Contraindications/Precautions/Warnings

Tropicamide is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation. It should not be used in patients with primary glaucoma as it can induce increases in intraocular pressure (IOP).³

Adverse Effects

Adverse effects reported in humans include sensitivity to bright light, burning and irritation upon application, and superficial punctate keratitis. Dry mouth, tachycardia, headache, nausea, vomiting, pallor, psychotic reactions, CNS disturbances, muscle rigidity, behavioral disturbance, and vasomotor and cardiorespiratory collapse have also been reported. Hypersalivation has been reported in cats and dogs due to bitter taste. Tear production may be decreased for several hours after application of tropicamide ophthalmic in cats and horses,^{4,5} and intraocular pressure (IOP) may increase in cats and dogs.^{3,6}

Dosages

Mydriasis (extra-label): 1 to 2 drops repeated in 5 minutes in the eye to be examined to achieve pupillary dilation⁷

Mydriasis post-cataract removal surgery (extra-label): 1 to 2 drops into the affected eye(s) once to twice daily postoperatively to facilitate pupil mobility

Monitoring

- Clinical efficacy
- Confirm continued pupillary dilation
- Intraocular pressure (IOP)

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medication. Do not touch the dropper (or ointment) tip or allow it to touch your animal's eye or any other surface to prevent contamination.
- Dogs and cats may drool excessively after administration of this medication due to the bitter taste.
- Wash your hands after application to prevent accidentally getting the medicine into your eyes, which can cause pupil dilation.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Keep the cap tightly closed when the medication is not in use. Store this medicine at room temperature away from moisture and sunlight; do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in solutions.
- Your veterinarian will need to monitor your animal's eye condition while taking this medicine. Do not miss these important follow-up visits.

Storage/Stability

Store at controlled room temperature (20°C-25°C [68°F-77°F]) away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Tropicamide should not be mixed directly with any other ophthalmic medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Tropicamide 0.5% Ophthalmic Solution, USP (containing tropicamide 5 mg/mL; benzalkonium chloride 0.01%; NaCl; edetate disodium; purified water; HCl and/or NaOH to adjust pH to 4 to 5.8), in 3 mL and 15 mL dropper bottles; *Tropicacyl*®, generic; (Rx)

Tropicamide 1% Ophthalmic Solution, USP (containing tropicamide 10 mg/mL; benzalkonium chloride 0.01%; NaCl; edetate disodium; purified water; HCl and/or NaOH to adjust pH to 4 to 5.8), in 3 mL and 15 mL dropper bottles; *Mydracyl*®, generic; (Rx)

HUMAN-LABELED COMBINATION PRODUCTS:

Hydroxyamphetamine hydrobromide 1% and Tropicamide 0.25% Ophthalmic Solution (containing hydroxyamphetamine 1%; tropicamide 0.25%; benzalkonium chloride 0.005%; edetate disodium 0.015%; NaCl; HCl and/or NaOH to adjust pH to 4.2 to 5.8; purified water), in sterile 15 mL ophthalmic dropper bottle; *Paremyd*®, (Rx)

References

For the complete list of references, see wiley.com/go/budde/plumb

Vancomycin Ophthalmic

(van-kohe-mye-sin)

Glycopeptide Antimicrobial Agent

For systemic use, refer to *Vancomycin*

Prescriber Highlights

- ▶ Topical use reserved for treatment of ocular surface and intra-ocular infections caused by methicillin-resistant gram-positive organisms
- ▶ Not effective against gram-negative bacteria
- ▶ Not approved in an ophthalmic dosage form in the United States, although ocular dosage forms are approved in other countries

Uses/Indications

Vancomycin can be used for treating ocular surface infections caused by methicillin-resistant microorganisms, such as methicillin-resistant *Staphylococcus aureus* (MRSA) or *Staphylococcus pseudintermedius* (MRSP). Vancomycin compounded into 0.3% to 1% ophthalmic ointment has been shown to be effective in treating methicillin-resistant *Staphylococcus aureus* keratitis in a dog,¹ rabbits,² and humans; however, it has also been demonstrated that vancomycin is not necessary for all MRSA ocular infections, as chloramphenicol, rifampicin, and tetracyclines, among others, show varied in vitro efficacy.^{3,4} Vancomycin can also be administered intravitreally for severe cases of methicillin-resistant endophthalmitis. Intracameral injection of vancomycin nanoparticles has also demonstrated good prophylactic activity against MRSA in rabbits with minimal adverse effects.⁵

The World Health Organization (WHO) has designated vancomycin as a Critically Important, Highest Priority antimicrobial for human medicine.⁶

Pharmacology/Actions

The bactericidal action of vancomycin results from inhibition of cell wall biosynthesis, which is induced by binding of the antibiotic to cell wall precursor molecules. Vancomycin is not active in vitro against gram-negative bacteria (due to different mechanisms for cell wall formation), mycobacteria, or fungi.

Contraindications/Precautions/Warnings

Vancomycin is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

Adverse Effects

Adverse effects reported in humans or animals include burning or stinging on application, ocular discharge, regional pruritus, and conjunctival hyperemia. Nephrotoxicity, ototoxicity, dermal toxicities, vasculitis, and severe gram-negative colitis have been associated with systemic use of vancomycin; however, these adverse effects are not likely to occur following ophthalmic use.

Dosages

Methicillin-resistant bacterial conjunctivitis or keratitis (extra-label): 1 drop of solution or ¼ inch strip of ointment applied to the affected eye(s) 4 to 5 times daily

Monitoring

- Clinical efficacy: corneal cytology and culture to evaluate for bacteria. Evaluate clinical signs suggestive of progressive infection.

Client Information

- Use this medicine as directed by your veterinarian.
- Use proper administration techniques to avoid contamination of the medicine. Keep the cap tightly closed when the medicine is not in use.
- Wait for 5 to 10 minutes after applying this medicine before applying any other medications to the eye.
- Store this product as directed on the package labeling away from moisture and sunlight. Do not freeze.
- Do not use this medicine if the color changes, if it becomes cloudy, or if there are particles in the solutions.

Storage/Stability

Store this product as directed on package labeling away from moisture and sunlight; do not freeze.

Compatibility/Compounding Considerations

Vancomycin preparations should not be mixed directly with any other medications.

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Vancomycin ophthalmic preparations are not commercially available in the United States, but vancomycin ointments and ophthalmic solutions (0.3% to 1%) have been prepared (compounded) by qualified compounding pharmacists.

References

For the complete list of references, see wiley.com/go/budde/plumb

Voriconazole Ophthalmic

(vohr-uh-**kon**-uh-zohl) VFend®

Azole Antifungal Agent

Prescriber Highlights

- ▶ Used as a treatment for ocular surface fungal infections
- ▶ Although this medication is not commercially available as an ophthalmic preparation, the commercially available IV form may be administered topically to the eye.
- ▶ Penetrates cornea efficiently and achieves effective concentrations in the aqueous humor
- ▶ Considerably more effective against filamentous fungal organisms than polyene or other azole antifungal agents
- ▶ The National Institute for Occupational Safety and Health (NIOSH) classifies voriconazole ophthalmic as a hazardous drug; use appropriate precautions when handling.

Uses/Indications

Ocular administration of voriconazole is primarily used to treat fungal keratitis, most commonly in horses but also in other species.^{1,2} Voriconazole 1% solution for IV injection, prepared according to the package instructions, achieves effective antifungal concentrations in the aqueous humor while causing minimal to no irritation when it is administered to normal eyes of horses.³ The same or higher concentrations have been associated with treatment success following intra-corneal and subconjunctival injection in horses.⁴⁻⁶ Voriconazole was associated with successful control of keratomycosis due to both filamentous and yeast organisms in dogs^{2,7,8} and filamentous organisms in a cat.^{1,9} Intravitreal administration is used in humans as treatment for fungal endophthalmitis.¹⁰

Pharmacology/Actions

Voriconazole interacts with 14-alpha-sterol demethylase, a cytochrome P-450 enzyme involved in the conversion of lanosterol to ergosterol. Ergosterol is an essential component of the fungal cell membrane and preventing its synthesis may increase cellular permeability and cause leakage of cellular contents. Other possible effects of voriconazole include inhibition of endogenous respiration, purine uptake and the transformation of yeasts to mycelial forms, and impairment of triglyceride and phospholipid biosynthesis. Because voriconazole specifically targets oxidative enzymes of fungal species, it has higher efficacy and lower host toxicity. Voriconazole has a broad antifungal spectrum against most fungi and yeasts of veterinary interest. Sensitive organisms include *Aspergillus* spp, *Candida* spp, *Fusarium* spp, and *Aspergillus* spp.¹¹⁻¹³

Contraindications/Precautions/Warnings

Voriconazole is contraindicated in patients that have a history of hypersensitivity to it or any component of the formulation.

The National Institute for Occupational Safety and Health (NIOSH) classifies voriconazole ophthalmic as a hazardous drug; personal protective equipment (PPE) should be used accordingly to minimize the risk for exposure.

Adverse Effects

Adverse effects reported in humans or animals include transient burning or stinging on application. Although not reported following

topical administration, neurologic signs have resulted in cats following oral administration of voriconazole.¹⁴

Dosages

Fungal keratitis (extra-label): 1 drop to the affected eye(s) or 0.1 mL in the subpalpebral lavage (SPL) catheter, every 2 to 3 hours initially, followed by tapering of frequency based on clinical response. Voriconazole may be administered through a SPL catheter; however, its use with constant-rate infusion pumps in conjunction with a SPL catheter is not advised due to crystallization of voriconazole within the unit.⁴

Monitoring

- Clinical efficacy: corneal scrapings with cytology and cultures to determine to clear fungal organisms; resolution of clinical signs of infection

Client Information

- Use this medication up to 10 to 12 times daily initially, with a reduction in frequency occurring as the infection resolves.
- Use proper administration techniques to avoid contamination of the medication. Keep the cap tightly closed when the medication is not in use.
- Wait 5 minutes after applying this medication before applying any other medications to the eye.
- Store this medication in the refrigerator; do not freeze.
- Do not use this medication if the color changes, if it becomes cloudy, or if there are particles in the solutions.
- This medication is considered to be a hazardous drug as defined by the National Institute for Occupational Safety and Health (NIOSH). Talk with your veterinarian or pharmacist about the use of personal protective equipment when handling this medicine.

Storage/Stability

Store this medication in the refrigerator (2°C-8°C [35.6°F-46.4°F]); do not freeze. Reconstituted solutions (see **Compatibility/Compounding Considerations**) are stable for 14 days (in the absence of sterility testing) when refrigerated.¹⁵

Compatibility/Compounding Considerations

Voriconazole ophthalmic preparations should not be mixed directly with any other ophthalmic medications. Voriconazole is not approved in an ophthalmic dosage form. A 1% ophthalmic solution can be created by aseptically reconstituting the powder in a 200 mg vial for injection with 19 mL of sterile water for injection or normal saline, according to the package instructions.¹⁶

Dosage Forms/Regulatory Status

VETERINARY-LABELED PRODUCTS: NONE

HUMAN-LABELED PRODUCTS: NONE

Voriconazole is available as a 200 mg lyophilized powder (VFend®, generic) for reconstitution to a 1% solution following the addition of 19 mL water for injection. This reconstituted solution can be safely used for topical ophthalmic administration. Voriconazole may be compounded to a 1% ointment by a qualified, licensed compounding pharmacist.

References

For the complete list of references, see wiley.com/go/budde/plumb